PL-200 Microsoft Power Platform Functional Consultant

Complete English Study Guide and Scenario-Based Exam Trainer

Prepared from the current official Microsoft Learn content

Research cut-off: 2026-08-17

About This Guide

> **Important retirement notice — current on 2026-08-17:** According to the official Microsoft Study Guide and certification page, **Exam PL-200, the certification, and the renewal assessment retire on 2026-08-31 at 11:59 PM Central Standard Time**. Before scheduling the exam, check the Microsoft exam page again for availability and the correct time zone. Because of the retirement, this guide clearly separates the current exam blueprint from the Functional Consultant knowledge that remains useful after the exam retires.

This is not a short summary. It is a textbook built from **40 unique modules and 319 units/pages** in the current official Microsoft Learn preparation collection. It preserves the official order and adds instructor explanations for PL-200 scenario reasoning. You can open a Microsoft Learn module and study it alongside the matching Module and Unit headings in this guide.

Sources and research method

- Primary blueprint: Official PL-200 Study Guide
- Exam details: Exam PL-200
- Certification: Power Platform Functional Consultant Associate
- Learning content: the current `study_guide` catalog mapping from the exam page. After duplicate modules were removed, the catalog contained 40 unique modules and 319 units.
- Technical behavior: official Microsoft Learn documentation for Power Platform, Dataverse, Power Apps, Power Automate, Power Pages, Power BI, and ALM.

Research cut-off: 2026-08-17. Feature names and user-interface navigation can change, so check the official pages again before your exam date. This guide gives a numeric limit only when current official documentation confirms it. A claim that could not be confirmed is labeled [Unverified].

Current exam blueprint

Official skill domain	Weight	Study priority
Configure Microsoft Dataverse	25–30%	Study the data model, data management, and security reasoning in depth
Create apps by using Microsoft Power Apps	25–30%	Model-driven, Canvas, Power Pages feature selection
Create and manage logic and process automation	25–30%	Cloud flows, BPF, classic workflows, Business Rules, Power Fx, low-code plug-ins

Official skill domain	Weight	Study priority
Manage environments	15–20%	ALM, Solutions, email/SharePoint/document interoperability

The English blueprint version has been current since **2024-12-26**. The official change log describes only **Minor** changes to Manage the data model, Build Microsoft Power Pages, and Implement low-code logic; Microsoft does not publish a line-by-line difference. For that reason, this guide does not invent an unverified list of added or removed topics.

How to understand the current Learn collection

The exam preparation area is not only one traditional linear Learning Path. It maps a collection of standalone modules together with a validation Learning Path. This guide uses the deduplicated official order as its Learning Journey. Current pages include modern units such as Prompt columns, Copilot-assisted creation, an AI-powered Power Pages agent, and conversion to an agent flow. They are taught because they are part of the current official preparation content. When a feature does not have a dedicated blueprint objective, it is labeled as having lower explicit exam priority.

AI Builder / Copilot Studio note: The current PL-200 Skills Measured list does not contain a dedicated AI Builder or Copilot Studio objective, and the current recommended module set does not contain a dedicated module for either product. Do not treat them as core PL-200 objectives. This guide covers Power Pages agent, Copilot, and maker-assistance units only to the extent that they appear in the official collection.

How to use this guide

1. From each Module heading, open the linked official Microsoft Learn module.
2. Read the Unit/Page headings in the official order. Every unit has a canonical link and a clear exam focus.
3. Use the Instructor Deep Dive to learn the decision logic, not only the feature definition.
4. Use ⚠ PL-200 Exam Trap sections and scenarios to practise eliminating plausible but incorrect answers.
5. Use Tiny Details and Exam Memory Notes for final revision.
6. Attempt each Module Knowledge Check before reading its Answers & Explanations.

Functional Consultant decision lens

In a PL-200 question, the best answer is often not merely the option that is technically possible. Microsoft normally expects the option that meets the requirement with the least code while remaining maintainable, secure, solution-aware, and supported.

Requirement signal	First feature to evaluate	Why
Data-centric, process-rich internal app	Model-driven App	Dataverse metadata-driven UI, forms/views/security
Pixel-specific or task-focused UX	Canvas App	Free-form layout, Power Fx behavior
External/anonymous or partner website	Power Pages	Dataverse-backed website security model
Immediate form validation/visibility	Business Rule or Power Fx	Client/server declarative logic; no asynchronous flow delay
Cross-service, background, approval, notification	Power Automate Cloud Flow	Connectors, triggers/actions, long-running automation
Guided human stages	Business Process Flow	Stage/step guidance; not a general automation engine
Synchronous Dataverse transaction logic without pro code	Low-code plug-in	Server-side Power Fx plug-in logic
Reusable deployment package	Solution	Components, dependencies, ALM transport

Official Learning Journey Map

This map provides clickable navigation. Unit-level links appear inside every Module.

Part I — Environments and Microsoft Dataverse Foundation

- Module 1 — Create and manage environments in Dataverse
- Module 2 — Create tables in Dataverse
- Module 3 — Create and manage columns within a table in Dataverse
- Module 4 — Work with choices in Dataverse
- Module 5 — Create a relationship between tables in Dataverse
- Module 6 — Create and define calculation or rollup columns in Dataverse
- Module 7 — Load/export data and create data views in Dataverse
- Module 8 — Get started with security roles in Dataverse
- Module 9 — Use administration options for Dataverse

Part II — Model-driven Apps

- Module 10 — How to build your first model-driven app with Dataverse
- Module 11 — Get started with model-driven apps in Power Apps
- Module 12 — Configure forms, charts, and dashboards in model-driven apps

- Module 13 — Use specialized components in a model-driven form
- Module 14 — Customize the command bar

Part III — Canvas Apps and Power Apps Cards

- Module 15 — Get started with Power Apps canvas apps
- Module 16 — Customize a canvas app in Power Apps
- Module 17 — Navigation in a canvas app in Power Apps
- Module 18 — Create formulas to change behaviors in a Power Apps canvas app
- Module 19 — Connect to other data in a Power Apps canvas app
- Module 20 — Introduction to Power Apps cards

Part IV — Microsoft Power Pages

- Module 21 — Core components of Power Pages
- Module 22 — Access Dataverse in Power Pages websites
- Module 23 — Explore Power Pages design studio
- Module 24 — Integrate Power Pages websites with Dataverse
- Module 25 — Integrate Power Pages with web-based technologies
- Module 26 — Explore Power Pages design studio data and security features
- Module 27 — Set up Power Pages security
- Module 28 — Authentication and user management in Power Pages

Part V — Logic and Process Automation

- Module 29 — Best practices for error handling in Power Automate flows
- Module 30 — Use Dataverse triggers and actions in Power Automate
- Module 31 — Introduction to expressions in Power Automate
- Module 32 — Define and create business rules in Dataverse
- Module 33 — Set up low-code plug-ins
- Module 34 — Use common low-code plug-in scenarios

Part VI — Power BI Foundation

- Module 35 — Get started building with Power BI

Part VII — Solutions, ALM, and Interoperability

- Module 36 — Introduction to solutions for Microsoft Power Platform
- Module 37 — Use Microsoft 365 services with model-driven apps and Microsoft Dataverse

Part VIII — Validation Learning Path and Capstone

Part I — Environments and Microsoft Dataverse Foundation

Official-order range: Modules 1–9. Sequence intentionally preserved from the current exam catalog.

Module 1 — Create and manage environments in Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 1 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Manage environments (15–20%) / Dataverse context
Official module	Create and manage environments in Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	9
Instructor focus	Choosing the Environment type and lifecycle boundary; provisioning Dataverse; configuring roles, settings, copy, reset, and backup

Official Unit-by-Unit Journey

Unit 1 — Environments in Microsoft Dataverse

Concept / Why: Learn how Environment types, Dataverse provisioning, roles, settings, copy, reset, and backup define an application lifecycle and administration boundary.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Developer environments

Concept / Why: Learn how Environment types, Dataverse provisioning, roles, settings, copy, reset, and backup define an application lifecycle and administration boundary.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Who can sign up for the Power Apps Developer Plan?; Manage Developer environments.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Create and manage an environment

Concept / Why: Learn how Environment types, Dataverse provisioning, roles, settings, copy, reset, and backup define an application lifecycle and administration boundary.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Cases for creating Dataverse environments; Exercise.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Create an instance of a Microsoft Dataverse database

Concept / Why: Learn how Environment types, Dataverse provisioning, roles, settings, copy, reset, and backup define an application lifecycle and administration boundary.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Add users and roles within an environment

Concept / Why: Learn how Environment types, Dataverse provisioning, roles, settings, copy, reset, and backup define an application lifecycle and administration boundary.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Create a custom security role.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Manage settings in an environment

Concept / Why: Learn how Environment types, Dataverse provisioning, roles, settings, copy, reset, and backup define an application lifecycle and administration boundary.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Environment operations

Concept / Why: Learn how Environment types, Dataverse provisioning, roles, settings, copy, reset, and backup define an application lifecycle and administration boundary.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Change the environment type; Add a Microsoft Dataverse datastore; Delete an environment; Recover an environment; Reset an environment; Copy an environment.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Environments and the Dataverse database

What is the concept, and why is it used?

An Environment is a tenant-level container that separates Power Apps, Power Automate, Dataverse data, connections, gateways, apps, flows, and solution components. Use an Environment to define a region, type, security boundary, capacity boundary, and ALM stage. When a Dataverse database is provisioned in an Environment, that Environment gains Dataverse capabilities such as tables, Security Roles, Business Units, Solutions, and auditing.

A Functional Consultant must not treat an Environment as a simple **app folder**. Dev/Test/Prod separation, data residency, security-group restriction, capacity, DLP, backup and restore, and deployment strategy all depend on this decision.

Environment types and selection logic

Type	Best use	Important behavior / trap
Production	Live workloads and permanent organisational work	Apply production discipline, backups, and change control; it is not the best place for maker experiments
Sandbox	Test, UAT, training, and isolation	Suitable for copy and reset operations; remember that reset is destructive
Developer	Individual development and learning	Intended for Developer Plan or individual maker work, not a shared production workload
Trial	Time-limited evaluation	Do not choose it for a durable solution because it can expire
Default	The shared Environment created automatically for the tenant	It has rename and delete restrictions and often a broad maker footprint; do not make it the automatic destination for a controlled production ALM design
Microsoft Teams	Dataverse for Teams scenarios	Do not assume it has the same capabilities and limits as a full Dataverse Environment

A Developer environment is intended for individual development. Check Developer Plan eligibility, maker ownership, and tenant policies. A Sandbox or Production Environment is usually better for shared UAT or production. The Default Environment supports tenant-wide collaboration, but building an unmanaged sensitive business solution there without governance is a common design mistake.

Reasoning for Environment creation and configuration

1. **Capture the requirement:** identify the workload owner, region and data-residency need, expected users, whether Dataverse is required, the lifecycle stage, security group, and capacity.
2. **Choose type and region:** select the Environment type and region. Do not treat region as a casual field that can always be changed later.
3. **Provision Dataverse:** configure the language, currency, URL or name, database option, and initial security context. The base-currency choice affects financial columns later.
4. **Set the access boundary:** if you assign a Microsoft Entra security group at Environment level, users who are not members can be blocked from the Environment. This is not a substitute for Dataverse table privileges.
5. **Assign roles:** assign Environment Admin or Environment Maker separately from Dataverse Security Roles.

6. **Apply settings and governance:** configure relevant product, privacy, email, auditing, search, DLP, and feature settings.

Environment role vs Dataverse data access

Control	Grants	Does not automatically grant
Environment Admin	Environment administration	Do not blindly assume that it grants every Dataverse row privilege needed by a business user; Dataverse roles and context still matter
Environment Maker	Creation of resources such as apps, connections, custom connectors, and flows	It does not automatically grant Read or Write access to business-table rows
Dataverse Security Role	Table privileges, access depth, miscellaneous privileges	Environment-wide maker/admin powers

If an exam question says that a user can create apps but cannot read Account rows, there is no contradiction. Environment Maker capability is not Dataverse data access.

Environment operations

- **Backup and restore:** verify the restore target and the conditions under which the operation is allowed. Overwriting production data is a high-impact action.
- **Copy:** choose a configuration-only or full copy according to the scenario; the target is normally a non-production Sandbox. If sensitive production data is copied to non-production, plan masking and governance.
- **Reset:** current documentation describes Sandbox and Developer reset as destructive operations that permanently remove content. Do not reset apps, flows, connections, connectors, or data without the required backup and approval.
- **Delete and recover:** verify the Environment type, status, and current recovery window in the admin documentation. The Default Environment cannot be deleted.
- **Change type:** verify supported transitions and capacity restrictions at the time of the operation.

When NOT to create another Environment

If table security or Business Units can safely solve a row-separation requirement, do not create many Environments and add unnecessary operational overhead. However, Business Units inside one Environment may be insufficient when the requirement needs a legally isolated region, a separate lifecycle, an incompatible DLP policy, or a strong administration boundary.

⚠ PL-200 Exam Trap

Requirement: Priya must create apps but must not access confidential Dataverse rows.

Likely design: Assign Priya Environment Maker but do not give her a Dataverse role for the confidential tables. Do not confuse Environment Maker with data access.

Requirement: Users in two Business Units share the same application lifecycle, but their row visibility differs.
First choice: Evaluate Dataverse Business Units and Security Roles before creating separate Environments. Use separate Environments for lifecycle, administration, or data-residency boundaries.

Scenario-based learning

Scenario: Contoso must separate maker experiments, UAT, and the live sales app. The team also needs a full copy of live data for UAT testing.
Question: What Environment design should be used?
Reasoning: Separate the lifecycle stages, avoid production change risk, and use a non-production copy target.
Correct concept: Developer (individual build), Sandbox (UAT/copy/reset), Production (live).
Why alternatives fail: The Default Environment is not a controlled ALM destination; a Trial expires; and one Production Environment would keep risky changes, testing, and live data together.

Scenario: Alex has Environment Maker but a canvas app shows a Dataverse permission error.
Correct reasoning: Check app-resource access, the connection, Dataverse table privileges, and row access separately. Environment Maker alone does not grant access to table rows.

Tiny Details You Should Remember

- An Environment can contain Power Platform resources without a Dataverse database. Provision a database when Dataverse capabilities are required.
- If a user is not a member of the Environment security group, sharing the app may still leave the access path blocked.
- The Default Environment cannot be deleted.
- Reset is destructive. Verify the Sandbox or Developer context and current prerequisites in the admin center.
- Base-currency and database choices have downstream effects on solution design.

Exam Memory Notes

- Environment = lifecycle/admin/region/resource boundary.
- A Business Unit is a Dataverse row-security boundary; it is not an Environment.
- Environment Maker ≠ Dataverse data access.
- Dev → Sandbox/UAT → Production is the safer ALM flow.

- Default environment ≠ default place for every production solution.

Module Knowledge Check — Questions

1. A maker must be able to create apps but must not be able to read Dataverse business rows. Which role should be assigned first?
2. Which Environment type is suitable when a UAT database must be reinitialised and tested from a production copy?
3. Why can a user with Environment Maker still receive a Dataverse table-access error?
4. The company has two regions with different legal data-residency requirements. Are Business Units alone sufficient?
5. Should the Default Environment automatically be selected as the controlled production ALM destination?

Answers & Explanations

1. **Environment Maker.** It grants resource-creation capability. Control Dataverse row access separately with Security Roles.
2. **Sandbox.** It is the intended non-production type for copy, reset, and UAT workflows. Do not use a Production target for test resets.
3. Environment Maker does not grant table privileges. Check app sharing, the connection, Environment boundary, Security Roles, and row access separately.
4. Usually **no**. Region and data residency are Environment-level decisions; a Business Unit is a row-security hierarchy inside one Dataverse database.
5. **No.** The Default Environment is a broad shared space. A dedicated Production Environment can be more appropriate after governance, DLP, lifecycle, and security requirements are evaluated.

Primary sources: Environment overview, Create an environment, Database security.

Module 2 — Create tables in Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 2 → 12 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Configure Microsoft Dataverse (25–30%)
Official module	Create tables in Dataverse
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	12
Instructor focus	Choosing correctly among standard, custom, Activity, and Virtual tables, including the ownership type

Official Unit-by-Unit Journey

Unit 1 — Introduction to Microsoft Dataverse

Concept / Why: Learn how standard, custom, Activity, and Virtual tables differ and how the ownership type changes security and behaviour.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Microsoft Dataverse; Tables.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Table characteristics

Concept / Why: Learn how standard, custom, Activity, and Virtual tables differ and how the ownership type changes security and behaviour.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Table properties; Row keys; Create a table in Dataverse; Start with Copilot; Start from a blank table; Import a SharePoint list.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Dual-write vs. virtual tables

Concept / Why: A Virtual Table exposes external rows through a Dataverse table interface without replicating the data.

Configuration / How: Configure the provider and data source, then test external security, unsupported features, availability, and performance.

Exam relevance: High/Medium — a Virtual Table may be a poor fit when auditing, BPF, search, rollup, or offline capability is mandatory.

Official page subtopics: Dual-write; Virtual tables; When to use Dual-write vs. virtual tables.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - Create a Microsoft Dataverse table

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Create a custom table; Add and customize columns; Customize a view; Customize the main form.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Import data into your Microsoft Dataverse database

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Create sample import data; Import data into your Dataverse table.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Table relationships

Concept / Why: 1:N and N:1 are two views of the same relationship; the Lookup is stored on the N side.

Configuration / How: Configure relationship behaviour, cascading, deletion, mappings, hierarchy, and Lookup properties.

Exam relevance: High/Medium — Cascade and junction-table traps.

Official page subtopics: Relationship types; Many-to-one vs. One-to-many; Lookup columns; Create table relationships manually; Many-to-many relationships.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Exercise - Create table relationships

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Create a custom table and add a column; Create a relationship by using a lookup column; Add a one-to-many relationship.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Dataverse logic and security

Concept / Why: Learn how standard, custom, Activity, and Virtual tables differ and how the ownership type changes security and behaviour.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Business rules; Actions; Differences between canvas and model-driven apps; Dataverse security; Column-level security.

Official Microsoft Learn page: Open Unit 8

Unit 9 — Exercise - Create a custom table and import data

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; Use Microsoft Dataverse to store data; Create a custom table; Add a business rule; Import data from an Excel file.

Official Microsoft Learn page: Open Unit 9

Unit 10 — Dataverse auditing

Concept / Why: Auditing is a compliance feature that captures configured data changes and history.

Configuration / How: Configure auditing at Environment, Table, and Column levels, together with audit-log access and retention.

Exam relevance: High/Medium — Audit vs backup vs Flow run history trap.

Official page subtopics: Key concepts; Configure tables and columns for auditing in Power Apps; Activity logging and Microsoft Purview.

Official Microsoft Learn page: Open Unit 10

Unit 11 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 11

Unit 12 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official page subtopics: What you've learned.

Official Microsoft Learn page: Open Unit 12

Instructor Deep Dive — Dataverse Tables

What is a Table?

A `Table` is the metadata-driven Dataverse structure that stores rows for a business entity. Columns, Relationships, Views, Forms, Business Rules, auditing, ownership, keys, and security behaviour all depend on table metadata. Choose between extending a standard table and creating a custom table by comparing the business meaning with reusable platform capabilities.

Table types and exam selection

Table type	Use	Security/behavior highlights
Standard table	Platform-provided business concepts such as Account and Contact	Reuse existing columns, relationships, and features instead of creating a duplicate custom equivalent
Custom standard table	Organisation-specific business data	Choose User or Team ownership, or Organisation ownership, when the table is created
Activity table	A time-bound interaction such as a call, task, or appointment-like record	It uses Activity and party behaviour and is User or Team owned; it is not a substitute for every ordinary table
Virtual table	External-source data exposed as a Dataverse table without replication	It uses an Organization-owned model; external security remains important and many platform features are unavailable
Elastic table	A high-scale or high-throughput scenario	The current PL-200 blueprint does not name it explicitly; treat it as awareness unless the scenario specifically requires an elastic design

Standard vs custom

When a requirement matches Account, Contact, or Activity semantics, extending the standard table can provide useful integration, model-driven UX, and prebuilt relationships. Creating a redundant `Customer2` custom table merely because customers need one extra property is usually poor design. Create a custom table when the standard-table meaning or lifecycle does not fit.

Ownership — create time decision

Ownership	Who owns each row?	Privilege depth	Best for
User or Team	User/Owner Team/Entra group team	User, BU, Parent:Child, Organization	Sales cases, requests, records that need owner/assign/share
Organization	No user/team owner for individual rows	Organization or None	Reference/config data everyone with privilege accesses uniformly

Current documentation states that a custom table's ownership type cannot be changed after creation. Activity tables are User or Team owned. An Organization-owned table does not use Assign and Share as row-ownership operations. Choose User or Team ownership when records need individual owners or row-by-row ownership boundaries.

Virtual table deep reasoning

A Virtual Table queries an external data source and presents it through a Dataverse table experience; the data is not copied into Dataverse. This avoids duplication, but performance and availability depend on the external provider. Current official restrictions exclude many capabilities, including auditing, duplicate detection, rollup, formula and calculated columns, BPF, offline, Dataverse search, charts and dashboards, change tracking, and Column Security. Virtual tables behave as Organization-owned tables, so the external data source must implement security correctly.

Use when: the source system must remain the system of record, near-current access is required, duplication should be avoided, and a supported provider exists.
Do not use when: Dataverse row ownership, audit, offline, BPF, rollups, or full search/analytics feature set requirement essential.

Important table properties

- Display name/plural name, schema/logical name, primary name column.
- Ownership type.
- Enable activities, notes/attachments, connections, queues, quick create, audit, duplicate detection, change tracking, offline, Dataverse search as supported.
- After you enable a table feature, it might be impossible to disable it later, or the table might gain many dependencies. Read the maker interface warning and check the current documentation before you enable it.
- The Primary Name column is the record label shown in lookups, views, and forms. It is not the same as the primary-key GUID.

Standard vs Activity vs Virtual decision example

For a help-desk interaction that needs a due date, owner, and activity timeline behaviour, evaluate a custom Activity table or an existing activity type such as Task. To show external ERP inventory without copying the data, evaluate a Virtual Table. For a shared reference list such as organization codes, where different row owners are unnecessary, use an Organization-owned custom table.

⚠ PL-200 Exam Trap

“Every employee with Read privilege should see every Country code, and rows never need Assign/Share.” → **Organization-owned table**.
“Sales reps should own and reassign Opportunities.” → **User/Team-owned table**.
“External ERP rows must appear without replication, but auditing and a BPF are mandatory.” → These requirements conflict with Virtual Table limitations. Revise the requirements or choose an integration design that stores the data in Dataverse.

Scenario-based learning

Scenario: A company needs a master list of tax rates. All permitted users see the same rows; no per-row owner.

Correct concept: Organization-owned custom standard table.
Why: Simple org-wide privilege. Owner Team is unnecessary.

Scenario: ERP inventory must stay in ERP. Model-driven users need current quantities, but no Dataverse audit/rollup/offline requirement.
Correct concept: Virtual table if a supported provider/implementation exists.
Wrong alternative: Periodic Excel import creates copies and stale data; ordinary custom table makes Dataverse system-of-record unless sync is added.

Tiny Details You Should Remember

- Primary name column ≠ GUID primary key.
- Custom table ownership type cannot be changed after creation.
- Activity tables are User/Team-owned.
- Virtual tables do not replicate data and have significant feature/security limitations.
- Feature checkboxes can create dependencies or be irreversible; treat table properties as architecture decisions.

Exam Memory Notes

- Reference data with no owner → Organization-owned.
- Row owner/Assign/Share needed → User/Team-owned.
- External/no replication → consider Virtual table; check unsupported features.
- Extend a fitting standard table before duplicating its business concept.
- Activity table = activity semantics, not just any timestamped row.

Module Knowledge Check — Questions

1. User A should own and assign a row to User B. Which table ownership is required?
2. Which table type exposes external data without storing it in Dataverse?
3. Can custom table ownership be changed from Organization-owned to User/Team-owned after creation?
4. Why is a custom `Customer` table often wrong when Account already fits?
5. A requirement demands virtual data plus Dataverse auditing and BPF. What is the correct response?

Answers & Explanations

1. **User/Team-owned.** It supports owner-based access, Assign, Share, and team ownership.
2. **Virtual table.** It presents external rows through Dataverse metadata without replication.

3. **No.** Current documentation treats ownership type as a creation-time, non-changeable choice.
4. It duplicates standard semantics and may lose existing relationships/integrations. Extend Account when it correctly represents the business concept.
5. Identify a requirements conflict. Those Dataverse capabilities are restricted for virtual tables; use persisted integration or relax the requirements.

Primary sources: Types of tables, Virtual tables.

Module 3 — Create and manage columns within a table in Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 3 → 11 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Configure Microsoft Dataverse (25–30%)
Official module	Create and manage columns within a table in Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	11
Instructor focus	Column data types, Date and Time behaviour, primary and alternate keys, Autonumber, and the current Prompt column behaviour.

Official Unit-by-Unit Journey

Unit 1 — Define columns in Microsoft Dataverse

Concept / Why: Understand column data types, Date and Time behaviour, primary and alternate keys, Autonumber, and the current Prompt column behaviour.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Column types in Microsoft Dataverse

Concept / Why: Understand column data types, Date and Time behaviour, primary and alternate keys, Autonumber, and the current Prompt column behaviour.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Text; Single line of text; Multiple lines of text; Number; Whole number; Decimal.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Add a column to a table

Concept / Why: Understand column data types, Date and Time behaviour, primary and alternate keys, Autonumber, and the current Prompt column behaviour.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Work with Prompt columns and row summaries

Concept / Why: A Prompt column stores AI-generated output. It is included in the current Microsoft Learn content.

Configuration / How: Define the inputs and prompt, choose the trigger, test the result, and review AI credits, governance, backfill, and audit behaviour.

Exam relevance: High/Medium — Current prep but lower explicit blueprint priority.

Official page subtopics: Prompt columns; Prerequisites; Create a Prompt column; Write effective prompts; Add input columns to a prompt; Test and refine a prompt.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Primary column

Concept / Why: Understand column data types, Date and Time behaviour, primary and alternate keys, Autonumber, and the current Prompt column behaviour.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Restrictions that apply to columns in a table

Concept / Why: Understand column data types, Date and Time behaviour, primary and alternate keys, Autonumber, and the current Prompt column behaviour.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Create an auto numbering column

Concept / Why: Autonumber creates a human-readable generated identifier. It is different from the primary-key GUID.

Configuration / How: Design the format, seed, and prefix carefully, and avoid making integrations depend on a fragile display format.

Exam relevance: High/Medium — Autonumber vs Alternate Key.

Official page subtopics: Seeding a starting value.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Create an alternate key

Concept / Why: An Alternate Key uses one or more business columns to identify a unique row and support upsert operations.

Configuration / How: Verify supported data types, uniqueness, null behaviour, special characters, and Column Security restrictions.

Exam relevance: High/Medium — External ID/upsert signal.

Official Microsoft Learn page: Open Unit 8

Unit 9 — Exercises

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Add columns to a custom table; Rename a primary column; Add an autonumber column; Create a key.

Official Microsoft Learn page: Open Unit 9

Unit 10 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 10

Unit 11 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 11

Instructor Deep Dive — Columns, data types, keys, Prompt columns

Functional Consultant logic for column design

A column type is not only a user-interface choice. It affects validation, sorting and filtering, calculations, delegation, integrations, storage, and future changes. Making every value Text may look easy at first, but it removes the correct meaning and behaviour of numbers, dates, choices, and lookups.

Requirement	Good column choice	Common wrong choice
Currency with exchange/base value behavior	Currency	Decimal/Text
Exact count, no fraction	Whole Number	Floating Point
High-precision decimal that is not money	Decimal	Floating Point when exact precision matters
Approximate scientific range	Floating Point	Currency
Date with user timezone conversion	Date and Time – User Local	Text date
Calendar date that must not shift	Date Only	User Local DateTime
Fixed date/time independent of user zone	Time Zone Independent	User Local
One value from managed list	Choice	Free text
Link to another record	Lookup	Choice or copied text
File/image content	File or Image	Long text/base64

Date and Time behavior

- User Local: stored/handled with timezone conversion so different users may see local representation.
- Date Only: calendar date; no time-zone shift should change the day.
- Time Zone Independent: value presented without per-user conversion.

Before changing an existing column's behaviour, review both current and new values and every dependency. A birthday, a policy effective date, and an appointment time have different Date and Time meanings.

Primary name, primary key, alternate key

- **Primary key:** Dataverse-generated unique GUID identifier.
- **Primary name column:** human-readable label shown in lookups/views.
- **Alternate key:** one or more business columns uniquely identify record for integration/upsert.

Current official documentation lists these supported Alternate Key data types: Decimal, Whole Number, Single Line of Text, Date and Time, Lookup, and Choice. A column with Column Security enabled cannot be used in an Alternate Key. Because `NULL` values do not enforce uniqueness in the same way, design a nullable business key carefully. Some special characters can interfere with Web API PATCH or upsert URLs, so test the integration key design.

Autonumber

Autonumber generates a display identifier for a record; it does not replace the primary GUID. A supported format can include static text, date parts, random parts, or a sequential part. Changing the seed affects the next sequence value. In an exam scenario, “automatically create a human-friendly ticket number” suggests Autonumber, while “use the existing natural key from an external system” suggests Alternate Key.

Column restrictions and dependencies

Changing a data type or format can be unsupported or can risk data loss. If the column is used by views, forms, rules, flows, keys, relationships, apps, or solutions, a dependency can block deletion or change. Use the solution dependency viewer and component usage information. Required levels such as `Optional`, `Business Recommended`, and `Business Required` affect the data-entry experience and validation; a required level is not Column Security.

Prompt columns — current modern Learn content

The current Module 3 includes the official unit `Prompt columns and row summaries`. A Prompt column stores an AI-generated response in a row column. According to the current documentation, it runs when a row is created or when a specified input column is updated. Existing rows are not automatically backfilled or generated on demand, Prompt output is not audited, and entitlement and AI-credit prerequisites apply. An effective prompt contains a clear instruction, the relevant input columns, the required output format, and repeated testing and refinement.

Exam priority: Know the concept because it appears in the currently recommended Microsoft Learn material. However, the current Skills Measured list does not name Prompt columns separately, so give formula, rollup, and general column creation a higher explicit priority.

⚠ PL-200 Exam Trap

- “Unique, human-visible identifier” can mean **Autonumber**. “An external integration must use an existing business key for upsert” means **Alternate Key**.
- Updating Choice options does not create a Lookup relationship. If each option needs its own metadata, reporting, security, or lifecycle, a separate table and Lookup are usually better.
- A Business Required column asks users to supply data. Column Security controls whether a user can see or update the value. They are not the same feature.

Scenario-based learning

Scenario: Supplier system sends `SupplierCode` and needs upsert without knowing Dataverse GUID.
Correct concept: SupplierCode alternate key (valid type/uniqueness/security conditions verify).
Why not Autonumber: Dataverse generates that value, so it does not solve matching against a natural key that already exists in the source system.

Scenario: "Contract expiry date" must appear as the same calendar day in Tokyo and London.
Correct concept: Date Only, not User Local.

Tiny Details You Should Remember

- Primary name is a label; primary key is GUID.
- Alternate key with NULL does not give the same uniqueness guarantee for null rows.
- Secured columns cannot form alternate keys.
- Formula/Prompt/rollup behaviors are not interchangeable with ordinary stored input.
- Prompt column result is stored but, per current docs, not audited and not automatically backfilled.

Exam Memory Notes

- Integration upsert key → Alternate Key.
- Human ticket sequence → Autonumber.
- Date Only prevents timezone day shift.
- Lookup links records; Choice stores option value.
- Pick the semantic type at design time; type changes create risk/dependencies.

Module Knowledge Check — Questions

1. External system only knows a unique employee number. Which Dataverse feature supports upsert without GUID?
2. A case number should be generated automatically and be readable by agents. Alternate Key or Autonumber?
3. Which Date and Time behavior is best for a birthday?
4. Can a Column Security-enabled column be used in an alternate key?
5. Does a Prompt column automatically backfill every existing row when created?

Answers & Explanations

1. **Alternate Key.** It lets a supported business column combination identify the row.

2. **Autonumber.** It generates a display identifier. A key is about uniqueness/addressing, not necessarily sequence generation.
3. **Date Only.** Birthday is a calendar date and should not shift with timezone.
4. **No.** Current alternate-key documentation excludes secured columns.
5. **No.** Current Prompt column documentation says creation/update triggers do not automatically backfill existing rows; plan an explicit population approach if needed.

Primary sources: Column types, Date/time behavior, Alternate keys, Prompt columns.

Module 4 — Work with choices in Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 4 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Configure Microsoft Dataverse (25–30%)
Official module	Work with choices in Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	6
Instructor focus	Local and Global Choice, single-select and multi-select, and choosing between Choice and Lookup.

Official Unit-by-Unit Journey

Unit 1 — Define choice column

Concept / Why: A Choice stores a predefined option value. A Global Choice can be reused.

Configuration / How: Configure Local or Global, single-select or multi-select, labels, values, default, and display order.

Exam relevance: High/Medium — Choice vs Lookup based on whether option needs its own data.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Choice columns

Concept / Why: A Choice stores a predefined option value. A Global Choice can be reused.

Configuration / How: Configure Local or Global, single-select or multi-select, labels, values, default, and display order.

Exam relevance: High/Medium — Choice vs Lookup based on whether option needs its own data.

Official page subtopics: Local vs global choice; Local; Global; Choice vs Choices.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Standard choices column

Concept / Why: A Choice stores a predefined option value. A Global Choice can be reused.

Configuration / How: Configure Local or Global, single-select or multi-select, labels, values, default, and display order.

Exam relevance: High/Medium — Choice vs Lookup based on whether option needs its own data.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - Create a new choice or modify an existing choice

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Create a new custom choice; Create a new custom table; Add the choice to the custom table; Define a choice when adding new columns.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Choice, Choices, local vs global

A `Choice` column stores one selected predefined option. A `Choices` column stores multiple selected options. A Choice value is option metadata and a numeric value; it is not a relationship to another record.

Local vs global choice

Design	Scope	Choose when	Limitation / trap
Local Choice	One column or table context	Use when the options are one-off and do not need reuse	Consistency with other columns must be maintained manually
Global Choice	Reusable option set	Use when the same options are needed in several columns or tables	The Global Choice is not a relationship; a column references or synchronises with the option set
Lookup	A row in a separate table	Use when each option needs extra columns, ownership, security, lifecycle, or relationships	It adds more data-model complexity than a Choice

Do not confuse ordinary Choice columns with the standard Status and Status Reason behaviour. Status and Status Reason follow system patterns for a record’s lifecycle and active or inactive state. Reuse a standard Global Choice only when its meaning truly matches the requirement.

Choice vs Lookup reasoning

If a country list needs only a stable label and value, with no extra security, a Choice can be appropriate. If each country needs an ISO code, region, tax rule, active period, translations, owner, or related rate rows, create a Country table and a Lookup. A Lookup provides referential integrity and navigation to related data; a Choice is lighter.

Configuration checklist

- Display name/schema name; single vs multi-select.
- Choose a Local Choice or an existing or new Global Choice.
- Labels, underlying numeric values, ordering, default.
- Add/remove option impact: existing rows, integrations, flows, reports, translations.
- Consider option changes and solution layering inside a Managed Solution.

⚠ PL-200 Exam Trap

Do not choose Choice merely because a requirement says “users need a drop-down.” A later sentence such as “each category has an approver and an SLA” shows that a Category table and Lookup are needed.

Scenario-based learning

Scenario: Low, Normal, and High priorities must be used identically in both the Cases and Requests tables. No extra metadata is required.
Correct: Global Choice reused by local Choice columns.
Why not Lookup: Separate table and relationship unnecessary.

Scenario: When a user selects a branch, the app must retrieve its branch manager, address, and cost centre.
Correct: Create a Branch table and a Lookup. A Choice label cannot safely model those extra related attributes.

Tiny Details You Should Remember

- Choice stores option value; label can be localized.
- A Global Choice provides reusable, consistent options. It does not create relational data.
- Do not assume that a multi-select `Choices` column behaves exactly like a single `Choice` in connectors, forms, and formulas.
- Deleting/revaluing options can break integrations and historical meaning.

Exam Memory Notes

- Small fixed list → Choice.
- Same list many columns → Global Choice.
- Option needs its own data/security/relationships → Lookup.
- Choice ≠ Lookup; dropdown appearance alone is not the decision.

Module Knowledge Check — Questions

1. Departments have manager, budget, and related employees. Choice or Lookup?
2. The same simple Risk Rating list is required in four tables. Local or global Choice?
3. Does a global Choice create a 1:N relationship?
4. Which design supports selecting several tags in one column?
5. Why should an integration avoid depending only on displayed Choice labels?

Answers & Explanations

1. **Lookup to a Department table.** Department is an entity with attributes and relationships.
2. **Global Choice.** Reuse avoids divergent option lists.
3. **No.** It reuses option metadata; a Lookup creates the relationship.
4. **Choices (multi-select Choice).** Verify connector/formula support for the scenario.
5. Labels can be localized/renamed; integrations should understand stable option values and solution changes.

Primary source: Create and edit choices.

Module 5 — Create a relationship between tables in Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 5 → 8 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Configure Microsoft Dataverse (25–30%)
Official module	Create a relationship between tables in Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	8
Instructor focus	1:N, N:1, N:N, the location of the Lookup, cascading behaviour, mappings, and junction-table reasoning.

Official Unit-by-Unit Journey

Unit 1 — Relate one or more tables - Introduction

Concept / Why: Understand 1:N, N:1, N:N, the location of the Lookup, cascading behaviour, mappings, and junction-table reasoning.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Relationship types that are available in Microsoft Dataverse

Concept / Why: 1:N and N:1 are two views of the same relationship; the Lookup is stored on the N side.

Configuration / How: Configure relationship behaviour, cascading, deletion, mappings, hierarchy, and Lookup properties.

Exam relevance: High/Medium — Cascade and junction-table traps.

Official page subtopics: One-to-many relationship; Many-to-one vs. One-to-many; Lookup columns and relationships; Many-to-many relationship.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Create a one-to-many relationship between tables

Concept / Why: 1:N and N:1 are two views of the same relationship; the Lookup is stored on the N side.

Configuration / How: Configure relationship behaviour, cascading, deletion, mappings, hierarchy, and Lookup properties.

Exam relevance: High/Medium — Cascade and junction-table traps.

Official page subtopics: One-to-many relationship; One-to-many relationship behaviors; Behaviors; Actions; Types of behavior; One-to-many relationship mappings.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Create a many-to-many relationship between tables

Concept / Why: An N:N relationship allows many associations on both sides. If the association itself needs attributes, use a junction table.

Configuration / How: Configure the relationship or association, required privileges, and solution dependencies.

Exam relevance: High/Medium — Native N:N cannot model Grade/Quantity-like association data.

Official page subtopics: Many-to-many relationship; Connections.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Edit or delete relationships

Concept / Why: 1:N and N:1 are two views of the same relationship; the Lookup is stored on the N side.

Configuration / How: Configure relationship behaviour, cascading, deletion, mappings, hierarchy, and Lookup properties.

Exam relevance: High/Medium — Cascade and junction-table traps.

Official page subtopics: Edit a relationship; Delete a relationship.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Exercise - Create two tables and relate them by using a one-to-many relationship

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 7

Unit 8 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 8

Instructor Deep Dive — Relationships and cascading

Relationship models

Relationship	Meaning	Dataverse implementation	Example
1:N	One parent, many children	Lookup column is on the N/child table	One Account → many Contacts
N:1	Same relationship viewed from child	Child row has lookup to parent	Many Contacts → one Account
N:N	Many on both sides	Intersect relationship table maintained by Dataverse	Many Students ↔ many Courses

1:N and N:1 are not two different relationships; they are two viewpoints of the same relationship. The Lookup column is always on the “many” side. Use a native many-to-many relationship for a simple association. If the association needs attributes such as Enrollment Date or Grade, create an explicit junction table with two 1:N relationships.

Relationship behavior and cascading

Relationship behaviour controls whether an operation on the parent row is propagated to related child rows. Depending on the relationship and table type, configurable actions can include Assign, Share, Unshare, Reparent, Delete, and Merge.

Behavior	Practical meaning	Select when
Parental	Broad cascade behavior; child follows parent for supported actions	True composition/lifecycle dependency
Referential	Relationship link maintained; delete generally removes link rather than deleting child	Child can survive parent deletion
Referential, Restrict Delete	Parent delete blocked while related children exist	Orphaning forbidden; manual cleanup/reassign needed
Configurable Cascading	Per action cascade choice	Business needs mixed behavior

Cascade modes such as Cascade All, Cascade Active, Cascade User-Owned, Cascade None, Remove Link, Restrict apply as available. Never select Cascade Delete without asking whether historical child records must remain.

Hierarchical and mappings

A self-referential 1:N relationship, such as manager to employee or parent account to child account, can be marked hierarchical where supported. A table can have only one hierarchical relationship. Relationship mappings can prepopulate child columns when a related row is created from the parent context; mapping is not continuous synchronisation. Changing the parent value later does not automatically update the mapped value in an existing child row.

Connections vs N:N

Use `Connections` for a flexible, ad hoc relationship that needs connection roles and metadata. For a fixed, strongly modelled relationship and consistent reporting, use a proper Lookup or N:N relationship. For native N:N association and disassociation, the official privilege documentation requires you to consider Append privileges on both tables.

Delete/edit dependencies

Before deleting a relationship, check Lookup columns, forms, views, Business Rules, apps, solutions, and stored-data dependencies. Managed component restrictions or solution layers can block the change. The relationship schema name and its Lookup column are solution components.

⚠ PL-200 Exam Trap

- “Each order line belongs to one order” → Lookup on **Order Line**, not Order.
- You cannot add a `Quantity` column directly to a native N:N relationship. If Quantity is required, create a junction table.
- If the requirement says “delete the parent but preserve child history,” Parental Cascade Delete is wrong. Choose an appropriate Referential or Restrict strategy.
- Relationship mapping = create-time defaulting, not ongoing sync.

Scenario-based learning

Scenario: Consultants work on many Projects. Assignment needs StartDate, Role, and AllocationPercent.
Correct: Project Assignment table; Lookup to Consultant and Project (two 1:N).
Why native N:N wrong: Intersect association cannot hold those business attributes in maker design.

Scenario: Account cannot be deleted while active Contracts exist.
Correct: 1:N Account→Contract with Restrict Delete or configurable delete restriction.
Why Cascade Delete wrong: It destroys contracts; Remove Link could orphan them.

Tiny Details You Should Remember

- Lookup is created on N side.
- 1:N and N:1 are two views of the same relationship.
- Native N:N is best for attribute-free association.
- Relationship mapping does not keep values synchronized.
- Cascading can expose/change many rows; treat Share/Assign/Delete cascades as security/data-lifecycle decisions.

Exam Memory Notes

- Parent 1 → child N; child holds Lookup.
- N:N + attributes → junction table.
- Preserve child → Referential / Restrict Delete.
- Composition lifecycle → Parental (after risk review).
- Append = current row can attach; Append To = other row can attach to current.

Module Knowledge Check — Questions

1. Where is the Lookup stored in Customer 1:N Orders?
2. Student–Course enrollment requires Grade. Native N:N or junction table?
3. Which delete behavior blocks deleting a parent while children exist?
4. Does relationship mapping continuously synchronize parent and child columns?
5. Are 1:N and N:1 separate data relationships?

Answers & Explanations

1. On the **Order** (N/child) table.
2. **Junction table.** Grade belongs to the association.
3. **Referential, Restrict Delete** (or matching configurable restriction).
4. **No.** It can default values during related-record creation; it is not continuous sync.
5. **No.** They are the parent and child perspectives of the same relationship.

Primary sources: Table relationships, 1:N relationship behavior, Lookup columns.

Module 6 — Create and define calculation or rollup columns in Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 6 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Configure Microsoft Dataverse (25–30%)
Official module	Create and define calculation or rollup columns in Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	9
Instructor focus	Evaluation behaviour and limitations of Calculated, Rollup, Formula, and Prompt columns.

Official Unit-by-Unit Journey

Unit 1 — Introduction to roll up columns

Concept / Why: Understand the evaluation behaviour and limitations of Calculated, Rollup, Formula, and Prompt columns.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Calculate rollup; Mass calculate rollup.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Create a rollup column

Concept / Why: A Rollup column aggregates values from related rows in a 1:N relationship through a scheduled or on-demand system calculation.

Configuration / How: Configure the aggregate function, related table, filters, optional hierarchy, and refresh behaviour.

Exam relevance: High/Medium — Not real-time, not N:N, no rollup-on-rollup.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Introduction to calculated columns

Concept / Why: Understand the evaluation behaviour and limitations of Calculated, Rollup, Formula, and Prompt columns.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Common Usage Scenarios; Calculated column limitations.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Create a calculation column

Concept / Why: Understand the evaluation behaviour and limitations of Calculated, Rollup, Formula, and Prompt columns.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Create a rollup column

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Exercise - Create a calculated column

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Introduction to Power Fx formula columns

Concept / Why: Understand the evaluation behaviour and limitations of Calculated, Rollup, Formula, and Prompt columns.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Calculated, Rollup, Formula, Prompt

Feature comparison

Feature	Evaluation/storage model	Best for	Key limits/traps
Calculated column (classic)	Defined condition/actions; calculated result behavior	Simple row/related-lookup calculations in classic designer	Related FLS may not protect derived output; limitations on chains/types
Rollup column	Periodic/on-demand aggregation over 1:N related rows	SUM, COUNT, MIN, MAX, AVG across children or hierarchy	Not real-time; 1:N not N:N; system context; refresh limits
Formula column	Power Fx evaluated at retrieval/query time	Modern row-level formula	Output type fixed at creation; unsupported functions/types/dependencies apply
Prompt column	AI-generated stored output on create/input update	Text summary/classification-like maker scenario	Credits/entitlement; non-deterministic; no automatic backfill; lower explicit blueprint priority

Feature	Evaluation/storage model	Best for	Key limits/traps
Business Rule	Conditions/actions on fields/forms/server scope	Validation, visibility, requiredness, set/clear	Not an aggregation engine
Cloud Flow	Triggered async cross-record/service automation	Notifications, approvals, complex updates	Not immediate computed display; latency/retries

Rollup behavior

A Rollup column aggregates related data. Depending on the data type, common functions include SUM, COUNT, MIN, MAX, and AVG. System jobs calculate rollups, and users can manually recalculate them where the interface allows. Therefore, a requirement such as “the value must update immediately before save” does not fit a Rollup column.

Current official limits verified:

- Default maximum **200 rollup columns per environment** and **50 per table**.
- Online rollup calculation processes up to **50,000 related rows** per calculation.
- Hierarchical rollup depth maximum **10**.
- Rollup-on-rollup is not supported; N:N aggregation is not supported directly.
- Rollup updates do not trigger workflows as ordinary user updates do.

Check the linked current documentation before memorising numerical limits because Microsoft can change service limits.

Rollup calculation executes under system context. Record/column security perception can surprise you: output visibility can be secured, but aggregation behavior does not simply mirror viewer’s readable child subset. Security requirement design explicitly.

Calculated column behavior and security

Classic Calculated columns use a condition-and-action formula builder. They can use the current row and supported values from a related parent Lookup. An important official warning is that Field Level Security on a related source column might not automatically protect the calculated output. If the derived result is also sensitive, enable Column Security on the output column separately.

Formula column behavior

For a Power Fx Formula column, you cannot change the output data type after it has been selected or determined. You can change the formula only while preserving a compatible output type. A Formula column is computed when the value is retrieved, so do not treat it like a stored workflow update. Check the current documentation for supported functions, query and delegation restrictions, Date and Time behaviour, currency behaviour, and related-table limitations.

Selection reasoning examples

- For `LineTotal = Quantity * UnitPrice`, where the value is derived immediately from the same row, choose a Formula column, or a classic Calculated column if the requirement and supported functionality make that necessary. Prefer the modern supported option.
- `Account.TotalOpenRevenue = SUM(open Opportunity.EstimatedRevenue)` → Rollup.
- Form field required if Type=Vendor → Business Rule.
- When total crosses threshold send approval → Cloud Flow can respond to appropriate change; don’t rely on rollup update trigger without design.
- Human-readable AI row summary → Prompt column, after credit/governance/reliability review.

⚠ PL-200 Exam Trap

“Calculate sum of all related invoices” → Rollup, not Formula column.
“Value must be current immediately while user edits Quantity” → Canvas Power Fx/form logic or appropriate formula behavior; scheduled rollup is wrong.
“Calculated result uses a secured salary field” → secure the calculated output too; source FLS is not automatically inherited.

Scenario-based learning

Scenario: Account form displays count of related active Cases. Minute-by-minute exact real-time is not required.
Correct: Rollup COUNT over 1:N with filter.
Why formula wrong: Formula column does not perform arbitrary child aggregation.

Scenario: DiscountedAmount = Amount × (1 - DiscountPercent) for each row.
Correct: Formula column if supported types/functions meet requirement.
Why flow weaker: Async update creates latency and extra automation.

Tiny Details You Should Remember

- Rollup = periodic/on-demand, not guaranteed real-time.
- Rollup supports 1:N aggregation, not native N:N.
- Rollup-on-rollup unsupported.
- Formula output type is fixed after creation.
- Calculated source FLS does not automatically make output safe.
- Prompt column is current official Learn content but not a named PL-200 objective.

Exam Memory Notes

- Child aggregate → Rollup.
- Same-row Power Fx derivation → Formula.

- Conditional field behavior → Business Rule.
- Cross-service action → Cloud Flow.
- Sensitive derived data → secure output as well.

Module Knowledge Check — Questions

1. Which feature calculates the sum of related Order rows?
2. Does a Rollup update synchronously on every child edit?
3. Can a Formula column output type be freely changed later?
4. Does source Column Security automatically protect a calculated output?
5. Which feature is best for a deterministic same-row multiplication?

Answers & Explanations

1. **Rollup column.** It aggregates 1:N related rows.
2. **No.** It uses scheduled/system calculation plus supported on-demand recalculation.
3. **No.** The output type is fixed; edits must keep a compatible type.
4. **No.** Secure the derived column separately when sensitive.
5. **Formula column** when its supported Power Fx/type behavior fits. A flow is unnecessarily asynchronous.

Primary sources: Rollup columns, Formula columns, Calculated columns, Prompt columns.

Module 7 — Load/export data and create data views in Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 7 → 10 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Configure Microsoft Dataverse (25–30%)
Official module	Load/export data and create data views in Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	10
Instructor focus	System, Personal, Lookup, and Quick Find views, plus one-time and repeatable import, export, and Dataflow options.

Official Unit-by-Unit Journey

Unit 1 — View data in a table

Concept / Why: A View stores query, column, filter, and sorting metadata.

Configuration / How: Configure the System, Personal, Lookup, or Quick Find view type and include the required view in the app.

Exam relevance: High/Medium — View does not grant row access.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Create or edit views of data in a table

Concept / Why: A View stores query, column, filter, and sorting metadata.

Configuration / How: Configure the System, Personal, Lookup, or Quick Find view type and include the required view in the app.

Exam relevance: High/Medium — View does not grant row access.

Official page subtopics: Open existing views; Create a new view.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Dataverse data import options

Concept / Why: Import maps source data to Dataverse columns, Lookups, Choices, and owners in a one-time or managed data-load pattern.

Configuration / How: Clean the source, map required columns and keys, decide how to handle duplicates, and test error-file processing.

Exam relevance: High/Medium — Import Wizard vs Dataflow selection.

Official page subtopics: Data import wizard; Import from Excel in model-driven apps; Import data from Excel in Power Apps maker portal; Dataflows.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Load data into a table

Concept / Why: Understand System, Personal, Lookup, and Quick Find views and choose the correct one-time or repeatable import, export, or Dataflow option.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Dataverse data export options

Concept / Why: Export and Excel patterns differ depending on whether the requirement is a snapshot, online editing, a reusable template, or an integration.

Configuration / How: Configure the view, filters, permissions, file format, and expectations for a round trip back to Dataverse.

Exam relevance: High/Medium — Static export ≠ sync pipeline.

Official page subtopics: Export to Excel in model-driven apps; Open in Excel Online; Export with Word and Excel templates; Export data in Power Apps maker portal; Power Automate; Azure Synapse Link.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Export

Concept / Why: Export and Excel patterns differ depending on whether the requirement is a snapshot, online editing, a reusable template, or an integration.

Configuration / How: Configure the view, filters, permissions, file format, and expectations for a round trip back to Dataverse.

Exam relevance: High/Medium — Static export ≠ sync pipeline.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Add, update, or delete data in a table by using Excel

Concept / Why: Export and Excel patterns differ depending on whether the requirement is a snapshot, online editing, a reusable template, or an integration.

Configuration / How: Configure the view, filters, permissions, file format, and expectations for a round trip back to Dataverse.

Exam relevance: High/Medium — Static export ≠ sync pipeline.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Import data using Power Query

Concept / Why: A Dataflow with Power Query is a repeatable transform-and-load pipeline.

Configuration / How: Configure source credentials or a gateway, transformations, mappings, keys, schedule, and error handling.

Exam relevance: High/Medium — Repeatable ETL vs one-off import.

Official page subtopics: Create a dataflow.

Official Microsoft Learn page: Open Unit 8

Unit 9 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 9

Unit 10 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official page subtopics: References.

Official Microsoft Learn page: Open Unit 10

Instructor Deep Dive — Views, import/export, Excel, Dataflows

Views

A View is a saved query for table rows, together with selected columns and sort and filter definitions. Model-driven grids, Lookups, charts, dashboards, and exports can use View metadata.

View type	Ownership/availability	Use
System or Public View	A solution component that can be published for app users	Use for organization-wide standard operational views

View type	Ownership/availability	Use
Personal View	Owned by an individual user and shareable when privileges allow	Use for personal analysis and filtering; it is not an ALM solution component
Lookup View	Controls the presentation of Lookup search results	Use for the experience in which a user selects a related record
Quick Find view	Quick Find search columns/results	Search behavior and displayed fields
Associated view	Related records grid	Parent form related navigation/subgrid context

In the System View designer, configure columns, order, width, sorting, and filters, and then publish the View. A View never bypasses Security Role access; it returns only rows that the user is allowed to read. Sharing a Personal View does not grant access to the rows returned by that View.

Data movement option selection

Requirement	Preferred option	Reason
One-time small CSV/Excel load	Import wizard / maker import	Mapping, basic transformation, error file
User edits a manageable dataset	Export/Open in Excel where supported	Familiar grid; permissions still apply
Repeatable transform/load from sources	Dataflow + Power Query	Reusable ETL, mapping, scheduled refresh
Cross-service orchestration/event process	Power Automate	Trigger/action business process
Complex enterprise integration/high volume	Data Factory/API/integration architecture	Monitoring, scale, custom logic
Data must remain external	Virtual table	No replication, with limitations

Import planning: unique identifier/alternate key, required columns, Lookup mapping, Choice values, date formats/time zones, owners, duplicate behavior, row errors, and security. Data import does not magically fix poor source quality.

Excel patterns

- Export to Excel creates data file; static export does not stay synchronized.
- Open in Excel Online/editing patterns can update rows if permissions and supported columns allow.
- Excel/Word templates create formatted outputs, not a live ETL pipeline.
- Import/export honors user privileges and column/row access; exporting sensitive data is governance issue.

Dataflows and Power Query

Dataflow connects source, transforms with Power Query, maps target tables/columns, and refreshes. Use for repeatable ingestion. Decide append vs upsert pattern, alternate key, relationship order, error monitoring, ownership, gateway/credentials, and solution/environment deployment implications. Dataflow is not a UI form rule and not a per-record instant workflow.

⚠ PL-200 Exam Trap

- “Users need a company-standard view in every environment” → System View in Solution, not a Personal View.
- “Every night clean and load supplier CSV” → Dataflow/Power Query, not manual Import Wizard.
- “Share my Personal View so colleague can read hidden rows” → View sharing does not override row security.

Scenario-based learning

Scenario: Operations needs a saved `Open High Priority Cases` grid deployed Dev→Test→Prod.
Correct: System View in an unmanaged solution, export managed/unmanaged according to ALM target.
Why Personal View wrong: User-owned and not the standard solution component path.

Scenario: ERP extract needs type cleanup and scheduled upsert into Dataverse.
Correct: Dataflow using Power Query, keys, mapping, refresh monitoring.

Tiny Details You Should Remember

- View controls query/presentation; Security Role controls access.
- Personal View ownership/sharing does not make it a system solution component.
- Lookup and Quick Find views serve different UI purposes.
- Static Excel export is a snapshot.
- Dataflow needs keys/mapping/error strategy; refresh success must be monitored.

Exam Memory Notes

- Standard/deployable query → System View.
- User-specific query → Personal View.
- Repeatable ETL → Dataflow + Power Query.
- One-off small load → import wizard.
- View never grants hidden row access.

Module Knowledge Check — Questions

1. Which view type should be transported in a Solution?
2. Does sharing a Personal View grant access to rows the recipient cannot Read?
3. Which feature is best for scheduled transformation and load?
4. Is a normal Excel export continuously synchronized?
5. What design element helps repeatable upsert avoid duplicates?

Answers & Explanations

1. **System/Public View.** It is solution-aware metadata.
2. **No.** Security is evaluated separately.
3. **Dataflow with Power Query.** It is reusable ETL.
4. **No.** Treat it as a snapshot unless using a specific supported online/edit integration.
5. A stable **Alternate Key** (or GUID mapping) plus correct upsert/mapping strategy.

Primary sources: Import/export data, Create/edit views, Dataflows.

Module 8 — Get started with security roles in Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 8 → 10 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Configure Microsoft Dataverse (25–30%)
Official module	Get started with security roles in Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	10
Instructor focus	Privileges, access depth, Business Units, ownership, Teams, Sharing, Column and Hierarchy Security, and the effective-access algorithm.

Official Unit-by-Unit Journey

Unit 1 — Introduction to environment roles

Concept / Why: Understand privileges, access depth, Business Units, ownership, Teams, Sharing, Column and Hierarchy Security, and the effective-access algorithm.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Understand environment roles

Concept / Why: Understand privileges, access depth, Business Units, ownership, Teams, Sharing, Column and Hierarchy Security, and the effective-access algorithm.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Understand role types; Environment Admin role; Environment Maker role; Environments with a Dataverse datastore; Summary of resources available for predefined security roles.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Adding or disabling an environment user

Concept / Why: Understand privileges, access depth, Business Units, ownership, Teams, Sharing, Column and Hierarchy Security, and the effective-access algorithm.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Add a single user to the Environment; Add user to Security role.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Understand user security roles and security role defaults

Concept / Why: A Security Role is a cumulative set of privileges that grants table actions at defined access depths.

Configuration / How: Configure Create, Read, Write, Delete, Append, Append To, Assign, and Share privileges and their access depths.

Exam relevance: High or Medium — Having a privilege does not prove that the user can reach a particular row.

Official page subtopics: Default user security roles; Table privileges; Access levels; Permission settings; Privacy related privileges; Miscellaneous privileges.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Create a custom role

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Create a custom security role and assign to tables and users; Part 1 - Create a security role; Part 2 - Assign users to your security role.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Check the roles that a user belongs to

Concept / Why: Understand privileges, access depth, Business Units, ownership, Teams, Sharing, Column and Hierarchy Security, and the effective-access algorithm.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Run diagnostics; Access checker.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Configure Dataverse teams for security

Concept / Why: A Team is a security principal that groups users and can provide roles, ownership, or record-specific collaboration.

Configuration / How: Choose Owner Team, Access Team, or Microsoft Entra ID Group Team; then configure Business Unit, membership, Security Roles, and any record ownership.

Exam relevance: High/Medium — Access Team cannot own or hold Security Roles.

Official page subtopics: Types of teams; Team operations; Access your team’s page; Create a new team.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Configure Dataverse group teams for security

Concept / Why: A Microsoft Entra ID Group Team combines group-based membership with Owner Team capabilities.

Configuration / How: Verify the supported group type, the Team’s Business Unit, Security Roles, app sharing, and runtime membership.

Exam relevance: High/Medium — Entra Group Team vs Owner Team vs Access Team.

Official page subtopics: Using Microsoft Entra ID groups to manage a user’s app and data access; Provision and deprovision users; Remove user access at run time; Administer user security role; Secure user access to environments; Share Power Apps to team members of a Microsoft Entra ID group.

Official Microsoft Learn page: Open Unit 8

Unit 9 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 9

Unit 10 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 10

Instructor Deep Dive — Dataverse Security Masterclass

The core model for solving security questions

Always solve a Dataverse access question in two stages:

1. **Privilege check:** Does the user have the table-action privilege through a directly assigned role or a Team role? Check `Create`, `Read`, `Write`, `Delete`, `Append`, `Append To`, `Assign`, and `Share` as required.
2. **Record access check:** Based on privilege depth, ownership, sharing, Teams, and hierarchy security, does that privilege reach this specific row?

Security grants are **cumulative**. Direct roles, Team roles, sharing, and other grants combine to give the user their greatest effective access. If one role grants Organization-level Read, adding a more restrictive role does not subtract that access.

Business Units

Every Dataverse user belongs to one Business Unit. The root Business Unit represents the environment's Dataverse organization, and child Business Units can form a hierarchy for security boundaries and data segmentation. A Security Role belongs to a Business Unit context. How role copies and assignments are managed can depend on current modernised Business Unit feature settings.

Access depth	Row reach for user-owned table
User (Basic)	User's own rows, rows owned by teams they belong to, plus shared/delegated paths
Business Unit (Local)	Rows owned by users/teams in user's BU
Parent: Child Business Units (Deep)	User's BU and descendant BUs
Organization (Global)	All rows in environment/table
None	Privilege not granted

Organization-owned tables only meaningful Organization/None privilege depth because rows have no per-user owner.

Table privileges

Privilege	Meaning	Common trap
Create	New row create	Create alone does not guarantee later Read/Write
Read	View row	App access still needs table/row reach
Write	Modify existing row	Does not include Assign or Share
Delete	Delete row	Cascade behavior can affect related rows
Append	**Current row** can be attached/associated to another row	Mnemonic: “Append this”
Append To	**Other row** can be attached to current row	Mnemonic: “Append to this”
Assign	Change row owner	User/Team-owned tables/rows context
Share	Grant another principal selected rights to row	Sharing cannot exceed meaningful access/privileges and creates record-specific grant

Example: To set an Account Lookup on a Contact, the user generally needs Append on the Contact and Append To on the Account. For native N:N association and disassociation, the official privilege guidance requires you to consider Append privileges on both tables.

Users and role assignment

User must exist/provision in environment, licensed/access-eligible, enabled, and meet security group restrictions. Role assignment gives privileges; app sharing/role assignment/Dataverse access are separate checks. A user’s effective privileges combine direct roles and team roles.

New security roles have team privilege inheritance option:

- Team privileges only: User-level privileges obtained through team apply to team-owned/team-access context.
- Direct User (Basic) access level and Team privileges: team role’s User-level privileges can also give direct basic access behavior; current docs identify this as default for new roles.

Read the scenario wording carefully; team inheritance configuration can change outcomes.

Team comparison

Team type	Can own rows?	Security Roles?	Membership	Best use
Owner Team	Yes	Yes	Manually managed	Stable group ownership and role-based access
Access Team	No	No	Record-specific; manual or auto-created template	Collaborate on selected rows without ownership
Microsoft Entra ID Group Team	Yes	Yes	Group membership dynamically maintained	Central identity group controls app/data access; team owns rows

Access Team rights are granted through sharing for particular row; official team overview describes record rights such as Read/Write/Append. Access Team cannot be assigned a Security Role and cannot own rows. Owner Team can own records and get roles. Entra group team acts like an owner team with membership coming from supported Entra group type (Security or Microsoft 365; assigned/dynamic user groups supported; dynamic device group not supported in current docs).

Owner Team → Access Team conversion requires records reassigned and removes roles; official docs describe conversion as one-way. Treat as redesign, not a casual toggle.

Ownership, sharing, and access teams

- Owner automatically has access subject to baseline table privilege evaluation.
- Security Role depth provides broad rule-based reach.
- `Share` gives selected rights to selected row/principal; record-by-record administrative overhead.
- Access Team is organized record-sharing collaboration, not ownership.
- Owner Team is suited when the group collectively owns work and same access repeats across many rows.

Requirement: “Users must access records without becoming owners.”
If ad-hoc single recipient/small number → direct Sharing.
If changing group collaborates per record → Access Team.
If stable group should collectively own and role grants apply → Owner Team (but requirement says not owners, so wrong unless team ownership is acceptable).
If all rows in BU should be visible → Security Role depth, not thousands of shares.

Column Security / Field Security Profiles

Enable Column Security on eligible column, then a Field Security Profile/Column Security Profile grants per column `Read`, `Create`, `Update` as supported. It is organization-wide layer and applies across apps/API. System Administrator can bypass. Not every system column can be secured. Column Security supplements row/table security; it cannot grant row access. User must first reach row/table.

Recent masking capabilities exist in current docs, but exam core remains secure column + profile permissions. Calculated derived fields need separate review because source FLS may not carry over.

Hierarchy Security

Hierarchy Security extends—not replaces—role security. It is disabled by default. Manager hierarchy uses manager relationship; Position hierarchy uses configured positions. Users need at least User-level Read on table; hierarchy can expand access. Official docs describe direct reports potentially Read/Write/Append/Append To and nondirect reports read-only subject to configured depth/model. Exact scenario must check model, depth, relationship, and base privilege.

App access

Model-driven app share/security role access is another gate. User might have row privileges but app not shared/role not assigned. Canvas app sharing doesn't automatically grant data source privileges. Security solve checklist:

1. Environment eligibility/security group.
2. License and enabled user.
3. App shared / model-driven app role access.
4. Table privilege and correct depth.
5. Row access path: owner, BU depth, team, share, hierarchy.
6. Column Security.
7. Related record Append/Append To and cascading dependencies.

John/Sara worked scenario

Facts: John owns Case C1. John and Case C1 are in Sales BU context. Sara is member of Owner Team T in Support BU. T has a Security Role with Case: Read/Write at Business Unit depth; Delete/Assign/Share = None. No record share. Sara has no other Case role.

Step 1 — privilege: Through T, Sara has Read/Write privileges. Delete/Assign/Share absent, so those actions fail immediately.
Step 2 — record reach: C1 is owned by John, not team T, and outside T/Sara's BU. BU-depth role does not reach Sales BU. No share/hierarchy/organization grant. Therefore Sara cannot Read/Write C1 despite possessing the action names.

If C1 is assigned to Team T, team ownership gives members access consistent with role/inheritance: Sara can Read/Write, still not Delete/Assign/Share. If C1 is shared with T granting Read only, Sara can Read but not Write—assuming baseline role check permits Read. If role depth becomes Organization for Read only, Sara can Read every Case but still cannot Write C1 because Write remains BU depth. Broad Read cannot be removed by another restrictive role.

⚠ PL-200 Exam Traps

- User has Write at BU depth ≠ user can edit every row. Need row reach.
- Share is not a replacement for baseline table privilege; effective access evaluation still considers privileges.
- Access Team cannot own records or have Security Roles.
- Security Role cannot “deny” access granted by another role; grants union together.
- Business Unit is not a group ownership container; Team is not a BU hierarchy node.
- Column Security can restrict a column but cannot make an otherwise inaccessible row visible.

Scenario-based learning

Scenario 1: Sales reps see own Opportunities; managers see all opportunities in their BU; regional director sees descendant BUs.
Correct: User-level for reps, BU depth for managers, Parent:Child BU depth for regional director, with correct BU hierarchy.
Why sharing wrong: Repeated broad pattern; per-row shares are fragile.

Scenario 2: A rotating five-person legal group needs Read/Write on one Case but must not own it.
Correct: Access Team/template with appropriate record rights.
Why Owner Team wrong: It introduces ownership/role semantics. Direct sharing five times works but Access Team better manages changing collaboration.

Scenario 3: HR users can read Employee rows but only payroll specialists can see Salary.
Correct: Row Read role + Column Security enabled on Salary + profile granting Read to payroll specialists.
Why separate form wrong: Forms are UX, not data security; API/views could expose it.

Tiny Details You Should Remember

- Access is cumulative; greatest grant wins.
- Organization-owned tables have Organization/None depth, not owner scopes.
- Append/Append To both often required to set a Lookup.
- Owner/Entra group teams can own rows; Access Teams cannot.
- Dynamic device Entra group is not supported for Dataverse group team membership in current docs.
- Hierarchy Security is disabled by default and needs base user-level Read.
- Main form security by role affects form selection, not row authorization.

Exam Memory Notes

- First privilege, then record access.
- BU = security boundary; Team = group of users/owner/access mechanism.
- Owner Team → owns + roles.
- Access Team → selected rows, no ownership, no roles.
- Entra Group Team → group membership + owner-team capabilities.
- Append = this row attaches; Append To = others attach here.
- Column Security restricts fields after row access is established.

Module Knowledge Check — Questions

1. Sara has Case Write at BU depth, but Case is owned in another unrelated BU. No share/team ownership. Can she edit it?
2. Which Team type gives changing collaborators rights to a single row without ownership?
3. Can a restrictive Security Role remove Organization Read granted by another role?
4. What privileges are commonly required to set a Contact's Account lookup?
5. A user can query rows by API but cannot open the model-driven app. What extra gate should be checked?
6. Can Column Security grant access to a row the user cannot Read?
7. Which team type uses Microsoft Entra group membership and can own rows?
8. What minimum base privilege is required before Hierarchy Security can expand Read access?

Answers & Explanations

1. **No.** She has the action privilege but no access path to that row.
2. **Access Team.** It receives record-specific shared rights and cannot own the row.
3. **No.** Grants are cumulative; the broader grant wins.
4. `Append` on Contact and `Append To` on Account, plus necessary Read/Write/row access.
5. Model-driven **app sharing / associated Security Role access**.
6. **No.** Column Security restricts columns after table/row access; it does not grant the row.
7. **Microsoft Entra ID Group Team.** It can have roles and own rows like an owner team.
8. At least **User-level Read** on the table, per current hierarchy security documentation.

Primary sources: Dataverse security concepts, Security roles and privileges, How record access is determined, Manage teams, Group teams, Column Security, Hierarchy Security.

Module 9 — Use administration options for Dataverse

Your current position: Part I — Environments and Microsoft Dataverse Foundation → Module 9 → 11 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Configure Microsoft Dataverse (25–30%)
Official module	Use administration options for Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	11
Instructor focus	Admin centre, search and indexing, auditing, Duplicate Detection, Bulk Delete, capacity, and retention.

Official Unit-by-Unit Journey

Unit 1 — Introduction to Microsoft Power Platform Admin Center portal

Concept / Why: Understand admin centre tasks, search and indexing, auditing, Duplicate Detection, Bulk Delete, capacity, and retention.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Use Microsoft Power Platform Admin Center portal

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Tenant storage capacity

Concept / Why: Understand admin centre tasks, search and indexing, auditing, Duplicate Detection, Bulk Delete, capacity, and retention.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Advanced Customization options in Power Apps Portal

Concept / Why: Understand admin centre tasks, search and indexing, auditing, Duplicate Detection, Bulk Delete, capacity, and retention.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Dataverse Search

Concept / Why: Search indexing, relevance, and table or View search experiences help users find records; they never bypass security.

Configuration / How: Enable search, select searchable tables and columns, review index capacity and status, and test with realistic security roles.

Exam relevance: High or Medium — Distinguish a View from Search and allow for indexing delay.

Official page subtopics: Single table search; Advanced find; Categorized search; Dataverse search; Manage search index.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Auditing

Concept / Why: Auditing is a compliance feature that captures configured data changes and history.

Configuration / How: Configure auditing at Environment, Table, and Column levels, together with audit-log access and retention.

Exam relevance: High/Medium — Audit vs backup vs Flow run history trap.

Official page subtopics: Environment settings; Audit settings; Audit summary; Enabling auditing on tables and columns; Configure multiple columns; Deciding when to enable or disable auditing.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Duplicate detection

Concept / Why: Duplicate Detection rules identify or warn about possible matches. They are not the same as a unique Alternate Key contract.

Configuration / How: Enable Duplicate Detection, create and publish a rule, and configure the required jobs or data operations.

Exam relevance: High or Medium — If “prevent duplicates” means strict enforcement, evaluate an Alternate Key rather than only a warning rule.

Official page subtopics: Enable duplicate detection; Configure duplication detection rules; Duplicate detection jobs.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Bulk delete

Concept / Why: Bulk Delete is an asynchronous, query-based deletion job.

Configuration / How: Define the query and filters, choose any schedule, verify Delete privileges and cascade behaviour, and review backup and retention needs.

Exam relevance: High/Medium — Destructive cleanup vs retention trap.

Official Microsoft Learn page: Open Unit 8

Unit 9 — Long term data retention

Concept / Why: Long-term data retention moves inactive data from the active-data policy into read-only long-term retention. It is not a backup and it is not Bulk Delete.

Configuration / How: Verify policy and table support, capacity and licensing, and expected access behaviour, and then test the policy.

Exam relevance: High or Medium — Distinguish long-term retention, Bulk Delete, and backups.

Official page subtopics: Business application data lifecycle; Data retention policies.

Official Microsoft Learn page: Open Unit 9

Unit 10 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 10

Unit 11 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 11

Instructor Deep Dive — Administration, search, auditing, duplicate detection, bulk delete, retention

Power Platform admin center and environment settings

A Functional Consultant should know where to find and troubleshoot environment details, capacity, users and roles, analytics, product and feature settings, backups, updates, and Dataverse settings. Do not confuse maker-portal solution customisation with tenant and environment governance in the Power Platform admin centre.

Dataverse search options

Search experience	Scope	Configuration focus
Dataverse search	Relevance-based multi-table search, indexed	Enable/select searchable tables/columns; index capacity and update delay
Quick Find	Table-oriented searchable columns/result view	Quick Find view definitions
Advanced Find / modern advanced filtering	User-defined multi-condition query	Saved views/query analysis
Categorized search / table search	Context-dependent legacy/modern experience	Know terminology; current UX may evolve

Index configuration has capacity/performance implications. Newly configured data/index changes may not appear instantaneously. Search does not bypass table/row/column security.

Auditing

Auditing usually requires levels:

1. Environment/organization audit enabled.
2. Table auditing enabled.
3. Relevant column auditing enabled.

Audit logs show configured data changes/access events depending settings; auditing every noisy column increases storage. Security, retention, access to audit history, and compliance requirements matter. Turning auditing on now does not recreate past history.

Auditing ≠ version control, backup, or flow run history. Solution history tracks solution operations; Power Automate run history tracks flow executions; Dataverse audit tracks data/configured events.

Duplicate detection

Enable duplicate detection environment/settings and applicable operations; create/publish duplicate detection rules; run scheduled/system duplicate detection jobs where needed. Rules can warn/identify potential duplicates but do not equal a database alternate key. If duplicates must be technically rejected for integration upsert, use unique alternate key/design—not only a warning rule.

Bulk Delete

Bulk Delete job defines target query and schedule/recurrence, runs asynchronously, and records success/failure. It needs Delete privileges and can trigger cascading consequences. Preview/count/filter, retention/legal hold, dependencies, and backup approval before deleting. It is not the same as long-term retention.

Long-term data retention

Current Dataverse retention lifecycle lets qualifying application data move from active storage to inactive long-term retained, read-only/immutable access pattern, and later delete according to policy. Use when compliance requires preservation but active database performance/capacity should be managed. Feature/table/support constraints and licensing/capacity must be verified. Retention is not ordinary backup; records are intentionally moved under policy.

Capacity

Tenant capacity includes database/file/log categories and add-ons/current entitlement details. Numeric allocations are licensing-dependent and change; this guide intentionally does not invent fixed capacities. Admin center Capacity area is source of actual tenant figures.

⚠ PL-200 Exam Trap

- Need to prevent exact duplicate integration keys → Alternate Key, not only duplicate detection rule.
- Need history of who changed Salary → Auditing, not Power Automate run history.
- Need find relevant rows across tables → Dataverse search; a System View is a saved table query, not an index substitute.
- Need retain read-only historical data → Retention policy; Bulk Delete would destroy it.

Scenario-based learning

Scenario: Users complain newly added searchable column isn’t immediately in relevance search.
Reasoning: Verify Dataverse search table/column configuration, security, index status/delay, and capacity—not app form alone.

Scenario: Import must never create two records with same ExternalCustomerId.
Correct: Alternate Key/upsert strategy. Duplicate Detection may assist users but is not the same enforcement.

Tiny Details You Should Remember

- Search index and security are both evaluated.
- Auditing is layered environment → table → column.
- Audit enabled today does not backfill yesterday’s changes.
- Duplicate rules must be configured/published and relevant detection enabled.
- Bulk Delete is async and security/cascade aware.
- Long-term retention data is not normal active editable data.

Exam Memory Notes

- Search = indexed discovery; View = saved query presentation.
- Auditing = change/history evidence.
- Alternate Key = uniqueness/addressing.
- Duplicate Detection = identify/warn/process possible duplicates.
- Bulk Delete = destructive async cleanup.
- Retention = policy-based inactive preservation.

Module Knowledge Check — Questions

1. Which feature records configured changes to Dataverse rows?
2. Which feature should enforce a unique external ID?
3. Does enabling table auditing automatically audit every column if column auditing is off?
4. Which operation is appropriate for recurring deletion of old test rows?
5. Does Dataverse search expose rows a user lacks Read privilege for?
6. Is long-term retention equivalent to a backup?

Answers & Explanations

1. **Dataverse auditing.** Configure environment/table/column scope as needed.
2. **Alternate Key.** Duplicate Detection is a detection/warning mechanism, not the same uniqueness contract.
3. **No.** Required audit levels/settings must all be enabled for the intended detail.
4. **Bulk Delete job**, with carefully validated query, privileges, and cascade impact.
5. **No.** Search respects Dataverse security.
6. **No.** It is a managed inactive/read-only retention lifecycle, not a general restore copy.

Primary sources: Dataverse search, Auditing, Duplicate detection, Bulk Delete, Long-term retention.

Part II — Model-driven Apps

Official-order range: Modules 10–14. Sequence intentionally preserved from the current exam catalog.

Module 10 — How to build your first model-driven app with Dataverse

Your current position: Part II — Model-driven Apps → Module 10 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Model-driven
Official module	How to build your first model-driven app with Dataverse
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	6
Instructor focus	The Dataverse metadata-first architecture of a Model-driven App and its comparison with a Canvas App.

Official Unit-by-Unit Journey

Unit 1 — Introduction to Dataverse

Concept / Why: Understand the Dataverse metadata-first architecture of a Model-driven App and how it differs from a Canvas App.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Introduction to model-driven apps

Concept / Why: Understand the Dataverse metadata-first architecture of a Model-driven App and how it differs from a Canvas App.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Robust design capability; Pages in Model-driven apps; Dataverse as your data source; Automation using Business Process Flows; Responsive apps with a similar UI across various devices from desktop to mobile.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Model-driven apps, powered by Microsoft Dataverse

Concept / Why: A Model-driven App is a data and process application generated from Dataverse metadata.

Configuration / How: Compose the required tables, forms, views, navigation, security, and solution components.

Exam relevance: High/Medium — Data-first vs Canvas UX-first.

Official page subtopics: Approach to model-driven app making.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Explore sample apps

Concept / Why: Understand the Dataverse metadata-first architecture of a Model-driven App and how it differs from a Canvas App.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Model-driven App mindset

A Model-driven App does not begin with individual UI pixels. It generates a responsive application from Dataverse data-model, relationship, form, view, process, and security metadata. It is the first option to evaluate for a relational, data-centred, process-heavy internal application.

Model-driven strength	Why it matters
Metadata-driven	Changes to tables, forms, and views remain in the same platform metadata system
Consistent responsive shell	Works across supported desktop and mobile clients without the maker positioning every pixel manually
Dataverse security	Roles, ownership, sharing, column security apply
Process integration	Business Rules, BPF, flows, commands, charts, dashboards
Navigation/app composition	Selected pages/tables/dashboards/custom pages

Build order

1. Model business tables, columns, relationships, ownership.
2. Configure security model.
3. Create forms/views/charts.
4. Create app in Solution; add pages/navigation.
5. Add automation/process and commands.
6. Validate role access, responsive behavior, and ALM dependencies.

Dataverse as data source is mandatory for core model-driven table pages. External data can appear through virtual tables/integration, but a pure SharePoint-list app with free-form design is usually Canvas App territory.

Canvas App vs Model-driven App

Dimension	Model-driven App	Canvas App
Starting point	Data model/Dataverse	User experience/screen layout
Layout	Metadata-driven responsive shell	Free-form/control-based
Data	Dataverse-centered	Many connectors/data sources
Best for	Complex related business data/process	Task-specific/mobile/pixel-controlled UX
Security	Dataverse roles/row/column + app access	App sharing + each data source’s security
Logic	Forms, Business Rules, BPF, commands, flows	Power Fx/control events, flows

⚠ PL-200 Exam Trap

“App must work on phone” alone does not mean Canvas. Model-driven apps are responsive too. Ask whether UX must be pixel-specific or data/process metadata should drive it.

Scenario-based learning

Scenario: 30 related tables, role-based forms, ownership, audit, views, and guided case stages.
Correct: Model-driven App.
Why Canvas weaker: Possible, but recreates metadata-driven forms/navigation/security UX with more work.

Tiny Details You Should Remember

- App designer controls app composition; table maker defines underlying metadata.
- Adding a table page to app does not grant Security Role privileges.
- Sample apps help learn patterns but do not replace requirement-driven design.
- Model-driven UI can include custom pages and embedded canvas for targeted experiences.

Exam Memory Notes

- Data-first + relational + process → Model-driven.
- UX-first + custom layout → Canvas.
- Dataverse security still needs explicit roles/app access.
- Build model before decorating screens.

Module Knowledge Check — Questions

1. Which app type starts from Dataverse metadata?
2. Does adding a table to app navigation grant Read access?
3. Is a model-driven app responsive?
4. When should an embedded canvas/custom page be considered?
5. Which comes first: data/security design or command styling?

Answers & Explanations

1. **Model-driven App.** Forms/views/navigation are generated from metadata.
2. **No.** Security Role and row access are separate.
3. **Yes.** The shell is designed for responsive supported clients.
4. For a focused custom experience inside the model-driven shell, without rebuilding the entire app as Canvas.

5. **Data/security design.** UX must follow correct architecture.

Primary source: Model-driven app overview.

Module 11 — Get started with model-driven apps in Power Apps

Your current position: Part II — Model-driven Apps → Module 11 → 13 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Model-driven
Official module	Get started with model-driven apps in Power Apps
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	13
Instructor focus	App Designer and navigation, form types, views, custom pages, embedded Canvas Apps, and current Copilot-assisted units.

Official Unit-by-Unit Journey

Unit 1 — Introducing model-driven apps

Concept / Why: Understand App Designer and navigation, form types, views, custom pages, embedded Canvas Apps, and current Copilot-assisted units.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: The approach to making model-driven apps; Model your business data; Define your business processes; Compose the app; Configure security roles; Share the app.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Components of model-driven apps

Concept / Why: A Model-driven App is a data and process application generated from Dataverse metadata.

Configuration / How: Compose the required tables, forms, views, navigation, security, and solution components.

Exam relevance: High/Medium — Data-first vs Canvas UX-first.

Official page subtopics: Data components; UI components; Logic components; Visualizations.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Design model-driven apps

Concept / Why: A Model-driven App is a data and process application generated from Dataverse metadata.

Configuration / How: Compose the required tables, forms, views, navigation, security, and solution components.

Exam relevance: High/Medium — Data-first vs Canvas UX-first.

Official page subtopics: Business requirements; Data model; User Interface (UI) and User Experience (UX); Business logic; Output; Industry accelerators.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - Create a model-driven app

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Create a Solution and add an existing data table; Create a model-driven app; Add a page to your app; Edit your form; Edit views; View a chart.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Build pages with generative AI

Concept / Why: Understand App Designer and navigation, form types, views, custom pages, embedded Canvas Apps, and current Copilot-assisted units.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: What is a generative page?; Create a generative page; Iterate on your page.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Exercise - Control security when sharing model-driven apps

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; More on Security roles; Create a security role; Assign security roles to users; Assign a security role to the pet grooming technicians; Assign a security role to the Pet Grooming Schedulers.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Incorporate business process flows

Concept / Why: A Business Process Flow guides users through stages and steps.

Configuration / How: Configure stages, steps, branches, Security Roles and order, enabled tables, the BPF table, and solution inclusion.

Exam relevance: High/Medium — BPF vs Cloud Flow vs Multistep Form.

Official page subtopics: Use business process flows; System business process flows; Multiple tables in business process flows; Multiple business process flows are available per table.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Add logic to commands with Power Fx

Concept / Why: A Command Bar button gives the user an action. The modern designer can use Power Fx for `OnSelect` and visibility.

Configuration / How: Configure the command location, selection context, action, visibility, publish state, and required privileges, and then test it.

Exam relevance: High/Medium — Hidden button is not security; Command ≠ BPF.

Official page subtopics: What is commanding?; OnSelect; Visible; The Selected property; Notify the user.

Official Microsoft Learn page: Open Unit 8

Unit 9 — Exercise - Create a model-driven app

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; Create the model-driven app for the Prospects table; Create a chart; Create the form.

Official Microsoft Learn page: Open Unit 9

Unit 10 — Enable Copilot chat for app users

Concept / Why: Understand App Designer and navigation, form types, views, custom pages, embedded Canvas Apps, and current Copilot-assisted units.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: What can Copilot chat do?; Enable Copilot chat; Disable Copilot chat for a specific app.

Official Microsoft Learn page: Open Unit 10

Unit 11 — Add agents to your model-driven app

Concept / Why: A Model-driven App is a data and process application generated from Dataverse metadata.

Configuration / How: Compose the required tables, forms, views, navigation, security, and solution components.

Exam relevance: High/Medium — Data-first vs Canvas UX-first.

Official page subtopics: Types of agents in model-driven apps; Autonomous agents and the agent feed; Add an autonomous agent to an app; App assistant agent.

Official Microsoft Learn page: Open Unit 11

Unit 12 — Module assessment

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 12

Unit 13 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official page subtopics: References.

Official Microsoft Learn page: Open Unit 13

Instructor Deep Dive — App Designer, navigation, forms, views, custom pages, Copilot

App Designer and pages

In the modern App Designer, add supported table pages, dashboard pages, custom pages, URLs, or web resources, and arrange them into navigation groups and areas. Creating the app in a Solution makes dependencies and transport easier to manage. Unpublished changes might not appear to users.

`Sitemap` is the classic term for the navigation definition. Although the modern designer presents it as navigation and page arrangement, exam material can use both “sitemap” and “navigation.”

Form types

Form type	Purpose	Editable?	Role assignment?	Key limits
Main Form	Full record create/edit/read experience	Yes	Yes, Main forms only	Tabs, sections, subgrids, timeline, controls
Quick Create Form	Fast minimal create dialog	Yes	No	One section/3-column layout; no subgrid, Quick View, web resource, iframe, notes, Bing Maps
Quick View Form	Related Lookup row fields embedded in another form	No	No	Read-only; blank if Lookup blank; no scripts/header/footer/nav
Card Form	Compact record presentation in supported contexts	Limited/contextual	No	Do not confuse with Power Apps Cards product

Table and app must enable Quick Create; one Quick Create form is used by everyone for the table, chosen by form order—not role-specific. Main forms can map to Security Roles. Every table must have fallback Main Form; form order decides which accessible form opens first, but form access does not grant row access.

Multiple forms, role-based forms, form order

Use multiple Main Forms when user groups need materially different layout/process. Assign form access by Security Role and set order. Minimize duplicates because every form change becomes maintenance. Sensitive fields must use Column Security, not merely hide them on a form.

Views

App can include selected System Views. Set default view, filter/sort/columns, lookup/associated experiences. Personal views remain user-owned. View security always intersects with row privileges.

Custom Page vs embedded Canvas App

Custom Page is a Canvas-based page integrated in model-driven app navigation/dialog experience, solution-aware, using modern component model. Embedded Canvas App is placed on a Main Form and gets current record context via `ModelDrivenFormIntegration`. Choose Custom Page for app-level page/dialog; embedded Canvas for form-context experience. Avoid adding multiple overlapping solutions when a standard form component meets requirement.

Current generative/Copilot units

Current recommended Module 11 includes generative AI page creation, Copilot chat enabling, and agents. Treat maker Copilot as assistance, not security/ALM bypass: review generated table/form/logic, data permissions, licensing/region/tenant settings, and publish changes intentionally. These are current Learn units but not individually named PL-200 blueprint objectives.

⚠ PL-200 Exam Trap

- Form role assignment is not data security. A user with API access could still reach permitted columns/rows.
- Quick View looks like fields but is read-only; edit related record through its own form/navigation.
- Quick Create cannot be customized per Security Role.
- Custom Page and embedded Canvas App are not synonyms.

Scenario-based learning

Scenario: Call center agents need a six-field fast-create popup for Contact; no role-specific version.
Correct: Enable and configure Quick Create Form.
Why Main Form wrong: More navigation/fields than required.

Scenario: On Opportunity form, show read-only Account credit rating from Account lookup.
Correct: Quick View Form, plus correct Account/column privileges.
Why copied column wrong: Duplicates data and sync requirement.

Tiny Details You Should Remember

- Only Main Forms can be assigned to Security Roles.
- Fallback Main Form always exists for users without an assigned accessible form.
- Quick Create uses one form for everyone and has component restrictions.

- Quick View is read-only and requires populated Lookup.
- Role-based form order chooses UX; it does not authorize data.

Exam Memory Notes

- Main = full experience + role assignment.
- Quick Create = minimal fast create.
- Quick View = related read-only fields.
- Custom Page = app page/dialog; Embedded Canvas = form context.
- Sitemap = navigation composition.

Module Knowledge Check — Questions

1. Which form type can be assigned to Security Roles?
2. Which form shows related lookup data read-only?
3. Can a Quick Create form contain a subgrid?
4. Does form order grant row access?
5. Which feature suits a custom model-driven navigation page?
6. What must be checked after Copilot generates an app artifact?

Answers & Explanations

1. **Main Form.** Quick Create/Quick View/Card forms do not use role assignment.
2. **Quick View Form.** It embeds related record data read-only.
3. **No.** Current documented Quick Create limits exclude subgrids.
4. **No.** It only decides presentation among accessible Main Forms.
5. **Custom Page.** An embedded Canvas App is tied to a form component/context.
6. Data model, formulas, security, licensing/tenant availability, dependencies, and publish/ALM readiness.

Primary sources: Form types, Control access to forms, Form order, Quick Create, Quick View.

Module 12 — Configure forms, charts, and dashboards in model-driven apps

Your current position: Part II — Model-driven Apps → Module 12 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Model-driven
Official module	Configure forms, charts, and dashboards in model-driven apps
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	9
Instructor focus	Forms, tabs, sections, controls, Subgrids, Timeline, charts, dashboards, and choosing Power BI when appropriate.

Official Unit-by-Unit Journey

Unit 1 — Forms overview

Concept / Why: A View stores query, column, filter, and sorting metadata.

Configuration / How: Configure the System, Personal, Lookup, or Quick Find view type and include the required view in the app.

Exam relevance: High/Medium — View does not grant row access.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Form elements

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Form organization; Form designer; Table columns and properties; Components; Adding components; Form settings.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Configure multiple forms

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Access to forms; Assign security roles to a form; Set form order; Set the fallback form; Form types and behaviors; Miscellaneous form details.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Use specialized form components

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Grid controls; Display controls; External website (iframe); Quick view; Timeline control; Input controls.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Extend model-driven apps with custom pages

Concept / Why: A Model-driven App is a data and process application generated from Dataverse metadata.

Configuration / How: Compose the required tables, forms, views, navigation, security, and solution components.

Exam relevance: High/Medium — Data-first vs Canvas UX-first.

Official page subtopics: Custom pages; Custom pages vs. embedded canvas apps; Generative pages; Create a generative page; Availability and limitations.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Configure views overview

Concept / Why: A View stores query, column, filter, and sorting metadata.

Configuration / How: Configure the System, Personal, Lookup, or Quick Find view type and include the required view in the app.

Exam relevance: High/Medium — View does not grant row access.

Official page subtopics: Create and edit public views; Edit system views; Configure the grid control.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Configure charts overview

Concept / Why: A View stores query, column, filter, and sorting metadata.

Configuration / How: Configure the System, Personal, Lookup, or Quick Find view type and include the required view in the app.

Exam relevance: High/Medium — View does not grant row access.

Official page subtopics: Dashboards; Dashboard types; Multi-stream dashboards; Single-stream dashboards; Interactive tiles; Configuring interactive dashboard columns.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Module assessment

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official page subtopics: More information.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Forms, controls, charts, dashboards

Main Form composition

Form → Header + Tabs → Sections → Columns/controls. Use tabs for major task groups, sections for readable grouping. Required/frequently used fields place early; avoid overloaded forms. Business Rules can change visibility/required state; multiple Main Forms can support roles.

Common components

- Subgrid: related or selected table rows; configure view, relationship, commands, editable behavior where supported.

- Timeline: activities, notes, posts/attachments in chronological interaction context; configure record types/filters/actions.
- Quick View: related Lookup fields read-only.
- Reference Panel, tabs, sections, spacers, web resources/iframes and custom controls as supported.
- Controls can change input rendering (choice, toggle, date, rich text, etc.) but do not change underlying column type/security.

Charts and dashboards

System chart is solution-aware shared visualization based on table/view/query. Personal chart is user-owned. Dashboard combines charts, lists, web resources/other components. Power BI dashboard/report adds richer analytics and external semantic model; users may need Power BI permissions/license plus model-driven access.

Need	Select
Simple operational aggregate on Dataverse view	Model-driven Chart
Several operational lists/charts one page	Dashboard
Rich cross-source analytics, measures, drilldown	Power BI report/dashboard

Embedding does not grant dataset permission. Security trimming applies at Dataverse/Power BI layers.

Form performance/usability

Minimize unnecessary subgrids/scripts/iframes; use tabs/sections and relevant controls. Configure responsive layouts; test at supported resolutions/mobile. Do not use hidden fields as security. Publish and add dependencies to Solution.

⚠ PL-200 Exam Trap

Chart filters do not grant access. Power BI embed requires report/dataset access. A control can render a Text column differently but does not turn it into a secure Choice/Lookup.

Scenario-based learning

Scenario: Account form needs related Contacts in a grid with Add Existing command.
Correct: Subgrid configured for relationship/view, then security Append/Append To.

Scenario: Executives need measures from Dataverse and ERP in interactive visualizations.
Correct: Power BI report embedded, with dataset permissions. Model-driven chart is less suited to cross-source semantic modeling.

Tiny Details You Should Remember

- Form component visibility ≠ security.

- System charts/dashboards can move in Solutions; personal versions are user-owned.
- Subgrid operation may need related table privileges and Append/Append To.
- Timeline is optimized for activity/history interactions, not arbitrary child tables.

Exam Memory Notes

- Tabs organize; Sections group; Columns hold controls.
- Related rows grid → Subgrid.
- Interaction history → Timeline.
- Basic operational visual → Chart; rich BI → Power BI.

Module Knowledge Check — Questions

1. Which component displays related rows in a grid?
2. Does hiding Salary section secure Salary data?
3. Which artifact combines several lists/charts on one page?
4. What extra permission is commonly required for embedded Power BI?
5. Which component suits activities and notes chronologically?

Answers & Explanations

1. **Subgrid.** Configure relationship and view.
2. **No.** Use Column Security and row privileges.
3. **Dashboard.** It composes operational components.
4. Power BI report/dataset/workspace sharing and licensing as applicable, in addition to app access.
5. **Timeline.** It presents activities/notes/posts in context.

Primary sources: Design model-driven forms, Create charts, Dashboards.

Module 13 — Use specialized components in a model-driven form

Your current position: Part II — Model-driven Apps → Module 13 → 8 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Model-driven
Official module	Use specialized components in a model-driven form
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	8
Instructor focus	Business Process Flows, embedded Canvas Apps, Timeline and report components, and BPF compared with automation.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Understand Business Process Flows, embedded Canvas Apps, Timeline and report components, and BPF compared with automation.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Create business process flows

Concept / Why: A Business Process Flow guides users through stages and steps.

Configuration / How: Configure stages, steps, branches, Security Roles and order, enabled tables, the BPF table, and solution inclusion.

Exam relevance: High/Medium — BPF vs Cloud Flow vs Multistep Form.

Official page subtopics: User interface of a business process flow.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Exercise - Create a business process flow

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Prerequisites; Create a business process flow; Add another stage with a related table; Branching; Add the business process flow to a model-driven app; Review business process flow.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Embed a canvas app in a model-driven form

Concept / Why: A Model-driven App is a data and process application generated from Dataverse metadata.

Configuration / How: Compose the required tables, forms, views, navigation, security, and solution components.

Exam relevance: High/Medium — Data-first vs Canvas UX-first.

Official page subtopics: Embed a canvas app by using the classic experience; Embed a canvas app by using the modern experience; ModelDrivenFormIntegration control properties and actions.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Add a timeline in a model-driven form

Concept / Why: A Timeline shows activities, notes, and posts in chronological context.

Configuration / How: Configure record types, commands, filters, and placement on the form.

Exam relevance: High/Medium — Timeline is not an arbitrary child-table grid.

Official page subtopics: Add the timeline control; Set up the timeline control.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Create a report in a model-driven form

Concept / Why: A Model-driven App is a data and process application generated from Dataverse metadata.

Configuration / How: Compose the required tables, forms, views, navigation, security, and solution components.

Exam relevance: High/Medium — Data-first vs Canvas UX-first.

Official page subtopics: Create a report by using the report wizard; Performance tips for reports.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 7

Unit 8 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 8

Instructor Deep Dive — BPF, embedded Canvas, Timeline, Reports

Business Process Flow (BPF)

A Business Process Flow gives users guided stages and steps. It provides **human process guidance** and is not a replacement for a Cloud Flow. Understand stage transitions, required steps, branches, cross-table stages, Security Role access and process order, and the separate process table.

Each activated BPF creates a BPF table that stores process instances/stage state. Add this table/process component to Solution explicitly where required. BPF table custom forms are not supported. Table must have BPF enabled; current docs say custom table's BPF enable setting cannot be undone.

Current verified limits:

- Up to **10 active BPFs per table**.
- Up to **30 stages per process**.
- A process can span up to **5 tables**.

On the last stage, a `Stage Exit` transition does not occur because there is no next stage; use appropriate global process completed/abandoned trigger logic if needed.

BPF vs Cloud Flow

BPF	Cloud Flow
Guides user through visible stages	Automates background/event/manual/scheduled actions
Stores process instance/stage	Executes trigger/action run
Can require stage steps	Can send email, update many systems, approvals
User can progress process	System/user triggers run; no stage bar by default

BPF stage can include workflow/flow step capabilities where supported, but do not treat the process bar itself as notification/integration engine.

Embedded Canvas App

Embed on Main Form for custom task UI. `ModelDrivenFormIntegration` provides current row context and actions. Share embedded app and data source permissions; include app/dependencies in Solution. Keep one primary embedded experience per form where current limitations/recommendations apply; test record save/context refresh.

Timeline and reports

Timeline configuration selects activity/note/post types, filters, command actions. Report Wizard creates classic-style operational reports; for rich analytics consider Power BI. Report performance depends on query/filters; large unbounded reports are poor design.

⚠ PL-200 Exam Trap

Requirement “show current stage and require fields before moving” → BPF.
Requirement “when approved, send Teams message and create SharePoint folder” → Cloud Flow.
One answer can use both: BPF guides; Cloud Flow automates.

Scenario-based learning

Scenario: Loan officers must follow Qualify→Assess→Approve, while an approval completion emails customer.
Correct: BPF for stages + Cloud Flow for email/automation.
Why one alone weaker: BPF does not provide full integration; Flow does not natively give stage bar.

Tiny Details You Should Remember

- BPF creates a backing table for process instances.
- Add process/BPF table components to Solution as required.
- Last stage has no Stage Exit transition.
- Embedded Canvas App needs sharing and data permissions.

- BPF/security role order influences process availability.

Exam Memory Notes

- Guided human stages → BPF.
- Background/cross-service actions → Cloud Flow.
- BPF table stores stage/process state.
- Embedded Canvas uses ModelDrivenFormIntegration.

Module Knowledge Check — Questions

1. Which feature shows a stage bar to users?
2. Which feature sends a cross-service notification after approval?
3. Where is BPF instance state stored?
4. Does the final stage fire a normal Stage Exit transition?
5. What must be shared besides the model-driven app when using an embedded Canvas App?

Answers & Explanations

1. **Business Process Flow.** It guides stages/steps.
2. **Cloud Flow.** It handles connectors/actions.
3. In the **BPF table** created for that process.
4. **No.** There is no next-stage transition; use completed/abandoned logic.
5. The embedded Canvas App and its data source permissions/connections as applicable.

Primary sources: BPF overview, Embed Canvas App, Timeline control.

Module 14 — Customize the command bar

Your current position: Part II — Model-driven Apps → Module 14 → 7 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Model-driven
Official module	Customize the command bar
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	7
Instructor focus	Modern Command Bar, Power Fx OnSelect and visibility, command context, and awareness of classic commands.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Understand the modern Command Bar, Power Fx `OnSelect` and visibility, command context, and relevant classic command behaviour.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Command bar locations; Command bar composition; Command bar designer; Options for implementing actions.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Create or edit modern commands

Concept / Why: A Command Bar button gives the user an action. The modern designer can use Power Fx for `OnSelect` and visibility.

Configuration / How: Configure the command location, selection context, action, visibility, publish state, and required privileges, and then test it.

Exam relevance: High/Medium — Hidden button is not security; Command ≠ BPF.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Work with classic commands

Concept / Why: A Command Bar button gives the user an action. The modern designer can use Power Fx for `OnSelect` and visibility.

Configuration / How: Configure the command location, selection context, action, visibility, publish state, and required privileges, and then test it.

Exam relevance: High/Medium — Hidden button is not security; Command ≠ BPF.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Use Power Fx

Concept / Why: Understand the modern Command Bar, Power Fx `OnSelect` and visibility, command context, and relevant classic command behaviour.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Configuring OnSelect; Configuring visibility; Currently active data; Data from data sources; Common scenarios.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Customize the command bar

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Task - Prepare your environment; Task - Import solution; Task - Add command button; Task - Add button action.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 6

Unit 7 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 7

Instructor Deep Dive — Modern commanding with Power Fx

Command Bar buttons perform record/grid actions. Modern Command Designer lets makers add/edit commands and use Power Fx for `OnSelect` action and visibility. Locations include main grid, main form, subgrid/associated view contexts depending app/table.

Modern vs classic commands

Modern commanding	Classic commanding
App-scoped command design; Power Fx action/visibility	Ribbon/command metadata, often classic tools/JavaScript
Low-code supported scenarios	Existing legacy/custom complex commands
Easier Solution-aware maker experience	Maintain only where modern feature gap/legacy dependency exists

Power Fx command formulas can act on `Self.Selected`, selected records/context, `Patch`, `Notify`, `Navigate`, or call supported actions/flows according to context. Visibility formula should be performant and security-safe. Hidden command is not authorization; Dataverse privileges still enforce operations.

Command Bar vs BPF

Command = action user invokes (`Close Case`, `Create follow-up`). BPF = sequence/stage progress. A command can trigger an action that changes data/process, but it is not stage guidance.

Configuration reasoning

1. Open Model-driven App in Solution/App Designer.
2. Select table/page and command bar location.
3. Create/edit command, label/icon/order.
4. Configure Power Fx `OnSelect` and visibility.
5. Test no selection/one/multiple selection and privileges.
6. Publish and transport dependencies.

⚠ PL-200 Exam Trap

"Hide Delete button for users" does not revoke Delete privilege. Remove privilege for security; command visibility is UX. If both required, configure both.

Scenario-based learning

Scenario: Show Escalate button only when Priority=High and user has appropriate business process, then update status and call flow.
Correct: Modern command visibility/OnSelect Power Fx plus server security/flow.
Why Business Rule alone wrong: Business Rule cannot add command bar button.

Tiny Details You Should Remember

- Command bar location/context changes available record selection.
- Visibility formula is not a permission boundary.
- Modern commanding is app-scoped; verify same table in other apps.
- Classic commands remain relevant for legacy features/gaps, but prefer modern supported configuration.

Exam Memory Notes

- Button action/visibility → Modern Command + Power Fx.
- Security → Security Role, never button hiding alone.
- Stage guidance → BPF, not Command Bar.
- Test grid multi-select and form single-row context.

Module Knowledge Check — Questions

1. Which property controls a modern command's action?
2. Does hiding a Delete command remove Delete privilege?
3. Which feature represents a multi-stage guided process?
4. Why test command context on grid and form separately?
5. When might a classic command remain?

Answers & Explanations

1. Power Fx **OnSelect**.
2. **No.** Revoke Delete in Security Role for enforcement.
3. **Business Process Flow.** Command is an action, not process progression.
4. Selection/context can differ: none, one, or multiple selected records and different command locations.
5. Existing legacy customization or a modern commanding feature gap, after supported-path review.

Primary source: Modern commanding overview.

Part III — Canvas Apps and Power Apps Cards

Official-order range: Modules 15–20. Sequence intentionally preserved from the current exam catalog.

Module 15 — Get started with Power Apps canvas apps

Your current position: Part III — Canvas Apps and Power Apps Cards → Module 15 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Canvas
Official module	Get started with Power Apps canvas apps
Catalog last modified	2026-06-03T23:10:00+00:00
Official units	9
Instructor focus	Canvas App structure, Screens, Controls, Galleries, Forms, Data Cards, and source-system security.

Official Unit-by-Unit Journey

Unit 1 — Introduction to Power Apps

Concept / Why: Understand Canvas App structure, Screens, Controls, Galleries, Forms, Data Cards, and source-system security.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Start Power Apps

Concept / Why: Understand Canvas App structure, Screens, Controls, Galleries, Forms, Data Cards, and source-system security.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Exercise - Create your first app in Power Apps

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Get the data; Build a three-screen app; Create an app from a template; Create a Canvas app.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Power Apps data sources

Concept / Why: Understand Canvas App structure, Screens, Controls, Galleries, Forms, Data Cards, and source-system security.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: SharePoint; Excel; Dataverse; SQL; Evaluate and recommend connectors.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Create an app from Excel using Copilot

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Create a canvas app from a conversation; Evaluate field suggestions; Upload a file to create an app.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Use Power Apps with Power Automate and Power BI

Concept / Why: A Power BI semantic model, report, or dashboard provides rich analytics.

Configuration / How: Configure workspace and content sharing, semantic-model or dataset Row-Level Security, refresh or gateway, and any host embedding.

Exam relevance: High/Medium — Embed does not grant dataset permission.

Official page subtopics: Power Automate; Power Automate Desktop; When to use Power Automate; Power BI; Embed Power BI tiles in a canvas app; Embed a canvas app in a Power BI dashboard.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Designing a Power Apps app

Concept / Why: Understand Canvas App structure, Screens, Controls, Galleries, Forms, Data Cards, and source-system security.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Understanding user needs; Business requirements; Data model; User experience (UX); User interface (UI).

Official Microsoft Learn page: Open Unit 7

Unit 8 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you learned..

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official page subtopics: References.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Canvas App structure, controls, data

A Canvas App is a free-form app built as a tree of screens and controls. Formulas run on control properties and events. Understand the structure of the App, Screens, Containers, Components, Controls, Data Sources, Media, Variables and Collections, and Formulas.

Core controls

Control	Purpose	Exam clue
Gallery	Repeats a template to display multiple records	Use when the requirement is “show a list or cards of rows”

Control	Purpose	Exam clue
Display Form	One record read-only	Item property selects record
Edit Form	Create/edit one record	NewForm, EditForm, SubmitForm
Data Card	Form field container	Unlock carefully; DataField, Update, validation
Text input/Dropdown/Combo box	User input/select	Default/Items/Selected properties
Button/Icon	Action via OnSelect	Navigate, submit, patch, collect
Container	Responsive layout/grouping	Prefer layout responsiveness over fixed coordinates

App creation paths

Blank app, data-generated app, template, and Copilot-assisted creation are current maker options. Generated content must be reviewed: table choice, relationships, formulas, accessibility, security, delegation, and ALM.

Data source and connector model

Connector wraps service operations; connection stores identity/credentials; data source is app instance reference. Sharing Canvas App does **not** automatically share every underlying data source. Dataverse permissions, SharePoint list rights, SQL gateway/connection, and connector consent may each matter.

⚠ PL-200 Exam Trap

Canvas App freedom does not remove source security. A user can open app but still receive permission error. Conversely, hiding a control does not stop direct data access if source permission exists.

Scenario-based learning

Scenario: Warehouse staff need a phone-first barcode task with large buttons and two screens.
Correct: Canvas App, responsive containers, task-specific controls.
Why Model-driven weaker: Free-form task UX is primary requirement.

Tiny Details You Should Remember

- Gallery = many rows; Form = one selected row.
- Data Card `Update` determines submitted value.
- App sharing and data sharing are separate.
- Generated app is starting point, not validated production design.

Exam Memory Notes

- Canvas = UX-first.
- Gallery.Items returns table.

- Form.Item returns record.
- Button.OnSelect changes behavior.
- Source permissions still enforce data access.

Module Knowledge Check — Questions

1. Which control displays many records using a repeated template?
2. Which control edits one selected row?
3. Does app sharing grant Dataverse table Read automatically?
4. Which property usually identifies a Form’s row?
5. Why use responsive containers?

Answers & Explanations

1. **Gallery.** Its template repeats for each record.
2. **Edit Form.** Use its mode and SubmitForm.
3. **No.** Assign Dataverse Security Roles/data permissions separately.
4. Item.
5. To adapt layout across screen sizes instead of relying on rigid pixel positions.

Primary sources: Canvas app overview, Control reference.

Module 16 — Customize a canvas app in Power Apps

Your current position: Part III — Canvas Apps and Power Apps Cards → Module 16 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Canvas
Official module	Customize a canvas app in Power Apps
Catalog last modified	2026-06-03T23:10:00+00:00
Official units	9
Instructor focus	Gallery and Form customisation, the SubmitForm lifecycle, Data Cards, responsive design, and accessibility.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Understand Gallery and Form customisation, the `SubmitForm` lifecycle, Data Cards, responsive design, and accessibility.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Scenario; What you learn; Prerequisites.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Improve your app by making basic customizations

Concept / Why: Understand Gallery and Form customisation, the `SubmitForm` lifecycle, Data Cards, responsive design, and accessibility.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Modify Gallery fields; Modify Form fields.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Explore controls and screens in canvas apps

Concept / Why: Understand Gallery and Form customisation, the `SubmitForm` lifecycle, Data Cards, responsive design, and accessibility.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Controls in Power Apps; Gallery controls and properties; Form controls and properties; Adding screens; Add AI capabilities with Microsoft 365 Copilot; Enable Microsoft 365 Copilot in your app.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Apply Power Fx formulas to canvas app controls

Concept / Why: A Power Fx formula is a declarative expression that defines a value or behaviour.

Configuration / How: Verify data type, evaluation context, supported functions, errors, and delegation.

Exam relevance: High/Medium — Formula vs Rollup vs Flow/Business Rule selection.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Work with Power Fx formulas in canvas apps

Concept / Why: A Power Fx formula is a declarative expression that defines a value or behaviour.

Configuration / How: Verify data type, evaluation context, supported functions, errors, and delegation.

Exam relevance: High/Medium — Formula vs Rollup vs Flow/Business Rule selection.

Official page subtopics: How Power Fx works in canvas apps; Filter data with Filter(); Test a condition with If(); Navigate between screens with Navigate(); Pass data between screens with context variables; Write data with Patch().

Official Microsoft Learn page: Open Unit 5

Unit 6 — Create screen navigation in a canvas app

Concept / Why: `Navigate` and `Back` manage screen flow and can pass optional context values.

Configuration / How: Configure the destination, transition, local context, and `OnVisible` behaviour, and then test navigation paths.

Exam relevance: High/Medium — Hidden screens are not security.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Build reusable components and formulas in canvas apps

Concept / Why: A Power Fx formula is a declarative expression that defines a value or behaviour.

Configuration / How: Verify data type, evaluation context, supported functions, errors, and delegation.

Exam relevance: High/Medium — Formula vs Rollup vs Flow/Business Rule selection.

Official page subtopics: Named formulas; User-defined functions; Component libraries; Create a component; Use a component from a library; When to use each approach.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official page subtopics: References.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Customize galleries, forms, controls, themes

When customising a generated Canvas App, preserve the screen hierarchy, selected Gallery record, Form modes, validation, and accessibility.

Gallery pattern

`Gallery.Items = Filter(...)` or `Search(...)`; template controls use `ThisItem.Column`. `Gallery.Selected` feeds detail/form. Sorting and search should remain delegable for large sources. Avoid controls that modify same data source inside gallery in ways that cause excessive network calls or ambiguous selection.

Form pattern

- `ViewForm(Form1)` → read-only mode.
- `EditForm(Form1)` → edit existing `Form1.Item`.

- `NewForm(Form1)` → create mode.
- `SubmitForm(Form1)` → validates and submits.
- `ResetForm(Form1)` → discard/reset.
- `Form1.OnSuccess` → navigate/notify after successful save.
- `Form1.OnFailure` and `Form1.Error` → user feedback/troubleshooting.

`SubmitForm` honors required fields and validation, then fires success/failure. For standard edit/create, Form is simpler than hand-coded Patch. Unlocking Data Cards gives flexibility but maker becomes responsible for preserving `DataField`, `Default`, `Update`, `DisplayMode`, and error message wiring.

Control properties and behavior

Value properties (`Text`, `Items`, `Default`, `Visible`, `DisplayMode`) recalculate. Behavior formulas (`OnSelect`, `OnChange`, `OnVisible`) perform actions. Use semicolon to sequence actions according to locale/formula syntax. Avoid putting side effects in value properties.

Accessibility and UX

Meaningful `AccessibleLabel`, keyboard focus/order, contrast, non-color cues, responsive layout, error feedback. Functional Consultant testing includes both business success and accessible use.

⚠ PL-200 Exam Trap

Patch every Data Card manually when a Form already meets requirement is unnecessary complexity. Choose SubmitForm for ordinary record editing; Patch for complex multi-record/partial/no-form scenarios.

Scenario-based learning

Scenario: On save, required fields must validate, errors display, and user moves to success screen only if save works.
Correct: Edit Form + SubmitForm; Navigate in `OnSuccess`, error message in `OnFailure`.
Wrong: `Navigate` immediately after SubmitForm in same OnSelect can navigate before outcome handling.

Tiny Details You Should Remember

- `ThisItem` = current gallery/form template record.
- `Gallery.Selected` = selected gallery record.
- Form `OnSuccess` is correct place for post-save navigation.
- Unlocking card doesn't change underlying data type/security.

Exam Memory Notes

- NewForm/EditForm/ViewForm set mode.
- SubmitForm validates and saves.
- OnSuccess after confirmed save.

- ThisItem within record template.
- Patch only when Form pattern is insufficient.

Module Knowledge Check — Questions

1. Which formula changes a form to create mode?
2. Where should navigation occur only after successful save?
3. What does `ThisItem` mean inside a gallery template?
4. Why can unlocking Data Cards be risky?
5. Which function resets unsaved form changes?

Answers & Explanations

1. `NewForm(FormName)`.
2. The Form's **OnSuccess** property.
3. The record for the current repeated gallery item.
4. Automatic metadata/validation wiring can be accidentally broken; preserve DataField/Update/error properties.
5. `ResetForm(FormName)`.

Primary source: Edit form control.

Module 17 — Navigation in a canvas app in Power Apps

Your current position: Part III — Canvas Apps and Power Apps Cards → Module 17 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Canvas
Official module	Navigation in a canvas app in Power Apps
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	6
Instructor focus	Navigate, Back, screen transitions, selected-record context, and state scope.

Official Unit-by-Unit Journey

Unit 1 — Understanding navigation

Concept / Why: `Navigate` and `Back` manage screen flow and can pass optional context values.

Configuration / How: Configure the destination, transition, local context, and `OnVisible` behaviour, and then test navigation paths.

Exam relevance: High/Medium — Hidden screens are not security.

Official page subtopics: Navigate function; Back function; Hidden screens.

Official Microsoft Learn page: Open Unit 1

Unit 2 — The Navigate and Back functions

Concept / Why: `Navigate` and `Back` manage screen flow and can pass optional context values.

Configuration / How: Configure the destination, transition, local context, and `OnVisible` behaviour, and then test navigation paths.

Exam relevance: High/Medium — Hidden screens are not security.

Official page subtopics: Navigate function; Back function; Screen transitions; Examples.

Official Microsoft Learn page: Open Unit 2

Unit 3 — More ways to use the Navigate function

Concept / Why: `Navigate` and `Back` manage screen flow and can pass optional context values.

Configuration / How: Configure the destination, transition, local context, and OnVisible behaviour, and then test navigation paths.

Exam relevance: High/Medium — Hidden screens are not security.

Official page subtopics: OnSuccess navigation; OnTimerEnd navigation; OnChange navigation.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - App navigation practice

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Module assessment

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Knowledge check for navigation in a canvas app in Power Apps.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official page subtopics: References.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Navigation, screen context, state

`Navigate(TargetScreen, Transition, {ContextName: Value})` moves forward and optionally sets screen context variables. `Back()` returns to previous screen. `Screen.OnVisible` runs when screen becomes visible; use carefully because repeated navigation can reinitialize state.

Need	Feature
Move to a known screen	Navigate
Return to previous screen	Back
Pass screen-local value while navigating	Navigate context record / UpdateContext
App-wide state	Set global variable
Temporary tabular data	Collection

Hidden screen is still part of app and formulas can reference it; setting `Visible=false` or removing a navigation button is not access security. Avoid deeply tangled navigation; define start screen/role routing carefully.

⚠ PL-200 Exam Trap

`UpdateContext` screen variable is not available everywhere like a global `Set` variable. Use narrowest scope. `Back` follows navigation history, not necessarily a hard-coded screen.

Scenario-based learning

Scenario: User selects Order in Gallery and opens Detail screen.
Correct: `Navigate(scrDetail, ScreenTransition.Fade, {locOrder: ThisItem})`; Detail Form Item=`locOrder`.
Alternative: Global variable works but unnecessarily broad if only detail screen needs it.

Tiny Details You Should Remember

- Context variables are screen-scoped.
- `Back()` uses prior navigation stack.
- `OnVisible` may run every time screen is entered.
- Hidden controls/screens are not security controls.

Exam Memory Notes

- Navigate = known destination.
- Back = previous destination.
- Pass local record in Navigate context.
- Use narrowest state scope.

Module Knowledge Check — Questions

1. Which function returns to the previous screen?
2. How can a selected record be passed to a detail screen without a global variable?
3. Does hiding a screen secure its data?
4. When can `OnVisible` run?
5. What is the scope of a context variable?

Answers & Explanations

1. `Back()`.
2. Pass a context record in `Navigate(..., {locRecord: ThisItem})`.
3. **No.** Enforce security at the data source.
4. Each time the screen becomes visible, depending navigation.
5. The screen where it is created/updated.

Primary source: Navigate and Back.

Module 18 — Create formulas to change behaviors in a Power Apps canvas app

Your current position: Part III — Canvas Apps and Power Apps Cards → Module 18 → 8 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Canvas
Official module	Create formulas to change behaviors in a Power Apps canvas app
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	8
Instructor focus	Power Fx functions, variables, Collections, Patch, SubmitForm, error handling, and delegation correctness.

Official Unit-by-Unit Journey

Unit 1 — Formulas and functionality

Concept / Why: A Power Fx formula is a declarative expression that defines a value or behaviour.

Configuration / How: Verify data type, evaluation context, supported functions, errors, and delegation.

Exam relevance: High/Medium — Formula vs Rollup vs Flow/Business Rule selection.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Understanding true and false

Concept / Why: Understand Power Fx functions, variables, Collections, `Patch`, `SubmitForm`, error handling, and delegation correctness.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Understanding control behaviors and actions

Concept / Why: Understand Power Fx functions, variables, Collections, `Patch`, `SubmitForm`, error handling, and delegation correctness.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Performing multiple actions in a formula

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Combine functions.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Control the display mode through a formula

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: DisplayMode.Edit; DisplayMode.Disabled; DisplayMode.View; Try it out.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Use controls and functions to create a dynamic formula

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Module assessment

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Author a basic formula to change behaviors in a Power Apps canvas app knowledge check.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 8

Instructor Deep Dive — Power Fx formulas, variables, collections, errors, delegation

Core formula functions

Function	Returns/does	Typical PL-200 use
Filter(Table, condition)	All matching rows (table)	Gallery rows
Search(Table, text, columns...)	Case-insensitive substring matches (table)	User search
LookUp(Table, condition [, reduction])	First matching row/value	Find one record
Sort / SortByColumns	Sorted table	Grid/gallery order
If	Conditional result/action	Two/few branches
Switch	Match one expression against cases	Many discrete branches
Set	Global variable	App-wide state
UpdateContext	Screen context variable	Screen-local state
Collect	Add records to collection/source	Cache/worklist; side effect
ClearCollect	Clear then collect	Refresh local collection
Patch	Create/update records flexibly	Partial/complex/multi-record updates
SubmitForm	Validate and submit form	Standard create/edit
IfError / Errors	Handle/report data errors	Resilient UX

Variables and collections

- Global variable: `Set(varUser, User())`; available throughout app.
- Context variable: `UpdateContext({locEdit:true})`; screen-local.
- Collection: in-memory table `ClearCollect(colCart, ...)`; useful offline-like staging/transform, but it is not automatically synchronized source or security layer.

Use variables only when formula cannot be directly derived; too much mutable state causes stale data.

Patch vs SubmitForm

`SubmitForm` recommended for form-driven standard validation. `Patch(DataSource, Defaults(DataSource), {...})` creates; `Patch(DataSource, BaseRecord, {...})` updates. Patch can update selected fields or multiple rows but maker must handle validation/error and relationship types.

Delegation — correctness issue

Delegable formula is sent to data source; nondelegable query is processed locally on a limited subset. Current Canvas docs: default nondelegation record limit **500**, maker can configure up to **2,000**. A nondelegation warning on 100,000-row table can return **wrong results**, not merely slow results.

Delegability depends on connector, function, operator, and column type. Blue underline/warning inspect; redesign with delegable Filter, Dataverse views, server-side logic, or narrower dataset. Setting limit to 2,000 does not solve larger data.

Error handling

Use `IfError(Patch(...), Notify(...), success action)` and `Errors(DataSource)`/Form.OnFailure. Validate input, handle blank/permission/conflict/network errors, and avoid showing raw sensitive details. `Notify` gives user message; monitoring can inspect issues.

⚠ PL-200 Exam Trap

- `LookUp` returns first record, while `Filter` returns table.
- `Collect` into local collection does not grant access or write back automatically.
- Nondelegation warning can produce incomplete answer even though app displays records.
- `If` vs `Switch`: choose clarity; neither substitutes for data security.

Scenario-based learning

Scenario: Gallery must find all open Dataverse cases for selected account in 200,000 rows. Formula produces delegation warning.
Correct reasoning: Use delegable predicates/functions supported by Dataverse or server-side view; don’t raise limit and accept incomplete data.

Scenario: User edits standard Contact form and needs validation.
Correct: SubmitForm + OnSuccess/OnFailure. Patch only if special update pattern needed.

Tiny Details You Should Remember

- Filter/Search return tables; LookUp returns first matching record/value.
- Set/global, UpdateContext/screen, Collection/table.
- Delegation support is data-source specific.
- 500 default / 2,000 configurable nondelegation cap is not a total datasource limit; it is local processing subset.
- Formula suggestions/Copilot must be reviewed for delegation and security.

Exam Memory Notes

- Filter = many; LookUp = one.
- SubmitForm = normal form save; Patch = flexible record update.
- Delegation warning = possible wrong result.
- IfError/Errors + OnFailure for resilience.
- Keep state scope minimal.

Module Knowledge Check — Questions

1. Which function returns every matching record?
2. Which variable type is app-wide?
3. What is the current default nondelegation row limit?
4. Does increasing it to 2,000 make a 100,000-row nondelegable query correct?
5. Which function is simplest for ordinary Edit Form save?
6. Which function handles a formula error and provides fallback behavior?

Answers & Explanations

1. `Filter` (or Search for text search). `LookUp` only returns the first match.
2. A global variable created with `Set`.
3. **500** rows, configurable up to **2,000** per current docs.
4. **No.** Records beyond the local subset remain unprocessed.
5. `SubmitForm`.
6. `IfError`; use Errors/Form.OnFailure for detailed data-operation handling.

Primary sources: Delegation, Filter/Search/LookUp, Patch, Forms, Variables, IfError.

Module 19 — Connect to other data in a Power Apps canvas app

Your current position: Part III — Canvas Apps and Power Apps Cards → Module 19 → 5 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Canvas
Official module	Connect to other data in a Power Apps canvas app
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	5
Instructor focus	Connector, Connection, and Data Source; gateways, DLP, app sharing, and underlying data permissions.

Official Unit-by-Unit Journey

Unit 1 — Overview of the different data sources

Concept / Why: A View stores query, column, filter, and sorting metadata.

Configuration / How: Configure the System, Personal, Lookup, or Quick Find view type and include the required view in the app.

Exam relevance: High/Medium — View does not grant row access.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Work with action data

Concept / Why: Understand Connector, Connection, and Data Source; gateways, DLP, app sharing, and underlying data permissions.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Add the Office 365 Users data source; Display a list of users in a gallery; Find the email address of the logged in user’s manager; Update the logged-in user’s profile info.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Power Automate is a companion to Power Apps

Concept / Why: Understand Connector, Connection, and Data Source; gateways, DLP, app sharing, and underlying data permissions.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Business logic; Data connections.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Module assessment

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 4

Unit 5 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 5

Instructor Deep Dive — Connectors, connections, on-premises data, sharing

Canvas App can connect to Dataverse, SharePoint, SQL, Microsoft 365, and many services. Connector defines API contract; Connection is an authenticated instance; Data Source is app reference. Custom connector wraps a custom API. On-premises Data Gateway bridges supported on-prem sources; gateway availability, credentials, region, and sharing matter.

Data-source selection

- Dataverse for relational security/process/ALM.
- SharePoint for document/collaboration list scenarios with its own limits/security; do not emulate complex relational Dataverse model blindly.

- SQL for existing relational system; connector/gateway/delegation/licensing.
- Microsoft 365 connectors for user profile/mail/calendar actions; least privilege.

Share checklist

1. Share app with user/group (User or Co-owner as supported).
2. Grant underlying data permissions/Dataverse roles.
3. Share or configure connections/connection references where applicable.
4. Confirm premium licensing/connector policy/DLP.
5. Confirm gateway and environment access.

⚠ PL-200 Exam Trap

App co-owner can edit app, but this does not mean they can administer every connected system. Run/test identities and shared connections must be reviewed.

Scenario-based learning

Scenario: App opens but SQL query fails only for users.
Reasoning: Check connection authentication model, SQL permissions, gateway access, licensing, DLP—not only app share.

Tiny Details You Should Remember

- Connector ≠ Connection ≠ Data Source.
- DLP can block connector combinations.
- Gateway handles supported on-prem connectivity; it does not grant database authorization.
- Canvas app sharing does not automatically grant source access.

Exam Memory Notes

- API definition → Connector.
- Authenticated credential/session → Connection.
- App-bound table/service → Data Source.
- Always test as ordinary user.

Module Knowledge Check — Questions

1. What stores service authentication: Connector or Connection?
2. Does an on-premises gateway grant SQL row permissions?
3. What can block combining business and non-business connectors?
4. Why test with a non-maker account?
5. Which platform is first choice for complex Power Platform relational security?

Answers & Explanations

1. **Connection.** Connector defines operations.
2. **No.** Database authorization remains required.
3. A **Data Loss Prevention (DLP) policy**.
4. Makers often have broader roles/connections; ordinary-user test reveals real access gaps.
5. **Dataverse**, when requirements fit its capabilities/licensing.

Primary sources: Add connections, Share a Canvas App.

Module 20 — Introduction to Power Apps cards

Your current position: Part III — Canvas Apps and Power Apps Cards → Module 20 → 7 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Current official preparation content — lower explicit blueprint priority
Official module	Introduction to Power Apps cards
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	7
Instructor focus	Power Apps Cards as lightweight Teams/conversation experiences and their limitations.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Power Apps Cards as lightweight Teams/conversation experiences and their limitations.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Overview; Limitations.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Design Power Apps cards

Concept / Why: A Power Apps Card is a lightweight interactive experience for a host or conversation.

Configuration / How: Design a small UI, configure the data connection and Teams sharing, and verify current limitations.

Exam relevance: High/Medium — Power Apps Card ≠ model-driven Card Form.

Official page subtopics: Card design considerations.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Exercise - Create a card to capture employee suggestions

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; Exercise steps (video); Exercise; Optional exercise.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Send Power Apps cards

Concept / Why: A Power Apps Card is a lightweight interactive experience for a host or conversation.

Configuration / How: Design a small UI, configure the data connection and Teams sharing, and verify current limitations.

Exam relevance: High/Medium — Power Apps Card ≠ model-driven Card Form.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Share the employee suggestions card in Teams

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; Preparation; Exercise steps (video); Exercise; Optional exercise.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you learned..

Official Microsoft Learn page: Open Unit 6

Unit 7 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 7

Instructor Deep Dive — Power Apps Cards

Power Apps Cards are lightweight, interactive, data-connected cards that can be shared in Microsoft Teams/host contexts. They are not the same as Model-driven `Card Form` and not a replacement for a full Canvas App.

Use card when interaction should happen in conversation: approve/reject, collect a small suggestion, show concise status. Keep information/action minimal; responsive host size and supported controls/connectors/limits matter. Current module itself emphasizes limitations—verify current product support before design because Card capabilities evolve.

Card vs Canvas App vs Adaptive Card

Need	Likely choice
Small maker-built interactive Teams experience with Power Platform data	Power Apps Card
Multi-screen rich custom application	Canvas App
Message payload authored by flow/code for a host	Adaptive Card pattern

Share Card and ensure connection/data permissions. Do not expose sensitive data merely because conversation participant can see the Teams chat.

⚠ PL-200 Exam Trap

`Card Form` in model-driven apps is a compact record form type; `Power Apps Cards` is a separate product experience.

Scenario-based learning

Scenario: Employees submit one-line improvement idea directly in Teams message.
Correct: Power Apps Card if current supported data/action requirements fit.
Why Canvas App may be excessive: Full app navigation unnecessary.

Tiny Details You Should Remember

- Cards are lightweight and host-based.
- Data/connection/security still apply.

- Current PL-200 blueprint does not name Cards explicitly; it is in current official prep modules, so moderate/low explicit priority.

Exam Memory Notes

- Conversation micro-experience → Card.
- Full app → Canvas.
- Card Form ≠ Power Apps Card.

Module Knowledge Check — Questions

1. Are Power Apps Cards the same as model-driven Card Forms?
2. Which scenario best fits a Card?
3. Does Teams chat membership automatically grant Dataverse row access?
4. Should a Card hold a complex ten-screen process?
5. What must be checked before production use?

Answers & Explanations

1. **No.** They are separate features.
2. A concise in-conversation interaction such as one approval or small input.
3. **No.** Dataverse/security permissions still apply.
4. **No.** Use Canvas or Model-driven App for complex workflows.
5. Current limitations, host support, licensing, connector/data access, sharing, and governance.

Primary source: Power Apps cards overview.

Part IV — Microsoft Power Pages

Official-order range: Modules 21–28. Sequence intentionally preserved from the current exam catalog.

Module 21 — Core components of Power Pages

Your current position: Part IV — Microsoft Power Pages → Module 21 → 7 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Power Pages
Official module	Core components of Power Pages
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	7
Instructor focus	Power Pages architecture, pages, templates, Lists and Forms, Contact and Web Role, security layers, and extensibility.

Official Unit-by-Unit Journey

Unit 1 — Introduction to Power Pages

Concept / Why: Understand Power Pages architecture, pages, templates, Lists and Forms, Contact and Web Role, security layers, and extensibility.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Power Pages capabilities; Simplified authoring experience for makers; AI-powered features with Copilot; Advanced development capabilities for pro developers; Design and responsive rendering; Security and governance.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Get started with Power Pages

Concept / Why: Understand Power Pages architecture, pages, templates, Lists and Forms, Contact and Web Role, security layers, and extensibility.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Provision a website; Dynamics 365 templates; Provision a Dynamics 365 site; Create a site with Copilot.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Core tools and components of Power Pages

Concept / Why: Understand Power Pages architecture, pages, templates, Lists and Forms, Contact and Web Role, security layers, and extensibility.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Tools; Core components; Web Pages; Working with data; Tables; Lists.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Overview of Power Pages security

Concept / Why: A View stores query, column, filter, and sorting metadata.

Configuration / How: Configure the System, Personal, Lookup, or Quick Find view type and include the required view in the app.

Exam relevance: High/Medium — View does not grant row access.

Official page subtopics: Site visibility; Authentication; Authorization.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Overview of Power Pages extensibility

Concept / Why: A View stores query, column, filter, and sorting metadata.

Configuration / How: Configure the System, Personal, Lookup, or Quick Find view type and include the required view in the app.

Exam relevance: High/Medium — View does not grant row access.

Official page subtopics: Integration with other Microsoft Power Platform components; Power Pages Extensibility; Liquid; Web templates; Code editor; JavaScript.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 6

Unit 7 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official page subtopics: Resources.

Official Microsoft Learn page: Open Unit 7

Instructor Deep Dive — Power Pages architecture

Power Pages is a Dataverse-backed platform for external-facing websites. It combines anonymous visitors, authenticated customers, partners or employees, responsive templates, Forms and Lists, Web Roles, Table and Page Permissions, Liquid, JavaScript and CSS extensibility, and Power Platform integration.

Core component map

Component	Role
Website/Site	Power Pages configuration container tied to environment
Web Page	URL/navigation/content page
Page Template / Web Template	Rendering/layout reuse; Liquid for advanced templates
List	Multiple Dataverse rows display/actions
Form	Dataverse row create/read/update
Multistep Form	Multi-page conditional data capture process
Web Role	Group of authenticated site users/contacts
Page Permission	Static page visibility/access
Table Permission	Dataverse data operation/scope authorization
Contact	Authenticated website user representation in Dataverse

Power Pages authorisation is not identical to the internal Dataverse Security Role model. An external user is normally represented by a linked Contact and receives Web Roles, Page Permissions, and Table Permissions. Public or private site visibility is not a substitute for authorisation.

Provisioning and templates

Blank/design templates and Dynamics 365 scenario templates can accelerate, but generated/site template assets review. Template = starting architecture; it does not decide your final data/security. Copilot site creation is current official content; validate generated pages/forms/data and governance.

Extensibility order

First Design Studio components/configuration. Then forms/lists metadata. Then Liquid/web templates, JavaScript, CSS only if requirements demand. Pro-code extension must preserve security; never trust client-side filter as authorization.

⚠ PL-200 Exam Trap

Website page being hidden from navigation does not secure it. Configure Page Permission. Dataverse rows on a form/list need Table Permission—adding component alone should not expose data.

Scenario-based learning

Scenario: Suppliers outside tenant submit onboarding data and track only their own requests.
Correct: Power Pages + authentication + Contact/Web Role + scoped Table Permissions.
Why model-driven wrong: Model-driven is internal Dataverse app access model, not public/external website experience.

Tiny Details You Should Remember

- Power Pages is Dataverse-backed.
- Contact represents authenticated site user.
- Page security and data security are separate.
- Liquid/JavaScript/CSS extend; client code is never the security boundary.

Exam Memory Notes

- External portal → Power Pages.
- Web Role groups site users.
- Page Permission secures page.
- Table Permission secures Dataverse rows/actions.

Module Knowledge Check — Questions

1. Which record generally represents an authenticated Power Pages user?
2. Which permission protects a Dataverse list’s rows?
3. Does removing a page from navigation secure its URL?
4. What is a template: final security model or starting point?
5. Which language is used for advanced server-rendered templates?

Answers & Explanations

1. **Contact.** Authentication identity links to a Contact.
2. **Table Permission.** Also assign it through Web Roles.
3. **No.** Use Page Permission/access control.
4. A **starting point**; validate/modify components and security.
5. **Liquid** (with web templates), plus HTML/CSS/JavaScript as applicable.

Primary source: Power Pages overview.

Module 22 — Access Dataverse in Power Pages websites

Your current position: Part IV — Microsoft Power Pages → Module 22 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Power Pages
Official module	Access Dataverse in Power Pages websites
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	6
Instructor focus	Dataverse Lists and Forms, modes/actions, View query vs Table Permission authorization.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Dataverse Lists and Forms, modes/actions, View query vs Table Permission authorization.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Use lists to display multiple Dataverse records

Concept / Why: A Power Pages List renders rows from a Dataverse View and can expose supported actions.

Configuration / How: Configure the View, search, filtering and actions, target Forms, and Table Permissions.

Exam relevance: High/Medium — View filter is not authorization.

Official page subtopics: Create a list; List rendering; Set up the list; Views; Configuration; Filter and search.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Use forms to interact with Dataverse data

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Common uses; Create a form; Mode; Set up the form; Form tab; Data tab.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - Use the list and form components

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Learning objectives; Prerequisites; Scenario; High-level steps; Create a table permission row; Create a webpage for the list.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Lists and Forms

List

List component renders Dataverse view rows. Configure table/view, page size, search/filter, create/read/edit/delete/detail actions, target forms/pages, and styling. The underlying view determines query/columns, while Table Permissions determine which rows/actions visitor can use.

Form

Power Pages Form uses a Dataverse model-driven form definition but site metadata config decides mode and behavior:

- Insert/Create mode creates new row.
- Edit updates target row.
- Read Only displays row.
- On submit: success message, redirect, or next action.
- Record source/query-string/relationship association must be secured; never trust arbitrary row ID from URL without Table Permission.

Form fields and requiredness may come from Dataverse metadata/form, but website validation should also consider server security. Hidden field/JavaScript cannot grant or deny row access.

Practical configuration flow

1. Dataverse table, columns, forms, views.
2. Web page.
3. Add List/Form component.
4. Create Table Permission with exact privileges and access type.
5. Assign permission to Web Role.
6. Test anonymous, authenticated allowed user, and authenticated disallowed user.

⚠ PL-200 Exam Trap

List uses a View, but View filter alone is not authorization. A user altering URL/query must still be blocked by Table Permission.

Scenario-based learning

Scenario: Customers see list of their own support cases and open one details form.
Correct: List + detail Form + Contact-scoped Table Permission via customer Web Role.
Why Global permission wrong: It could expose every customer’s rows.

Tiny Details You Should Remember

- List = multiple rows; Form = one row.
- Form mode and submit action are separate settings.

- Dataverse form/view reuse does not automatically configure portal authorization.
- Test URL tampering.

Exam Memory Notes

- List based on View.
- Form based on Dataverse form metadata.
- Table Permission always defines data scope/operations.
- Web Role connects users to permissions.

Module Knowledge Check — Questions

1. Which component displays multiple Dataverse records?
2. Which Form mode creates a row?
3. Does a View filter securely restrict each customer to their rows?
4. Where should success redirect be configured?
5. What three user types should security testing include?

Answers & Explanations

1. **List.** It uses a Dataverse view.
2. **Insert/Create** mode.
3. **No.** Use scoped Table Permission.
4. In the Power Pages Form/on-submit configuration.
5. Anonymous, authenticated allowed, and authenticated disallowed/tampered request.

Primary source: Use lists and forms.

Module 23 — Explore Power Pages design studio

Your current position: Part IV — Microsoft Power Pages → Module 23 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Power Pages
Official module	Explore Power Pages design studio
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	9
Instructor focus	Design Studio workspaces, responsive styling, Copilot-assisted content, and AI agent governance.

Official Unit-by-Unit Journey

Unit 1 — Introduction to Power Pages design studio

Concept / Why: Design Studio workspaces, responsive styling, Copilot-assisted content, and AI agent governance.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Power Pages design studio; Ask Copilot questions; Copilot governance.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Work with pages

Concept / Why: Design Studio workspaces, responsive styling, Copilot-assisted content, and AI agent governance.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Create a page; Configure site structure; Preview a page.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Page components

Concept / Why: Design Studio workspaces, responsive styling, Copilot-assisted content, and AI agent governance.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Source code; Copilot in the Pages workspace; AI-generated text; AI-generated form; AI-generated multistep form (preview); AI form fill assistance (preview).

Official Microsoft Learn page: Open Unit 3

Unit 4 — Site styling and templates

Concept / Why: A Word or Excel Template merges Dataverse row data into a formatted output.

Configuration / How: Configure the schema and relationships, document controls or spreadsheet formulas, upload and security, and sharing.

Exam relevance: High/Medium — Template ≠ live integration/document management.

Official page subtopics: Styling; Custom CSS; Templates.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Secure your Power Pages site

Concept / Why: Power Pages security depends on a chain of Page Permission, Table Permission, and Web Role.

Configuration / How: Configure scope, allowed operations, relationship paths, and test identities that should be denied as well as identities that should be allowed.

Exam relevance: High/Medium — Global scope and URL-tampering traps.

Official page subtopics: Monitor; Protect; Manage.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Add an AI-powered agent to your site

Concept / Why: Design Studio workspaces, responsive styling, Copilot-assisted content, and AI agent governance.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Prerequisites; Add an agent from the Set up workspace; Add an existing agent; Extend the agent.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Exercise - Edit pages

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; High-level steps; Detailed steps; Launch Power Pages design studio; Create a web page; Edit page.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Knowledge check

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Design Studio, pages, styling, AI agent

Design Studio workspaces commonly cover Pages, Styling, Data, and Set up/Security areas. Pages workspace builds hierarchy/navigation/sections/components. Styling controls themes/colors/fonts and custom CSS. Data workspace manages relevant Dataverse tables/views/forms. Set up handles site capabilities, identities/integrations.

Page design

- Create page, parent/child hierarchy, URL, navigation visibility.
- Use sections/columns/components for responsive layout.

- Preview desktop/tablet/mobile and browse site.
- Code editor/custom CSS only when designer cannot meet requirement.
- Accessibility: semantic headings, labels, alt text, contrast, keyboard use.

Copilot and current modern units

Current units cover Ask Copilot, AI-generated text/forms/multistep form (some experiences may be Preview), AI form-fill assistance, and AI-powered agent. Microsoft exam guide says Preview can appear if commonly used, but core objective names pages/forms/lists/security/authentication—not these individual AI features.

For AI-generated content:

1. Verify tenant/region/licensing/availability.
2. Review generated Dataverse changes and permissions.
3. Validate factual text, privacy, prompt input, and responsible AI.
4. Ensure agent knowledge/actions cannot bypass Table Permissions.
5. Mark Preview features as higher change risk.

⚠ PL-200 Exam Trap

Copilot can generate a page or form, but it does not remove Functional Consultant responsibility for data model, security, and ALM. “Generated successfully” is not “production-ready.”

Scenario-based learning

Scenario: Marketing needs brand-consistent pages; developers should only code gaps. **Correct:** Design Studio Styling/Page components first, custom CSS/web templates only for unmet requirements.

Tiny Details You Should Remember

- Navigation visibility and Page Permission are different.
- Custom CSS can affect maintainability/responsiveness.
- AI units in current Learn may include Preview labels; recheck before exam.
- Adding an agent requires governance, security, knowledge/action review.

Exam Memory Notes

- Pages workspace = structure/content.
- Styling = theme/CSS.
- Data = tables/views/forms.
- AI output always review.

Module Knowledge Check — Questions

1. Which workspace controls site page hierarchy?

2. Does custom CSS authorize data?
3. What is the first action after AI generates a form?
4. Why prefer standard components before code?
5. Is every AI unit a named PL-200 objective?

Answers & Explanations

1. **Pages** workspace.
2. **No.** Table/Page permissions enforce security.
3. Review data model, fields, validation, Table Permissions, privacy, and availability.
4. Better supportability, responsiveness, accessibility, and maintainability.
5. **No.** They are current prep content; the blueprint names broader Power Pages capabilities.

Primary source: Power Pages design studio.

Module 24 — Integrate Power Pages websites with Dataverse

Your current position: Part IV — Microsoft Power Pages → Module 24 → 8 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Power Pages
Official module	Integrate Power Pages websites with Dataverse
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	8
Instructor focus	Simple vs Multistep Forms, conditional steps, metadata, subgrids/notes/actions.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Simple vs Multistep Forms, conditional steps, metadata, subgrids/notes/actions.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Lists; Create a list; Set up the list; Views; Configuration; Actions.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Basic form configuration

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Create a form; Set up the form; Simplify form layout; Add a form to your website.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Multistep form processes

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Surveys; Complex data capture; Common practices; Create a multistep form.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Set up multistep forms

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Multistep form properties; Multistep form steps; Load Form and Load Tab steps; Redirect step; Condition step; Add a multistep form to your website.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Extend lists and forms

Concept / Why: A Canvas Form manages the lifecycle of creating, editing, or displaying one record, including validation.

Configuration / How: Configure `Item`, form mode, Data Cards, `SubmitForm`, `OnSuccess`, and `OnFailure`.

Exam relevance: High/Medium — SubmitForm vs Patch.

Official page subtopics: Link lists and forms; Advanced settings; Commands; Form metadata; Form subgrids; Notes.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Exercise - Extend forms with more actions

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Learning objectives; Prerequisites; Scenario; Create a workflow process; Create an action button; Test.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 7

Unit 8 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 8

Instructor Deep Dive — Forms, multistep forms, metadata, actions

Multistep Form

Use when data capture spans stages/pages, conditional branching, multiple related tables, or resumable process. It consists of Multistep Form definition + Steps such as Load Form/Load Tab, Redirect, or Condition, with session/record behavior.

Simple Form	Multistep Form
One page/record operation	Several steps/pages/conditions
Lower configuration	More metadata, state, security testing
Best for short create/edit	Complex applications/surveys/onboarding

Common practices: ask minimal data per step, preserve record identity securely, validate each stage, prevent skipping via URL, and configure Table Permissions for every table/relationship touched.

Extend lists/forms

- Link List row actions to Form/detail page.
- Configure commands/actions such as create/edit/delete/workflow where supported.
- Form metadata changes field labels, instructions, visibility, validation, subgrid/notes behavior at portal layer.
- Notes/attachments and subgrids need related Table Permissions and document/security review.
- Legacy workflow action may appear in current modules, but Cloud Flow is modern cross-service approach; classic workflow remains an explicit exam objective.

⚠ PL-200 Exam Trap

Multistep Form is not BPF. Both have “steps,” but Multistep Form is website data capture navigation; BPF is Dataverse/model-driven guided business process with a process table.

Scenario-based learning

Scenario: Grant application has Applicant, Project, Documents, Review screens with conditional extra evidence.
Correct: Multistep Form with conditional steps and Table Permissions for all rows.

Tiny Details You Should Remember

- Form metadata customizes portal rendering; underlying Dataverse form still matters.
- Related subgrid/notes need privileges on related tables.
- Redirect and Condition are step types, not data security.
- Users should not bypass a required step by direct URL.

Exam Memory Notes

- One page → Form.
- Multi-page/branching → Multistep Form.
- BPF ≠ Multistep Form.
- Every touched table needs permission design.

Module Knowledge Check — Questions

1. Which feature suits a four-page external application?
2. Is a Power Pages Multistep Form a BPF?
3. What protects a related Notes/Attachment table?
4. Which step can route user after completion?

5. Why test direct URLs to later steps?

Answers & Explanations

1. **Multistep Form.** It supports staged data capture.
2. **No.** Different component/runtime/purpose.
3. Appropriate **Table Permissions** and storage/document security.
4. A **Redirect** step/configuration.
5. To ensure authorization/state prevents step bypass and unauthorized row access.

Primary source: Multistep forms.

Module 25 — Integrate Power Pages with web-based technologies

Your current position: Part IV — Microsoft Power Pages → Module 25 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Power Pages
Official module	Integrate Power Pages with web-based technologies
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	6
Instructor focus	SharePoint documents, Power BI, charts, search, Liquid/JavaScript integration and separate security.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: SharePoint documents, Power BI, charts, search, Liquid/JavaScript integration and separate security.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Document management with SharePoint in Power Pages

Concept / Why: SharePoint integration exposes related collaborative documents and document libraries.

Configuration / How: Configure server-based integration, SharePoint sites, enabled tables, document locations, and permissions.

Exam relevance: High/Medium — Dataverse access ≠ SharePoint file access.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Integration with Power BI in Power Pages

Concept / Why: A Power BI semantic model, report, or dashboard provides rich analytics.

Configuration / How: Configure workspace and content sharing, semantic-model or dataset Row-Level Security, refresh or gateway, and any host embedding.

Exam relevance: High/Medium — Embed does not grant dataset permission.

Official page subtopics: Enable Power BI visualization; Parameters; Enable the Power BI Embedded service.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Model-driven charts in Power Pages

Concept / Why: Charts aggregate operational view data; dashboards combine multiple visuals/lists.

Configuration / How: Choose System or Personal, select the source View and filters, and include the required component in the app.

Exam relevance: High/Medium — Power BI vs native chart/dashboard.

Official page subtopics: Parameters.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — SharePoint, Power BI, charts, search, web technologies

SharePoint document management

Power Pages can surface Dataverse record-related SharePoint documents when server-based SharePoint integration/document management is configured. Dataverse stores document location metadata; files live in SharePoint. User must pass Power Pages authorization and underlying integration/configuration. Do not assume a Dataverse File column, Note attachment, and SharePoint document library are interchangeable.

Option	Data location	Best use
SharePoint document management	SharePoint	Collaborative document libraries/versioning
Dataverse File column	Dataverse file storage	File tied directly to row/column and Dataverse API
Note attachment	Dataverse annotation/attachment	Simple note-related files; not full document management

Power BI integration

Power Pages can embed Power BI visualization. Configure enablement, component/report, parameters/filter context, and appropriate authentication/embedding model. Power BI content permission/licensing/embedded capacity decisions are separate from Power Pages Web Roles. Never expose confidential report through insecure “publish” method.

Model-driven charts and site search

Model-driven charts can be surfaced with supported component/parameters for operational visualization. Power Pages search indexes configured website content; relevance and table/list search configuration should be tested. Search result visibility must respect page/data permissions; hiding from navigation alone is not security.

Web technologies

Liquid runs server-side template logic; JavaScript runs client-side; CSS styles; Web API lets supported site operations against Dataverse within Power Pages security/CORS/configuration. Client-side checks are UX only; server-side Table Permissions remain enforcement.

⚠ PL-200 Exam Trap

Embedding Power BI does not automatically share dataset/report access. Dataverse Web Role does not equal Power BI workspace role.

Scenario-based learning

Scenario: Partner page shows report filtered to partner Account.
Correct: Secure Power BI embedding configuration + reliable identity/filter model +

Power Pages authorization.
Wrong: URL parameter alone as security; user could alter it.

Tiny Details You Should Remember

- SharePoint integration needs document management configuration, not only a link.
- Power BI and Power Pages have separate permission layers.
- JavaScript filtering is not authorization.
- Site search configuration and security both matter.

Exam Memory Notes

- Collaborative docs → SharePoint integration.
- Rich analytics → Power BI.
- Server template → Liquid.
- Client behavior → JavaScript; never trust for security.

Module Knowledge Check — Questions

1. Where are files stored in SharePoint document management?
2. Does a Web Role grant Power BI dataset permission?
3. Can a URL filter be the only row-level security?
4. Which language handles Power Pages server templates?
5. Which file option is directly a Dataverse column?

Answers & Explanations

1. In **SharePoint**; Dataverse keeps related location metadata.
2. **No.** Configure Power BI permission/embedding separately.
3. **No.** Enforce with secure identity/RLS/permissions; parameters are not authorization.
4. **Liquid.** JavaScript is client-side.
5. A **Dataverse File column**.

Primary sources: SharePoint integration for Power Pages, Power BI integration.

Module 26 — Explore Power Pages design studio data and security features

Your current position: Part IV — Microsoft Power Pages → Module 26 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Power Pages
Official module	Explore Power Pages design studio data and security features
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	6
Instructor focus	Design Studio Data/Security configuration and negative security testing.

Official Unit-by-Unit Journey

Unit 1 — Introduction into Power Pages design studio data and security features

Concept / Why: Design Studio Data/Security configuration and negative security testing.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Data workspace in Power Pages design studio

Concept / Why: Design Studio Data/Security configuration and negative security testing.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Tables; Views; Forms.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Power Pages security features

Concept / Why: Power Pages security depends on a chain of Page Permission, Table Permission, and Web Role.

Configuration / How: Configure scope, allowed operations, relationship paths, and test identities that should be denied as well as identities that should be allowed.

Exam relevance: High/Medium — Global scope and URL-tampering traps.

Official page subtopics: Page permissions; Table permissions.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - secure Dataverse data access

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; High-level steps; Detailed steps; Launch Power Pages design studio; Table modifications; Create a web page.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Design Studio Data and Security workspaces

Data workspace lets maker inspect/create relevant tables, columns, views, and forms without leaving site design flow. Changes are real Dataverse metadata and can affect apps/solutions; use correct publisher/Solution and avoid uncontrolled production edits.

Security capabilities in Design Studio surface Page Permissions and Table Permissions. Convenience UI does not change underlying model:

1. Identify authenticated/anonymous audience.
2. Create Web Role.
3. Configure Page Permission for page tree/static access.
4. Configure Table Permission with operation and access scope.
5. Associate permission to Web Role; create parent permissions if related scope.
6. Clear site cache/restart where current behavior requires and test every identity.

Least privilege testing matrix

Test identity	Expected test
Anonymous	Public pages only; no unauthorized table data
Authenticated no Web Role	Sign-in works but protected data/page denied
Correct Web Role / owns relationship	Allowed exact Create/Read/Write/Delete
Wrong Account/Contact	Cannot change URL to read another user’s row
Site admin	Admin capability works; does not hide ordinary-user defect

⚠ PL-200 Exam Trap

Data workspace can create a Table Permission quickly, but selecting `Global` just to “make it work” is a classic data-leak trap. Choose narrow scope.

Scenario-based learning

Scenario: Test site works for maker but all visitors see authorization error.
Reasoning: Maker/admin has broad rights. Verify Web Role assignment, Table Permission operations/access type, Contact relationships, Page Permission, and site cache/config—not Dataverse internal role alone.

Tiny Details You Should Remember

- Design Studio security UI maps to persisted Power Pages configuration.
- Maker/admin test is insufficient.
- Page tree permission and table data permission solve different layers.
- Solution/ALM context matters for portal configuration and Dataverse metadata.

Exam Memory Notes

- Secure page + secure table.
- Web Role binds permission to site user.
- Test negative identities and URL tampering.
- Never choose Global only to fix an error.

Module Knowledge Check — Questions

1. Which scope is riskiest when only own rows are required?
2. Why can a maker test pass while customer test fails?
3. What secures static page content?
4. What secures row operations?
5. Which test detects insecure direct object reference?

Answers & Explanations

1. **Global** Table Permission.
2. Maker/admin has broader privileges or admin Web Role; ordinary user path is different.
3. **Page Permission**.
4. **Table Permission** with exact privileges/scope.
5. Change row ID/URL while signed in as another Contact and confirm denial.

Primary source: Power Pages security overview.

Module 27 — Set up Power Pages security

Your current position: Part IV — Microsoft Power Pages → Module 27 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Power Pages
Official module	Set up Power Pages security
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	6
Instructor focus	Web Roles, Page Permissions, Table/Column Permissions and Global/Contact/Account/Self/Parent scopes.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Web Roles, Page Permissions, Table/Column Permissions and Global/Contact/Account/Self/Parent scopes.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Microsoft Dataverse customer tables; Contact; Accounts; Web role.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Secure static content

Concept / Why: Power Pages security depends on a chain of Page Permission, Table Permission, and Web Role.

Configuration / How: Configure scope, allowed operations, relationship paths, and test identities that should be denied as well as identities that should be allowed.

Exam relevance: High/Medium — Global scope and URL-tampering traps.

Official page subtopics: Webpage content; Webpage access control rule; Access management in Power Pages design studio.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Table permissions

Concept / Why: A Security Role is a cumulative set of privileges that grants table actions at defined access depths.

Configuration / How: Configure Create, Read, Write, Delete, Append, Append To, Assign, and Share privileges and their access depths.

Exam relevance: High or Medium — Having a privilege does not prove that the user can reach a particular row.

Official page subtopics: Create a table permission; Use Power Pages design studio; Column permissions.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - Apply table permissions

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Learning objectives; Prerequisites; Scenario; High-level steps; Detailed steps; Launch Power Pages design studio.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Web Roles, Page Permissions, Table Permissions, Column Permissions

Authorization chain

Identity provider authenticates → Dataverse Contact identifies site user → Web Role groups user → Page Permission grants page access and/or Table Permission grants data operations/scope.

Page Permissions

Use for webpage access: restrict read, allow change/manage where supported, apply to children/inheritance according to configuration. Hidden navigation does not replace. Anonymous Users or Authenticated Users special/default web role patterns may apply; scope precisely.

Table Permission access types

Access type	Row scope	Typical use
Global	All rows in table	Public/reference data only after risk review
Contact	Rows related to current Contact through selected relationship	“My cases/requests”
Account	Rows related to Contact’s Account	Partner organization sees its rows
Self	Current Contact row	Profile self-service
Parent	Scope derived through parent Table Permission/relationship	Related child rows under an already scoped parent

Privileges Create, Read, Write, Delete, Append, Append To as exposed by portal permission configuration. Give only needed operations. Related create/attach requires matching permissions on both sides. Parent permission builds a chain; wrong relationship can expose or deny data.

Column Permissions

Where current Power Pages Column Permissions apply, use to further control individual column operations for portal; they supplement Table Permission and Dataverse model. Do not confuse with Dataverse Column Security Profiles for internal users—they are different configuration layers/audiences. Test Web API/forms/lists.

⚠ PL-200 Exam Trap

“Authenticated users can read their own Contact only” → Self, not Global.
“Employees of partner Account see all orders belonging to that Account” → Account access type.

"Contact can see cases directly related to them" → Contact access type.
Always choose relationship path explicitly.

Scenario-based learning

Scenario: A partner's staff need Account's Orders and Order Lines.
Correct: Account-scoped Order permission; Parent-scoped Order Line permission via relationship; exact privileges; Web Role.
Why Global wrong: Cross-partner data leak.

Tiny Details You Should Remember

- Authentication does not equal authorization.
- Table Permission must be associated to Web Role.
- Parent access depends on parent permission + relationship.
- Create may need Append/Append To for related record association.
- Dataverse role and Web Role are different.

Exam Memory Notes

- Global = all.
- Contact = current Contact-related.
- Account = Contact's Account-related.
- Self = current Contact row.
- Parent = child scope inherited through relationship.

Module Knowledge Check — Questions

1. Which access type gives a signed-in user their own Contact row?
2. Which access type lets partner employees see rows for their Account?
3. What connects a Table Permission to users?
4. Is sign-in alone enough to Read a Dataverse list?
5. What is needed for Order Line rows scoped through an allowed Order?
6. Are Power Pages Web Roles Dataverse Security Roles?

Answers & Explanations

1. **Self**.
2. **Account**.
3. A **Web Role** assignment.
4. **No.** Configure Web Role/Table Permission.
5. A **Parent** Table Permission using correct relationship and parent permission.
6. **No.** Separate authorization systems for external site vs internal Dataverse users.

Primary sources: Table permissions, Page permissions, Web roles.

Module 28 — Authentication and user management in Power Pages

Your current position: Part IV — Microsoft Power Pages → Module 28 → 8 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create apps using Power Apps (25–30%) — Power Pages
Official module	Authentication and user management in Power Pages
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	8
Instructor focus	Authentication provider, Contact linking, open vs invitation registration, and authorization troubleshooting.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Authentication provider, Contact linking, open vs invitation registration, and authorization troubleshooting.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Track users as contacts; Contact extensions; Administrator as a website user.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Power Pages authentication settings

Concept / Why: Authentication proves an identity and links it to a Contact; authorisation is decided by Web Roles and permissions.

Configuration / How: Configure the identity provider, registration or invitation process, Contact linking, redirects and settings, and Web Role assignment.

Exam relevance: High/Medium — Sign-in success but 403 → authorization.

Official page subtopics: Example of a setting; Authentication settings categories.

Official Microsoft Learn page: Open Unit 2

Unit 3 — User registration in Power Pages

Concept / Why: Authentication proves an identity and links it to a Contact; authorisation is decided by Web Roles and permissions.

Configuration / How: Configure the identity provider, registration or invitation process, Contact linking, redirects and settings, and Web Role assignment.

Exam relevance: High/Medium — Sign-in success but 403 → authorization.

Official page subtopics: Open registration; Invitation-based registration; Set up automation; Create Invitation; Redeem Invitation; Register user.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Authentication management for Power Pages users

Concept / Why: Authentication proves an identity and links it to a Contact; authorisation is decided by Web Roles and permissions.

Configuration / How: Configure the identity provider, registration or invitation process, Contact linking, redirects and settings, and Web Role assignment.

Exam relevance: High/Medium — Sign-in success but 403 → authorization.

Official page subtopics: External Identities.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Power Pages authentication providers

Concept / Why: Authentication proves an identity and links it to a Contact; authorisation is decided by Web Roles and permissions.

Configuration / How: Configure the identity provider, registration or invitation process, Contact linking, redirects and settings, and Web Role assignment.

Exam relevance: High/Medium — Sign-in success but 403 → authorization.

Official page subtopics: Local authentication; External authentication; Authentication and provider configuration; Azure Active Directory B2C; Migration to Microsoft Entra ID B2C.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Exercise - Use an invitation to register users

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Learning objectives; Prerequisites; Scenario; High-level steps; Create a test contact; Create an invitation flow.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 7

Unit 8 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 8

Instructor Deep Dive — Authentication and user management

Authentication proves identity; authorization decides access. Power Pages links external identity to Contact. Registration can be open/self-service or invitation-based depending business risk. External identity providers use standards/providers such as Microsoft Entra ID/B2C/external identity options current product supports; local authentication is legacy/not preferred in many modern designs—follow current Microsoft recommendations.

Registration patterns

Pattern	Use	Risk/control
Open registration	Public community/low-friction sign-up	Spam/identity assurance; minimal role by default
Invitation-based	Controlled partners/customers	Pre-create Contact/invitation, assign intended role/account
External provider	Enterprise/social/federated identity	Configure app registration, redirect URI, claims, secrets/certs, lifecycle

Invitation redemption should link correct Contact and Web Roles. Never assign privileged Web Role purely from untrusted self-asserted claim without validation. Provider sign-in may create/link Contact according to settings.

Authentication settings and migration

Site settings/provider configuration control registration, confirmation, password reset, identity provider behavior. Azure AD naming in older pages may now be Microsoft Entra ID; current Learn Module 28 references Microsoft Entra ID B2C migration terminology. Treat deprecated local/provider settings carefully and follow current provider docs.

User management/troubleshooting

- Verify Contact not disabled and external identity link correct.
- Verify invitation status/expiry/redemption and Account relationship.
- Verify Web Role assignments.
- Verify provider app registration/redirect URI/secret validity.
- Sign out/private session to avoid admin cache.
- Authentication success + 403 indicates authorization layer, not necessarily provider failure.

⚠ PL-200 Exam Trap

User can sign in but sees “not authorized” → do not reconfigure identity provider first. Check Web Role, Page Permission, Table Permission, Contact/Account relationship.

Scenario-based learning

Scenario: Only vetted distributors should access pricing portal; each belongs to an Account.
Correct: Invitation-based registration, Contact linked to Account, Distributor Web Role, Account-scoped permissions.
Why open registration wrong: Anyone could create identity; authorization still needed and vetting requirement unmet.

Tiny Details You Should Remember

- Authenticated site user is tracked as Contact.
- Invitation controls onboarding, not all subsequent authorization.
- Provider naming/features can be deprecated/renamed; use current docs.
- 401/sign-in failure and 403/authorization failure imply different troubleshooting paths.

Exam Memory Notes

- Authentication = who.
- Authorization = what.

- Controlled partner onboarding → Invitation.
- Web Roles/permissions after sign-in.
- Contact/Account relationship drives scoped access.

Module Knowledge Check — Questions

1. Which pattern best controls partner onboarding?
2. A user signs in but cannot open protected page. Which layer first?
3. What Dataverse row represents the site user?
4. Does invitation redemption alone grant all table data?
5. Which items must match for external provider sign-in?

Answers & Explanations

1. **Invitation-based registration.** It ties vetted identity to intended Contact/account.
2. **Authorization:** Web Role, Page/Table Permission, relationships.
3. **Contact**.
4. **No.** Web Roles and permissions govern access.
5. Provider/app registration, redirect URI, credentials, claims/configuration, and site settings per provider documentation.

Primary sources: Power Pages authentication, Authentication providers, Invite contacts.

Part V — Logic and Process Automation

Official-order range: Modules 29–34. Sequence intentionally preserved from the current exam catalog.

Module 29 — Best practices for error handling in Power Automate flows

Your current position: Part V — Logic and Process Automation → Module 29 → 5 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create/manage logic and process automation (25–30%)
Official module	Best practices for error handling in Power Automate flows
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	5
Instructor focus	Run After, Scopes, retry/idempotency, Run History, analytics and failure diagnosis.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Run After, Scopes, retry/idempotency, Run History, analytics and failure diagnosis.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Basic concepts; Key features and capabilities.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Configure run after option

Concept / Why: Run After, Scopes, retry/idempotency, Run History, analytics and failure diagnosis.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Use the Run after option to handle errors; Add a parallel branch for immediate notification; Use Parse JSON to get the current flow run information.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Power Automate analytics

Concept / Why: Run After, Scopes, retry/idempotency, Run History, analytics and failure diagnosis.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Built-in errors dashboard.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 4

Unit 5 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 5

Instructor Deep Dive — Flow error handling and troubleshooting

A production Cloud Flow must handle more than the successful path. Plan for failures, timeouts, skipped actions, retries, authentication problems, data conflicts, and partial completion.

Configure run after

You can configure an Action or Scope to run according to the outcome of its predecessor:

- is successful
- has failed
- is skipped
- has timed out

Use Scopes for a Try / Catch / Finally pattern. Put normal actions in Try; configure Catch to run after failure or timeout; configure Finally to run after all relevant outcomes. A notification or logging action can capture the flow-run URL and a safe version of the error body. Use secure inputs and outputs to hide sensitive payloads, while understanding that this reduces troubleshooting visibility.

Retry and idempotency

Retry policies can handle transient 429 and 5xx errors. Blindly retrying a permanent 400 data-validation error is not useful. A retry can repeat side effects, so design for idempotency with unique keys, an “already processed” marker, and safe upsert behaviour. Parallel branches and Apply to each concurrency can also create race conditions.

Monitoring

- Run History: trigger inputs/outputs, action status, duration, error.
- Analytics/errors dashboard: trend and failure patterns.
- Connection health/authentication.
- Trigger condition/filter and skipped runs.
- Dataverse/System Jobs for related classic processes separately.

Current troubleshoot documentation mentions 28-day Cloud Flow run history in relevant experiences; retention/licensing can evolve, so verify actual environment. Do not promise permanent audit from run history.

⚠ PL-200 Exam Trap

Email action after failed Create step will be skipped by default because default Run After is success. Explicitly configure failed/timed out path. “Retry everything” can create duplicates.

Scenario-based learning

Scenario: Invoice creation succeeds, email fails, and rerun creates duplicate invoice.
Correct reasoning: Separate idempotent transaction/state, use alternate key/upsert or processed flag, Scope error handling, retry only email or resubmit from safe point.

Tiny Details You Should Remember

- Default next action normally runs after success.
- Skipped may be consequence, not root failure; trace upstream.
- Secure Inputs/Outputs can hide data from run history.
- Authentication errors often require connection repair, not formula change.

Exam Memory Notes

- Try/Catch/Finally → Scopes + Run After.
- 429/5xx → transient/retry consideration.

- 400 → request/data problem.
- Design retries to be idempotent.

Module Knowledge Check — Questions

1. Which setting makes an alert run only after failure?
2. Why can blind retry create duplicate records?
3. Where do you inspect action inputs/outputs?
4. What does `skipped` often indicate?
5. Which pattern groups error logic cleanly?

Answers & Explanations

1. **Configure Run After** with `has failed`/`timed out` as needed.
2. Previous side effect may have succeeded before failure; rerun repeats it.
3. **Run History** (unless secure settings hide them).
4. An upstream branch/action did not meet condition or failed; find the root action.
5. **Scope-based Try/Catch/Finally**.

Primary sources: Fix flow failures, Error handling best practices.

Module 30 — Use Dataverse triggers and actions in Power Automate

Your current position: Part V — Logic and Process Automation → Module 30 → 7 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create/manage logic and process automation (25–30%)
Official module	Use Dataverse triggers and actions in Power Automate
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	7
Instructor focus	Dataverse row trigger filters/scope, query/actions, Lookups/relationships, pagination and loop prevention.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Dataverse row trigger filters/scope, query/actions, Lookups/relationships, pagination and loop prevention.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Learning objectives; Prerequisites; Authenticate; Set up the Dataverse environment; Name your flow steps.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Dataverse triggers

Concept / Why: A Trigger starts a Flow because of an event, manual request, or schedule condition.

Configuration / How: Configure scope, Select columns, Filter rows or trigger conditions, and the connection identity.

Exam relevance: High/Medium — Filter at source; avoid loops.

Official page subtopics: When a row is added, modified or deleted; Change type; Table name; Scope; Select columns; Filter rows.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Query data

Concept / Why: Dataverse row trigger filters/scope, query/actions, Lookups/relationships, pagination and loop prevention.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Select columns; Expand query; Use the Get a row by ID action; Use the List rows action; Retrieve more than 5,000 rows; Use OData style criteria.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Create, update, delete, and relate actions

Concept / Why: Dataverse row trigger filters/scope, query/actions, Lookups/relationships, pagination and loop prevention.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Create new rows; Update rows; Upsert rows; Relate data; Specify the ID for each row; Set lookup fields in the Add a new row and Update a row actions.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Create a cloud flow with a Dataverse connector

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Set up the contacts table (skip if table already exists); Create the cloud flow.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 6

Unit 7 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 7

Instructor Deep Dive — Dataverse triggers, queries, actions

Trigger configuration

When a row is added, modified or deleted includes:

- Change type: Added, Modified, Deleted.
- Table name.
- Scope: User, Business Unit, Parent:Child Business Units, Organization.
- Select columns: update trigger only; logical names; trigger if listed column included in update request even if same value is sent.
- Filter rows: OData condition evaluated after row saved; reduces flow runs.
- Delay until where relevant.

Relationship-only 1:N/N:N association changes do not trigger this row trigger as an ordinary row update per current docs; use supported relationship/action/event design.

Query actions

Action	Use
Get a row by ID	One known GUID
List rows	Filter/project/order/page multiple rows
Add a new row	Create
Update a row	Update by ID
Delete a row	Delete
Upsert a row	Create or update based on key
Relate/Unrelate rows	Establish/remove relationship association

Use Select columns to minimize payload; Filter rows OData; Expand query retrieves related data; FetchXML where supported for more complex query. Pagination settings are needed beyond default action result set; current Module mentions retrieving over 5,000 rows. Avoid hardcoding GUIDs; use dynamic content/Environment Variables.

Lookups

Set Lookup with expected OData bind/reference or connector field syntax; use correct related table plural/entity set and GUID. Relate rows establishes relationship; setting lookup and relating are not always the same for N:N.

Trigger loops

Flow triggers on update and then updates same row → infinite/duplicate loop risk. Use trigger conditions, Select columns, status flag, compare values, or separate table/action. Concurrency control and ownership connection identity matter.

⚠ PL-200 Exam Trap

- Select columns uses **logical names**, not display labels.
- Select columns triggers on presence in update request, not necessarily value difference.
- Filter rows reduces trigger runs; Condition after trigger still consumes a run.
- Scope in flow trigger is about changes visible under connection/user scope, not Dataverse Security Role replacement.

Scenario-based learning

Scenario: Flow should run only when statuscode changes to Approved, not every edit.
Correct: Select columns/status logical name + Filter rows/trigger condition for Approved; loop guard if flow updates row.

Scenario: Integrate customer by ExternalId, create if new/update if present.
Correct: Alternate Key + Upsert action/pattern.
Why Get then Add/Update weaker: Race conditions and extra calls.

Tiny Details You Should Remember

- Trigger scope values mirror user/BU hierarchy concepts.
- Select columns applies to update behavior.
- N:N association changes may need a relationship-specific design.
- List rows projection/filter/pagination improve scale.
- Flow connection identity determines connector authorization.

Exam Memory Notes

- Known GUID → Get row.
- Many rows → List rows.
- Business key create/update → Upsert.
- Relationship → Relate rows.

- Filter early at trigger/query.

Module Knowledge Check — Questions

1. Which trigger field limits Update events to specified logical columns?
2. Which action retrieves one known GUID?
3. Which action safely supports create-or-update by key?
4. Why is a Condition after trigger less efficient than Filter rows?
5. Can updating the same row cause a trigger loop?
6. Does an N:N association always fire the ordinary row update trigger?

Answers & Explanations

1. **Select columns**.
2. **Get a row by ID**.
3. **Upsert a row** with an Alternate Key/design.
4. The flow already started/consumed a run before the Condition.
5. **Yes.** Add a trigger condition/flag/value comparison.
6. **No.** Current docs state relationship changes do not trigger that row trigger as ordinary updates.

Primary source: Dataverse triggers/actions.

Module 31 — Introduction to expressions in Power Automate

Your current position: Part V — Logic and Process Automation → Module 31 → 8 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create/manage logic and process automation (25–30%)
Official module	Introduction to expressions in Power Automate
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	8
Instructor focus	Dynamic Content, Workflow expressions, variables, Condition/Switch/loops, null/date handling.

Official Unit-by-Unit Journey

Unit 1 — Introduction to expressions

Concept / Why: Dynamic Content, Workflow expressions, variables, Condition/Switch/loops, null/date handling.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Get started with expressions

Concept / Why: An Expression uses workflow function syntax to transform or compose dynamic output.

Configuration / How: Choose the function family, handle nulls, arrays, and Date and Time values, use Compose for clarity, and test representative cases.

Exam relevance: High/Medium — Dynamic Content vs expression; array vs record.

Official page subtopics: Auto suggest, hints, and links in the formula bar.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Notes make things easier

Concept / Why: Dynamic Content, Workflow expressions, variables, Condition/Switch/loops, null/date handling.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Defining text.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Types of functions

Concept / Why: Dynamic Content, Workflow expressions, variables, Condition/Switch/loops, null/date handling.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: String functions; Collection functions; Logical functions; Conversion functions; Math functions; Date and time functions.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Write complex expressions

Concept / Why: An Expression uses workflow function syntax to transform or compose dynamic output.

Configuration / How: Choose the function family, handle nulls, arrays, and Date and Time values, use Compose for clarity, and test representative cases.

Exam relevance: High/Medium — Dynamic Content vs expression; array vs record.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Exercise - Creating a manual flow and using expressions

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you learned..

Official Microsoft Learn page: Open Unit 7

Unit 8 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 8

Instructor Deep Dive — Expressions, dynamic content, variables, loops

Dynamic Content inserts outputs from prior trigger/actions. Expression (`fx`) composes functions and references when simple token insufficient.

Function families

- String: `concat`, `substring`, `replace`, `toLower`, `trim`.
- Collection: `first`, `last`, `length`, `join`, `contains`, `union`.
- Logical: `if`, `equals`, `and`, `or`, `not`, `coalesce`, `empty`.
- Conversion: `string`, `int`, `float`, `bool`, `json`.
- Date/time: `utcNow`, `formatDateTime`, `addDays`, `convertTimeZone`.
- Referencing: `triggerBody`, `outputs`, `body`, `items`, `variables`.

Function names/parameters use Workflow Definition Language style. Handle `null` vs empty string; use safe navigation/coalesce where needed. Time zones require explicit conversion; storing/display semantics differ.

Control actions

Control	Use
Condition	True/false branch
Switch	One value matched to multiple cases
Apply to each	Iterate array
Do until	Repeat until condition/limit
Scope	Group/run-after/error structure
Initialize/Set/Increment variable	Mutable run state

Apply to each can be auto-added when selecting array token. Concurrency can improve speed but order/race/shared variable concerns. Prefer Select/Filter array/data query when possible over huge loops.

Expression readability

Rename actions meaningfully so references remain understandable. Add Notes/comments. Break a long expression into Compose steps when troubleshooting/maintenance improves. Expression editor type hints/autosuggest help but test nulls and locale/date edge cases.

⚠ PL-200 Exam Trap

items('Apply_to_each') is current loop item; outputs('Get_row') refers action output. Mixing array and single record causes Apply-to-each surprise. Dynamic Content is not a literal string unless inserted correctly.

Scenario-based learning

Scenario: If ApproverEmail missing, use ManagerEmail; otherwise use ApproverEmail. **Correct expression:** coalesce(<ApproverEmail>, <ManagerEmail>) after checking connector null/empty behavior; perhaps if(empty(...), ..., ...) if empty string possible.

Tiny Details You Should Remember

- Flow date/time generally works with UTC; explicitly format/convert for recipients.
- Switch suits discrete cases; Condition suits boolean logic.
- Variables must be initialized before set/increment.
- Apply to each input is array.
- Notes/action renaming reduce maintenance errors.

Exam Memory Notes

- Dynamic Content = previous outputs.
- Expression = function/composition.
- Condition = two paths; Switch = many cases.
- Apply to each = array loop.
- Coalesce/empty = null defense.

Module Knowledge Check — Questions

1. Which control iterates an array?
2. Which function returns the first non-null value?
3. Which control best handles Status=A/B/C/D?
4. Why can concurrency with a shared variable be unsafe?
5. Which function formats a date/time string?

Answers & Explanations

1. **Apply to each**.
2. `coalesce` (account for empty string separately if needed).
3. **Switch**.
4. Parallel iterations may update/read variable in nondeterministic order.
5. `formatDateTime` (and convert timezone explicitly when needed).

Primary source: Workflow expressions, Use expressions in Power Automate.

Module 32 — Define and create business rules in Dataverse

Your current position: Part V — Logic and Process Automation → Module 32 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create/manage logic and process automation (25–30%)
Official module	Define and create business rules in Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	6
Instructor focus	Business Rule scope/actions and explicit classic Dataverse workflow objective bridge.

Official Unit-by-Unit Journey

Unit 1 — Define business rules - Introduction

Concept / Why: A Business Rule implements declarative field conditions and actions.

Configuration / How: Configure the scope, conditions, true and false actions, validate the rule, and activate it.

Exam relevance: High/Medium — Business Rule vs Power Fx/Flow/plug-in.

Official page subtopics: Scope; Business rules and model-driven app forms; Should I use a business rule or client-side scripting with JavaScript?; Server-side business rules.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Define the components of a business rule

Concept / Why: A Business Rule implements declarative field conditions and actions.

Configuration / How: Configure the scope, conditions, true and false actions, validate the rule, and activate it.

Exam relevance: High/Medium — Business Rule vs Power Fx/Flow/plug-in.

Official page subtopics: Conditions; Actions.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Create a business rule

Concept / Why: A Business Rule implements declarative field conditions and actions.

Configuration / How: Configure the scope, conditions, true and false actions, validate the rule, and activate it.

Exam relevance: High/Medium — Business Rule vs Power Fx/Flow/plug-in.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - Create a business rule

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Business Rules and the classic workflow bridge

Business Rule

A Business Rule uses declarative conditions and actions to implement table or form logic. Common actions include setting or clearing a value, setting Required or Recommended, showing or hiding, enabling or disabling, validating and showing an error, and displaying a business recommendation where supported.

Scope determines execution:

Scope	Behavior
Specific form	That Main Form client experience
All forms	Model-driven forms for table as supported
Table (Entity)	Server-side and form behavior for supported actions, so imports/API can be affected

Canvas Apps do not automatically execute model-driven form-scoped Business Rule UI behavior. Table-scope server validation/data action can apply at Dataverse layer, but test support/action. Business Rules cannot aggregate child rows, call arbitrary connectors, create complex loops, or replace all Power Fx/flow/plugin logic.

Business Rule vs Power Fx vs Power Automate vs low-code plug-in

Requirement	Best first choice	Why
Make field required/visible on model-driven form	Business Rule	Declarative form/table rule
Canvas control behavior/calculation	Power Fx in app	Immediate UI formula
Same-row computed column	Formula Column	Reusable Dataverse calculation
Send email/create Teams approval	Cloud Flow	Connector/background automation
Validate every Dataverse create synchronously, rollback invalid	Automated low-code plug-in pre-operation	Server transaction across all entry channels
Guided stages	BPF	Human process bar

Business Rule configuration

1. Create within table/Solution.
2. Set scope intentionally.
3. Add Conditions with AND/OR grouping.
4. Add True/False actions.
5. Validate, save, activate.
6. Test forms, imports/API, and conflicting rules.

Business Rules don’t guarantee execution order as a substitute for coordinated design. Avoid multiple rules fighting over same field. Deactivate before edit if required and publish/solution transport.

Objective bridge — Classic Dataverse workflows

Current Skills Measured explicitly requires configure/troubleshoot/log classic Dataverse workflows, even though current recommended module set has no dedicated standalone workflow module.

Classic workflow types:

- **Background workflow:** asynchronous System Job; recommended over real-time when immediate transaction not needed.
- **Real-time workflow:** synchronous; can run before/after operation and can stop/cancel with error; performance impact.

Triggers include row created/status/assigned/fields changed/on-demand/child where supported. Steps: check condition, conditional branch, wait condition (background only), create/update/assign/send email/start child workflow/change status. Convert background to real-time only when step compatibility permits; wait conditions aren't valid in real-time.

Troubleshooting:

- Background workflows create System Job entries; inspect status/message/owner/permissions and resume/cancel.
- Real-time workflow successful runs do not create normal System Job rows; error logging can be retained according to configuration.
- Check process activation, scope, trigger fields, owner, security, timeout, record status, and dependencies.
- Microsoft recommends considering Power Automate as modern alternative for many asynchronous/cross-service scenarios, but classic workflow remains exam objective and real-time transactional scenarios may still appear.

⚠ PL-200 Exam Trap

- Business Rule `All Forms` is not Canvas App behavior guarantee.
- Need cross-service email after update → Cloud Flow, not Business Rule.
- Need synchronous cancel of invalid save from all channels → real-time workflow or automated low-code plug-in depending stated requirement/current supported choice; not asynchronous Flow.
- Background workflow can Wait; real-time cannot use Wait condition.

Scenario-based learning

Scenario: On model-driven Case form, Resolution field becomes required only when Status Reason=Resolved.
Correct: Business Rule.
Why Flow wrong: Async notification cannot enforce immediate form requiredness.

Scenario: Legacy process waits seven days then creates follow-up.
Correct: Background workflow can contain wait, though modern Cloud Flow should be evaluated. Real-time workflow cannot wait.

Tiny Details You Should Remember

- Activate Business Rule before it runs.
- Scope determines client/server reach.

- Business Rule security does not replace Security Role.
- Background workflow logs as System Job; real-time success normally not.
- Wait condition incompatible with real-time workflow.

Exam Memory Notes

- Form field behavior → Business Rule.
- Cross-service/background → Cloud Flow.
- Background classic workflow = async + System Job.
- Real-time classic workflow = sync + can cancel; no wait.
- Table scope matters.

Module Knowledge Check — Questions

1. Which feature makes a model-driven field required based on another field?
2. Which Business Rule scope can apply server-side to supported actions?
3. Which classic workflow type supports a Wait condition?
4. Where do you troubleshoot failed background workflows?
5. Does a form-scoped Business Rule automatically run in Canvas App UI?
6. Which feature is best for a Teams approval?

Answers & Explanations

1. **Business Rule.** It is declarative immediate form logic.
2. **Table/Entity** scope.
3. **Background workflow.** Real-time cannot wait.
4. **System Jobs / workflow logs** with process/owner/security review.
5. **No.** Canvas uses Power Fx; test table-scope server effects separately.
6. **Power Automate Cloud Flow.** It uses connectors/approval actions.

Primary sources: Business Rules, Real-time workflows, Workflow steps, Monitor processes.

Module 33 — Set up low-code plug-ins

Your current position: Part V — Logic and Process Automation → Module 33 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create/manage logic and process automation (25–30%)
Official module	Set up low-code plug-ins
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	9
Instructor focus	Dataverse Accelerator, Instant vs Automated low-code plug-ins, parameters/scope/stages.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Dataverse Accelerator, Instant vs Automated low-code plug-ins, parameters/scope/stages.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Low-code plug-ins

Concept / Why: Low-code plug-in server-side Power Fx logic; Instant on-demand or Automated Dataverse event.

Configuration / How: In Dataverse Accelerator, configure the Solution, parameters, scope, event, stage, Connection References, and tests.

Exam relevance: High/Medium — Instant vs Automated; transactional rollback.

Official page subtopics: On-demand plug-ins; Automated plug-ins.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Create a low-code plug-in

Concept / Why: Low-code plug-in server-side Power Fx logic; Instant on-demand or Automated Dataverse event.

Configuration / How: In Dataverse Accelerator, configure the Solution, parameters, scope, event, stage, Connection References, and tests.

Exam relevance: High/Medium — Instant vs Automated; transactional rollback.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Instant plug-ins

Concept / Why: Low-code plug-in server-side Power Fx logic; Instant on-demand or Automated Dataverse event.

Configuration / How: In Dataverse Accelerator, configure the Solution, parameters, scope, event, stage, Connection References, and tests.

Exam relevance: High/Medium — Instant vs Automated; transactional rollback.

Official page subtopics: Choose good plug-in names; Set up input and output parameters; Use input parameters in your logic; Use output parameters in your logic; Set up scope; Test instant plug-ins.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Create instant plug-ins

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Task - Prepare your environment; Build a plug-in to calculate debt ratio; Task - Build plug-in to calculate contact debt ratio.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Automated plug-ins

Concept / Why: Low-code plug-in server-side Power Fx logic; Instant on-demand or Automated Dataverse event.

Configuration / How: In Dataverse Accelerator, configure the Solution, parameters, scope, event, stage, Connection References, and tests.

Exam relevance: High/Medium — Instant vs Automated; transactional rollback.

Official page subtopics: When your plug-in runs; Multiple automations on the same event; Avoid infinite loops.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Exercise - Create automated plug-ins

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Create automated plug-in; Test the plug-in.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you learned..

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Set up low-code plug-ins

A low-code plug-in is server-side Dataverse logic written with Power Fx. Create and test it in a Dataverse-provisioned Environment by using the Dataverse Accelerator app. As an ALM best practice, create a Solution first and associate the plug-in with that Solution.

Instant vs Automated

Instant low-code plug-in	Automated low-code plug-in
Caller explicitly invokes from Canvas App/Power Automate	Dataverse create/update/delete event invokes
Input/output parameters	Triggering row (ThisRecord/event context), no caller outputs
Global or Entity scope	Bound to table/event
Reusable on-demand function/action	Consistent validation/manipulation on every entry channel

Instant plug-in:

- Display name becomes publisher-prefixed action name and cannot be changed after save (current module).
- Input/output parameters; current module states inputs cannot be marked optional—caller supplies all.
- Scope default Global; Entity scope requires row context.
- Test from Dataverse Accelerator with request values.

Automated plug-in:

- Table + Create/Update/Delete event.
- Pre-operation can validate/change data before save; error stops and rolls back.
- Post-operation handles logic after main operation but still transaction-aware behavior as documented.
- Runs whether change comes from app, flow, agent, or API.
- Avoid infinite loops when logic updates triggering table.

Naming, Solution, security

Use clear verb-based action name, choose publisher prefix, parameter types, and least privilege connection. Put plug-in, connection references, and dependent components in same Solution. Test with calling user context and error behavior.

⚠ PL-200 Exam Trap

“Reusable calculation called only when button selected” → Instant.
“Reject invalid Account Number no matter which app/API writes” → Automated pre-operation.
Cloud Flow is async by default and cannot provide the same transaction rollback behavior.

Scenario-based learning

Scenario: Multiple Canvas Apps and flows need one DebtRatio calculation that returns Decimal.
Correct: Global Instant low-code plug-in with inputs/outputs.
Why duplicate Power Fx wrong: Logic drifts across apps.

Tiny Details You Should Remember

- Dataverse Accelerator is preinstalled in environments created on/after 2023-10-01 per current module; older environments may need update/install.
- Instant display name is immutable after save.
- All instant input parameters currently required.
- Default Instant scope is Global.
- Automated event logic sees row; no normal output parameters.

Exam Memory Notes

- On-demand + outputs → Instant.
- Event + every channel → Automated.
- Validate before save → Pre-operation.
- Build in Solution via Dataverse Accelerator.

Module Knowledge Check — Questions

1. Which plug-in type has input/output parameters?
2. Which type fires for Dataverse row events?
3. Which stage is best to reject invalid data efficiently?
4. What is the default Instant plug-in scope?
5. Can Instant input parameters be optional in current module behavior?
6. Where should the plug-in be associated for ALM?

Answers & Explanations

1. **Instant** low-code plug-in.
2. **Automated** low-code plug-in.
3. **Pre-operation**, using error to rollback invalid save.
4. **Global**.
5. **No.** Current module says all defined inputs require values.
6. A custom **Solution** with correct publisher/dependencies.

Primary source: Low-code plug-ins.

Module 34 — Use common low-code plug-in scenarios

Your current position: Part V — Logic and Process Automation → Module 34 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create/manage logic and process automation (25–30%)
Official module	Use common low-code plug-in scenarios
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	9
Instructor focus	Server-side Power Fx, transactional validation/manipulation, connectors, bound/unbound invocation.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Server-side Power Fx, transactional validation/manipulation, connectors, bound/unbound invocation.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Learning objectives; Prerequisites.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Use Power Fx with plug-ins

Concept / Why: Low-code plug-in server-side Power Fx logic; Instant on-demand or Automated Dataverse event.

Configuration / How: In Dataverse Accelerator, configure the Solution, parameters, scope, event, stage, Connection References, and tests.

Exam relevance: High/Medium — Instant vs Automated; transactional rollback.

Official page subtopics: Work with input/output parameters; Use ThisRecord with plug-ins; Reference tables in formulas; Variables; Use multiple lines in an expression.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Implement data validation

Concept / Why: Server-side Power Fx, transactional validation/manipulation, connectors, bound/unbound invocation.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Use instant plug-ins; Use automated plug-ins; Use Power Fx for validating data.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Data manipulation

Concept / Why: Server-side Power Fx, transactional validation/manipulation, connectors, bound/unbound invocation.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Change data before it’s saved; Create related data rows; Find and remove related data.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Use connectors in plug-in logic

Concept / Why: A Connector defines available operations, a Connection contains authentication, and a Data Source is the app’s binding to the service.

Configuration / How: Configure the Connection, gateway, DLP policy, permissions, sharing, and delegation behaviour.

Exam relevance: High/Medium — App share ≠ data share.

Official page subtopics: Use connectors in your logic; Exercise - Build an instant low-code plug-in.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Use instant plug-ins from Power Apps

Concept / Why: Low-code plug-in server-side Power Fx logic; Instant on-demand or Automated Dataverse event.

Configuration / How: In Dataverse Accelerator, configure the Solution, parameters, scope, event, stage, Connection References, and tests.

Exam relevance: High/Medium — Instant vs Automated; transactional rollback.

Official page subtopics: Use global instant plug-ins; Use entity instant plug-ins; Prepare your environment; Invoke the plug-in.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Use instant plug-ins from Power Automate

Concept / Why: Low-code plug-in server-side Power Fx logic; Instant on-demand or Automated Dataverse event.

Configuration / How: In Dataverse Accelerator, configure the Solution, parameters, scope, event, stage, Connection References, and tests.

Exam relevance: High/Medium — Instant vs Automated; transactional rollback.

Official page subtopics: Use an unbound low-code plug-in in a flow; Use a bound low-code plug-in in a flow.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Low-code plug-in scenarios and transaction logic

Power Fx context

Plug-in Power Fx is server-side and cannot use UI/device/control functions. Instant plug-ins use parameters; Entity Instant/Automated can use row context such as `ThisRecord`. Multiple statements/variables/tables follow supported subset.

Validation patterns

Pattern	Result
Instant returns IsValid + message	Caller decides UX/action
Automated pre-operation Error(...)	Dataverse operation stops and transaction rolls back
Automated writes ValidationStatus	Save allowed; follow-up remediation possible

Use hard rejection only when data must never persist invalid. For external validation service outage, blocking all saves may harm business; consider status/queue review.

Data manipulation and transaction

Pre-operation can change target row before saved (set defaults/normalized values). Automated plug-in can create related rows; current module notes data changes are part of original transaction, so failure can roll back primary and related changes. This distinguishes from Cloud Flow eventual async work.

Avoid recursion: plug-in updates same row/table and retriggers. Filter attributes/event and set guards where supported. Keep server transaction short; slow external connectors can affect user request and reliability.

Connectors and connection references

To use connector in plug-in logic, create Connection Reference in Environment/Solution. Connection Reference is deployable indirection; actual Connection supplies credentials. Include both plug-in and reference in Solution. DLP/licensing/connection owner/security apply.

Invoke from apps/flows

- Canvas App: Global Instant via Environment language object/action; Entity Instant via bound table/action and existing row ID.
- Power Automate: Dataverse Perform an unbound action for Global, Perform a bound action for Entity. Current official unit says Entity Instant plug-in is invoked by appropriate bound action, not “cannot use from Power Automate.”
- Outputs appear as result/dynamic content.

Automation selection matrix

Need	Choice
Immediate Canvas UI formula only	App Power Fx
Reusable server function with result	Instant low-code plug-in
Transactional validation across API/apps	Automated pre-operation plug-in
Cross-service long-running/approval	Cloud Flow
Simple field rule	Business Rule

Need	Choice
Pro-code complex SDK logic	C# plug-in (PL-400 territory; recognize escalation)

⚠ PL-200 Exam Trap

External connector call inside synchronous pre-operation can slow/block save. Even if technically supported, choose async Cloud Flow/status pattern when immediate transaction consistency isn't required.

Scenario-based learning

Scenario: New Project must atomically create three default Tasks; if any task fails, Project must not exist.
Correct: Automated plug-in transaction logic.
Why Cloud Flow wrong: Project may already commit before async task failure; compensating logic needed.

Scenario: User clicks Calculate Risk in Canvas App and needs result immediately without saving.
Correct: Instant low-code plug-in with inputs/output.

Tiny Details You Should Remember

- UI-oriented Power Fx functions are not available server-side.
- `Error` in automated validation rolls back.
- Pre-operation can change target before save.
- Connector needs Connection Reference in Solution.
- Global = unbound action; Entity = bound action/row context.

Exam Memory Notes

- Atomic/rollback → Automated plug-in.
- Reusable call/result → Instant plug-in.
- Connection Reference, not hard-coded connection.
- Keep synchronous transaction short.
- App UI behavior alone → app Power Fx.

Module Knowledge Check — Questions

1. Which function pattern rejects and rolls back an invalid automated operation?
2. Which stage can change target values before save?
3. What deployable component points plug-in connector logic to a connection?
4. Global Instant plug-in is bound or unbound?
5. Which feature best handles a two-day approval?

6. Why avoid slow external calls in pre-operation?

Answers & Explanations

1. `Error(...)` in automated pre-operation validation.
2. **Pre-operation**.
3. **Connection Reference**.
4. **Unbound/Global**; Entity is bound.
5. **Cloud Flow**. Long-running approval is not a database transaction.
6. It blocks/extends the synchronous save transaction and increases timeout/failure risk.

Primary source: Common low-code plug-in scenarios.

Blueprint Bridge — Business Process Flows complete objective checklist

Module 13 introduced BPF. Current blueprint additionally asks: configure BPF; add stages, workflows, and flow steps; manage BPF table. Remember:

- Activate BPF, associate Security Roles, order multiple BPFs.
- Stage holds Steps tied to columns; mark required for stage progression as supported.
- Branching conditions choose paths; cross-table stages require relationships and maximum table/stage limits.
- A stage can include workflow/flow step integrations where supported; Cloud Flow remains separate run.
- BPF table row tracks active stage/path/process instance; users need table privileges and component must be in Solution.
- Concurrent BPF instances and switching process can affect which instance is active; do not store business truth only in visual stage label.

Part VI — Power BI Foundation

Official-order range: Modules 35–35. Sequence intentionally preserved from the current exam catalog.

Module 35 — Get started building with Power BI

Your current position: Part VI — Power BI Foundation → Module 35 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create model-driven apps — Power BI integration / current prep
Official module	Get started building with Power BI
Catalog last modified	2026-06-15T23:15:00+00:00
Official units	6
Instructor focus	Power BI semantic models, reports, dashboards, workspaces, refresh and secure embedding.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Power BI semantic models, reports, dashboards, workspaces, refresh and secure embedding.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Use Power BI

Concept / Why: A Power BI semantic model, report, or dashboard provides rich analytics.

Configuration / How: Configure workspace and content sharing, semantic-model or dataset Row-Level Security, refresh or gateway, and any host embedding.

Exam relevance: High/Medium — Embed does not grant dataset permission.

Official page subtopics: Explore the flow of Power BI.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Building blocks of Power BI

Concept / Why: A Power BI semantic model, report, or dashboard provides rich analytics.

Configuration / How: Configure workspace and content sharing, semantic-model or dataset Row-Level Security, refresh or gateway, and any host embedding.

Exam relevance: High/Medium — Embed does not grant dataset permission.

Official page subtopics: Create a semantic model; Create visualizations in a report; Create a dashboard.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Tour and use the Power BI service

Concept / Why: A Power BI semantic model, report, or dashboard provides rich analytics.

Configuration / How: Configure workspace and content sharing, semantic-model or dataset Row-Level Security, refresh or gateway, and any host embedding.

Exam relevance: High/Medium — Embed does not grant dataset permission.

Official page subtopics: Organize items with workspaces; Explore sample reports; Distribute content; Explore template apps; Refresh a semantic model.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Knowledge check

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Power BI building blocks and embedding

Power BI flow: connect/transform data → semantic model → report pages/visuals → dashboard tiles/service distribution. A `Report` is multi-page interactive analysis; a `Dashboard` is single-page service canvas of tiles; a `Semantic model` contains relationships/measures/data.

For PL-200, focus on embedding/choosing Power BI in model-driven/Power Pages, not deep DAX engineering.

Need	Artifact
Define reusable measures/relationships	Semantic model
Interactive multi-page analysis	Report
At-a-glance pinned tiles	Dashboard
Team authoring/distribution	Workspace/App with permissions

Refresh method depends on source/DirectQuery/import/gateway. Dataverse security does not automatically translate to Power BI row-level security in every architecture. Embed permission, workspace/app access, dataset credential/refresh, and RLS must be designed.

⚠ PL-200 Exam Trap

Adding Power BI report to Model-driven App does not grant Power BI permission. Dashboard and report are not synonyms.

Scenario-based learning

Scenario: Users need cross-source profitability with drill-through and measures.
Correct: Power BI semantic model + report, then secure embed.
Why model-driven chart wrong: Limited operational visualization, not full semantic analytics.

Tiny Details You Should Remember

- Dashboard is one service page; report can have pages.
- Semantic model holds data relationships/measures.
- Refresh/gateway/credentials are operational dependencies.
- Embed and data permission are separate.

Exam Memory Notes

- Data model → Semantic model.
- Analysis → Report.
- Pinned overview → Dashboard.
- Secure at both host app and Power BI.

Module Knowledge Check — Questions

1. Which artifact holds measures and relationships?
2. Which artifact is multi-page?
3. Does model-driven app sharing grant dataset access?
4. What may be required for on-prem refresh?
5. Which is better for cross-source advanced analytics: chart or Power BI?

Answers & Explanations

1. **Semantic model**.
2. **Report**.
3. **No.** Share Power BI content and configure licensing/RLS.
4. An **on-premises data gateway** and credentials.
5. **Power BI**.

Primary source: Power BI concepts.

Part VII — Solutions, ALM, and Interoperability

Official-order range: Modules 36–38. Sequence intentionally preserved from the current exam catalog.

Module 36 — Introduction to solutions for Microsoft Power Platform

Your current position: Part VII — Solutions, ALM, and Interoperability → Module 36 → 6 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Manage environments (15–20%)
Official module	Introduction to solutions for Microsoft Power Platform
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	6
Instructor focus	Solutions, publishers, components/dependencies/layers, Managed/Unmanaged, versioning, Environment Variables and Connection References.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Solutions, publishers, components/dependencies/layers, Managed/Unmanaged, versioning, Environment Variables and Connection References.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Solution layering

Concept / Why: A Solution packages components and dependencies for ALM transport and solution layering.

Configuration / How: Configure the Publisher and version, Managed or Unmanaged export and import, dependencies, and Solution Checker.

Exam relevance: High/Medium — Dev unmanaged; downstream managed; active layer trap.

Official page subtopics: Managed layers; Unmanaged layer.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Solution architecture tools and techniques

Concept / Why: A Solution packages components and dependencies for ALM transport and solution layering.

Configuration / How: Configure the Publisher and version, Managed or Unmanaged export and import, dependencies, and Solution Checker.

Exam relevance: High/Medium — Dev unmanaged; downstream managed; active layer trap.

Official page subtopics: Managed solution updates and upgrades.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Use version control for solutions

Concept / Why: A Solution packages components and dependencies for ALM transport and solution layering.

Configuration / How: Configure the Publisher and version, Managed or Unmanaged export and import, dependencies, and Solution Checker.

Exam relevance: High/Medium — Dev unmanaged; downstream managed; active layer trap.

Official page subtopics: Microsoft Power Platform CLI; Solution unpack command.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you've learned..

Official Microsoft Learn page: Open Unit 5

Unit 6 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide's Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 6

Instructor Deep Dive — Solutions and ALM masterclass

Solution purpose

Solution groups Power Platform components and dependencies for lifecycle/deployment. Publisher supplies customization prefix and ownership identity. Build in a custom unmanaged Solution in development; export for downstream deployment.

Managed vs Unmanaged

Unmanaged Solution	Managed Solution
Source/customization container	Deployment artifact/layer
Editable components in development	Controlled changes; managed properties govern customization
Deleting solution container does **not** delete unmanaged customizations	Uninstall removes managed components (and can remove related data depending component)
Export as unmanaged for another dev source or managed for test/prod	Import/update/upgrade in downstream

Recommended path: Dev source = unmanaged; Test/UAT/Prod = managed unless specific organizational ALM policy says otherwise. Never “fix” managed production by making unmanaged active-layer edits; they override managed updates and create drift.

Default Solution and Common Data Services Default Solution

Default Solution contains all customizations and cannot be exported. Common Data Services Default Solution can be default preferred location when no custom preferred solution, but system/default solutions are poor separation and use default prefix. Create custom Solution with publisher; set preferred Solution where supported so new components land correctly.

Components and dependencies

Tables, columns, relationships, forms, views, apps, flows, connection references, environment variable definitions, BPFs, plug-ins, web resources, etc. Components depend on others. Before export, add required objects; use dependency checker. “Add table” may allow include all objects or selected metadata. Missing dependency causes import failure or runtime defect; including everything creates oversized coupling.

Solution layers

Managed solutions create layers. Active unmanaged layer sits highest and overrides managed layers. View Solution Layers to diagnose why imported change not visible. Removing active customization may reveal underlying managed layer. Managed properties restrict downstream customization/deletion.

Versioning, patch, update, upgrade

- Version convention supports major.minor.build.revision governance.

- Update imports new/changed components but may not remove components absent from new version as upgrade does.
- Upgrade can remove obsolete components through holding solution/apply upgrade pattern.
- Patch contains incremental changes, cannot delete components, tied to one parent. Current docs: after first patch, parent base is locked from changes until patches handled; clone solution consolidates patches into new base version.

Choose based on stated lifecycle. Do not use patch as permanent branch system.

Environment Variables

Environment Variable = deployable **Definition** + environment-specific **Current Value**. If current value exists it wins; otherwise Default Value is used. One current value per definition. Publisher update should not overwrite customer current value. Use for URL, email, site ID, feature flag—non-secret configuration (secret support/design requires current secure approach; never store plain credentials casually).

Connection References

Connector defines API; Connection stores credentials; Connection Reference is Solution component that points flow/app/plug-in to actual Connection. During import, map reference to target environment connection. Do not hard-code developer connection.

Environment Variable	Connection Reference
Stores/configures value	Points to authenticated connector connection
Example: target site URL, approval mailbox	Example: Outlook/SharePoint/Dataverse connection
Definition/default/current	Reference → Connection

Solution-aware flows and child flows

Create/add Flow in Solution to use connection references, environment variables, ALM, child flow capabilities and proper owner/dependency management. Child flows generally require solution context and are called by parent, reducing duplicated actions. Test connection mappings and turn-on state after import.

Checker tools

- App Checker: analyzes Canvas/Model-driven app formulas/accessibility/performance issues as supported.
- Solution Checker: static analysis of solution components for best practices/performance/reliability. Passing does not guarantee import or business correctness.

Dev → Test → Prod scenario

1. Dev: custom publisher + unmanaged Solution; environment variables/defaults; connection references; solution-aware apps/flows.

2. Validate dependencies, app/solution checker, tests; increment version.
3. Export Managed for Test; import; set Current Values/map Connections; run smoke/security/flow tests.
4. Promote same built artifact to Prod; map prod values/connections; use service account/ownership strategy.
5. Update/upgrade with versioned releases; avoid unmanaged production edits.

⚠ PL-200 Exam Traps

- Deleting unmanaged Solution ≠ undoing customizations.
- Managed import change not visible → check active unmanaged layer.
- URL/email varies per environment → Environment Variable.
- Credentials/connector connection varies → Connection Reference mapped to target Connection.
- Default Solution cannot be exported.
- Solution Checker success ≠ import success.

Scenario-based learning

Scenario: Flow uses SharePoint site URL and SharePoint connection; both differ across Dev/Test/Prod.
Correct: Environment Variable for site URL; Connection Reference for connector connection.
Why hard-code wrong: Import points to developer resources/credentials.

Scenario: Managed v2 imported but old field still appears; maker made direct production form change.
Correct reasoning: Inspect Solution Layers; active unmanaged layer overrides managed layer. Remove/reconcile unmanaged customization.

Tiny Details You Should Remember

- Unmanaged container deletion leaves customizations.
- Managed uninstall can remove managed components/data; assess before action.
- Active unmanaged layer is highest.
- Environment Variable Current Value overrides Default.
- `_upgrade` suffix/name is reserved by platform processes; avoid using it as normal Solution name.
- Patch cannot delete components.

Exam Memory Notes

- Dev = Unmanaged source.
- Test/Prod = Managed artifact.

- Config value = Environment Variable.
- Connector binding = Connection Reference.
- Layer issue = inspect Solution Layers.
- Static quality = Checker; still test import/runtime.

Module Knowledge Check — Questions

1. Which Solution type should be source in Dev?
2. What happens when an unmanaged Solution container is deleted?
3. Which layer normally overrides managed layers?
4. What stores a per-environment URL?
5. What points a Flow to an authenticated connection?
6. Can a Patch delete obsolete components?
7. Does Solution Checker guarantee successful import?
8. Why import same managed artifact to Test and Prod?

Answers & Explanations

1. **Unmanaged** custom Solution.
2. Its unmanaged customizations remain; only container is deleted.
3. The **active unmanaged layer**.
4. **Environment Variable**.
5. **Connection Reference**.
6. **No.** Use appropriate upgrade strategy.
7. **No.** It is static analysis; dependencies/config/runtime still require testing.
8. It preserves release integrity and reduces environment-specific drift.

Primary sources: Solution concepts, Solution layers, Update/upgrade/patch, Environment Variables, Connection References, Preferred Solution, Solution Checker.

Module 37 — Use Microsoft 365 services with model-driven apps and Microsoft Dataverse

Your current position: Part VII — Solutions, ALM, and Interoperability → Module 37 → 10 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Manage environments (15–20%)
Official module	Use Microsoft 365 services with model-driven apps and Microsoft Dataverse
Catalog last modified	2026-06-11T23:11:00+00:00
Official units	10
Instructor focus	Server-side email sync, mailboxes, SharePoint/document options, Outlook and Teams integration.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Server-side email sync, mailboxes, SharePoint/document options, Outlook and Teams integration.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Set up mailboxes

Concept / Why: Server-side synchronisation processes email and activities through Mailbox and Exchange integration.

Configuration / How: Configure the profile, Mailbox approval, Test and Enable process, tracking, and permissions.

Exam relevance: High/Medium — Approved ≠ enabled/test-success.

Official page subtopics: How email integration works; Microsoft Exchange Online; Advanced settings; Mailboxes; Set up mailboxes; Approve email.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Server-side sync

Concept / Why: Server-side synchronisation processes email and activities through Mailbox and Exchange integration.

Configuration / How: Configure the profile, Mailbox approval, Test and Enable process, tracking, and permissions.

Exam relevance: High/Medium — Approved ≠ enabled/test-success.

Official page subtopics: Email settings; Email tracking; Track email conversations; Folder-level tracking; Tracking between people.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Document management options

Concept / Why: Server-side email sync, mailboxes, SharePoint/document options, Outlook and Teams integration.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Microsoft SharePoint; Note attachments; Dataverse File column; Connectors to file storage.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Document management

Concept / Why: Server-side email sync, mailboxes, SharePoint/document options, Outlook and Teams integration.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Get started; Set up a new organization; Define tables to use with document management; Use document management.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Deploy the app for Outlook

Concept / Why: Server-side email sync, mailboxes, SharePoint/document options, Outlook and Teams integration.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Enable the app for users.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Set up Dynamics 365 App for Outlook

Concept / Why: Server-side email sync, mailboxes, SharePoint/document options, Outlook and Teams integration.

Configuration / How: On the official page, verify prerequisites, available options, notes and warnings, dependencies, security behavior, and the expected exercise result.

Exam relevance: Medium — this is supporting detail for the module objective.

Official page subtopics: Set up forms; Set up views.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Integrate with Microsoft Teams

Concept / Why: A Team is a security principal that groups users and can provide roles, ownership, or record-specific collaboration.

Configuration / How: Choose Owner Team, Access Team, or Microsoft Entra ID Group Team; then configure Business Unit, membership, Security Roles, and any record ownership.

Exam relevance: High/Medium — Access Team cannot own or hold Security Roles.

Official Microsoft Learn page: Open Unit 8

Unit 9 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 9

Unit 10 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 10

Instructor Deep Dive — Email, SharePoint, Outlook, Teams interoperability

Server-side synchronization and mailboxes

Dataverse email integration uses Mailbox rows, email server profile/Exchange Online configuration, approval where required, and Test & Enable Mailbox. Incoming/outgoing email and appointments/contacts/tasks synchronization behavior configured. A user Security Role alone does not make mailbox work; mailbox approval, address, server profile, method, and test status matter.

Email tracking options include manual tracking, correlation/conversation tracking, folder-level tracking as configured. Org-level email settings can affect maximum file size, processing, correlation, and mailbox behavior.

Document management options

Option	Best for	Key consideration
SharePoint	Collaborative documents, versioning, libraries	Server-based integration, site/location permissions
Notes attachments	Simple row notes/files	Dataverse file/log capacity; limited document management
Dataverse File column	Typed file directly on row	Dataverse storage/API/security
Connector to file storage	Automated integration	Flow ownership, connection, error handling

SharePoint setup: enable server-based integration, specify site, choose tables, create/validate document locations, permissions. SharePoint security and Dataverse security remain separate; document link visibility does not automatically grant file access.

Dynamics 365 App for Outlook / Teams

App for Outlook enables track/set regarding/create Dataverse rows in Outlook context; deploy eligibility, mailbox sync, roles, forms/views. Teams integration enables collaboration/record links/embedded experiences according to current capabilities. Blueprint’s explicit interoperability objectives focus email, SharePoint, document management, Word templates; Outlook/Teams units are official prep but relatively lower explicit weight.

⚠ PL-200 Exam Trap

Mailbox `Approved` does not mean `Test & Enable` succeeded. SharePoint document access requires SharePoint permission even if user can Read Dataverse record.

Scenario-based learning

Scenario: User can open Account but gets access denied opening related document.
Correct reasoning: Check SharePoint site/library/file permissions and document location/integration, not only Dataverse Security Role.

Tiny Details You Should Remember

- Mailbox approval and Test & Enable are distinct.
- Server-side sync configuration exists at org/profile/mailbox levels.
- Document location metadata can exist even when file permission denies.
- Note/File/SharePoint storage choices have different capacity and collaboration behavior.

Exam Memory Notes

- Email sync → Server-side sync + Mailbox.
- Collaborative docs → SharePoint integration.
- Simple attachment → Note/File after requirements.
- Two security systems: Dataverse and SharePoint.

Module Knowledge Check — Questions

1. Which row is tested/enabled for server-side email sync?
2. Does Dataverse record Read grant SharePoint file Read?
3. Which option provides SharePoint versioning/libraries?
4. What should be checked after mailbox approval?
5. Which app supports tracking Outlook email to Dataverse?

Answers & Explanations

1. The user/queue **Mailbox**.
2. **No.** SharePoint permissions apply.
3. **SharePoint document management**.
4. Server profile/method, then **Test & Enable Mailbox** result/errors.
5. **Dynamics 365 App for Outlook** (current naming/context).

Primary sources: Server-side synchronization, SharePoint document management.

Module 38 — Use Microsoft Word and Excel templates with Dataverse

Your current position: Part VII — Solutions, ALM, and Interoperability → Module 38 → 7 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Manage environments (15–20%)
Official module	Use Microsoft Word and Excel templates with Dataverse
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	7
Instructor focus	Word/Excel template generation, relationships, security and document-lifecycle boundaries.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Word/Excel template generation, relationships, security and document-lifecycle boundaries.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Use Word templates in Power Apps; Use Excel templates in Power Apps; Enable templates in Power Apps.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Create a dynamic Word template

Concept / Why: A Word or Excel Template merges Dataverse row data into a formatted output.

Configuration / How: Configure the schema and relationships, document controls or spreadsheet formulas, upload and security, and sharing.

Exam relevance: High/Medium — Template ≠ live integration/document management.

Official page subtopics: Create a dynamic Word template.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Exercise – Create a work order template

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; Prepare Dataverse; Prepare data tables; Add sample data to your tables; Create a document template.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Create a dynamic Excel template

Concept / Why: Export and Excel patterns differ depending on whether the requirement is a snapshot, online editing, a reusable template, or an integration.

Configuration / How: Configure the view, filters, permissions, file format, and expectations for a round trip back to Dataverse.

Exam relevance: High/Medium — Static export ≠ sync pipeline.

Official page subtopics: Create a dynamic Excel template.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise – Create a sales forecasting template

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Scenario; Prepare Dataverse; Prepare data; Create a spreadsheet template.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Check your knowledge

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Answer the following questions to see what you’ve learned..

Official Microsoft Learn page: Open Unit 6

Unit 7 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 7

Instructor Deep Dive — Word and Excel templates

Word template

Creates document based on selected Dataverse row and related data exposed in template. Download template schema, insert content controls/repeating sections as supported, upload and enable. Use for quotations, work orders, letters. Relationship data included in template design determines available fields; security applies when user generates.

Excel template

Creates spreadsheet with Dataverse data and can include formulas/charts/pivots depending supported template. Use for forecasts/analysis layouts. It is an output template, not an always-live data integration. Protect formulas/layout and verify refresh/update behavior before promising round-trip.

Word Template	Excel Template
Narrative document/print	Tabular analysis/calculation
Content controls/repeating rows	Tables/formulas/charts
Work order/letter	Forecast/budget

Templates can be organization or personal scope depending creation/sharing. Include in Solution where supported for ALM. Template does not bypass row/column security; missing data may be permission/relationship/template mapping issue.

⚠ PL-200 Exam Trap

Word Template generates document; it does not provide ongoing document management/version control. If collaborative files are required, combine with SharePoint.

Scenario-based learning

Scenario: Generate branded work order with Work Order lines.
Correct: Word Template with related repeating rows, permissions, and optionally save to SharePoint via process.

Tiny Details You Should Remember

- Templates use Dataverse schema/relationships available when created.
- Excel template is not Dataflow.
- Word Template is not SharePoint integration.
- User security influences generated data.

Exam Memory Notes

- Narrative merge → Word Template.
- Tabular forecast → Excel Template.
- Repeat child rows through relationship/template controls.
- Document lifecycle → SharePoint, not template alone.

Module Knowledge Check — Questions

1. Which template suits a formatted customer letter?
2. Which template suits a forecast workbook?
3. Does a Word Template store version history?
4. Can a user generate data they cannot Read?
5. Which feature handles scheduled transformed data load?

Answers & Explanations

1. **Word Template**.
2. **Excel Template**.
3. **No.** Use SharePoint/document management for lifecycle/versioning.
4. **No.** Dataverse security applies.
5. **Dataflow/Power Query**, not Excel Template.

Primary source: Word/Excel templates, Excel templates.

Part VIII — Validation Learning Path and Capstone

Official-order range: Modules 39–40. Sequence intentionally preserved from the current exam catalog.

Module 39 — Get started with Power Automate

Your current position: Part VIII — Validation Learning Path and Capstone → Module 39 → 9 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Create/manage cloud flows (25–30%) — validation path
Official module	Get started with Power Automate
Catalog last modified	2026-06-12T17:23:00+00:00
Official units	9
Instructor focus	Automated/Instant/Scheduled Cloud Flows, approvals, sharing, child flows, Canvas invocation and current agent-flow unit.

Official Unit-by-Unit Journey

Unit 1 — Introducing Power Automate

Concept / Why: Automated/Instant/Scheduled Cloud Flows, approvals, sharing, child flows, Canvas invocation and current agent-flow unit.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Learning objectives; What is Power Automate?; What can you do with Power Automate?; Where can I create and administer a flow?; A brief tour of Power Automate; What's next?.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Create a cloud flow

Concept / Why: A Cloud Flow uses a Trigger, Actions, and Connectors to run event-based, manual, or scheduled automation.

Configuration / How: Configure the Flow type, Trigger, Actions, controls, Connections, error handling, sharing, and Solution inclusion.

Exam relevance: High/Medium — Cloud Flow vs BPF/classic workflow/plug-in.

Official page subtopics: Create a flow with Copilot; Write effective prompts; Edit an existing flow with Copilot.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Exercise - Create recurring flows

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Prerequisites; Format an Excel Table; Create a scheduled flow.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - Monitor incoming emails

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Prerequisites; Specify the SharePoint site and library; Specify an event to start the flow; Specify an action.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Share flows

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Prerequisites; Create a shared flow; Add Microsoft Lists as a co-owner of a flow; Restrictions on changes to flows; Remove an owner; Embedded and other connections.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Troubleshoot flows

Concept / Why: A Cloud Flow uses a Trigger, Actions, and Connectors to run event-based, manual, or scheduled automation.

Configuration / How: Configure the Flow type, Trigger, Actions, controls, Connections, error handling, sharing, and Solution inclusion.

Exam relevance: High/Medium — Cloud Flow vs BPF/classic workflow/plug-in.

Official page subtopics: Identify the error; Authentication failures; Action configuration issues; Temporary issues; Issues with your pricing plan; Issues with data usage.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Convert a flow to an agent flow

Concept / Why: A Cloud Flow uses a Trigger, Actions, and Connectors to run event-based, manual, or scheduled automation.

Configuration / How: Configure the Flow type, Trigger, Actions, controls, Connections, error handling, sharing, and Solution inclusion.

Exam relevance: High/Medium — Cloud Flow vs BPF/classic workflow/plug-in.

Official page subtopics: What is an agent flow?; Before you convert; Convert a cloud flow to an agent flow; Capacity and billing.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Module assessment

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Knowledge check: Get started with Power Automate.

Official Microsoft Learn page: Open Unit 8

Unit 9 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 9

Instructor Deep Dive — Power Automate complete cloud-flow foundation

Cloud Flow types

Flow type	Trigger model	Scenario
Automated Cloud Flow	Event occurs	Dataverse row created, email arrives
Instant Cloud Flow	User/manual invocation	Button, Power Apps call, selected row
Scheduled Cloud Flow	Recurrence/time	Every weekday 08:00 summary

`Desktop Flow` automates UI/RPA on desktop. It is not explicitly named in current PL-200 Skills Measured; know the distinction only. `Business Process Flow` is Dataverse guided stages, not a Cloud Flow type despite "Flow" name.

Core components

Trigger starts run. Actions perform work. Connector exposes triggers/actions and authenticates through Connection. Dynamic Content passes outputs. Expressions transform. Controls branch/loop/scope. Run History monitors.

Create flow design checklist

1. Choose minimal correct trigger and connection identity.
2. Filter at trigger/source to reduce runs.
3. Add actions; rename them.
4. Add Condition/Switch/Apply to each/variables only as needed.
5. Add error Scopes, Run After, retry/idempotency.
6. Test with success, no-data, permission, duplicate, and failure cases.
7. Put production flow in Solution; use Connection References/Environment Variables.
8. Set ownership/support/monitoring.

Approvals

Approval action creates request, waits/continues based on response. Choose First to respond vs Everyone must approve as requirement dictates; handle rejection, timeout, cancellation, reassignment/ownership, links, and audit trail. Long-running approval is Cloud Flow fit, not real-time transaction or Business Rule.

Canvas App calls Flow

Use Power Apps trigger/input parameters; add Flow to app; invoke `FlowName.Run(...)`; handle returned result/error. Share app and grant flow run-only/connection/data access. If flow signature changes, refresh/re-add and test.

Sharing, ownership, run-only

- Co-owner can edit/manage/run/history/share broadly, but cannot remove original creator according to current sharing guidance.
- Run-only user can invoke but not edit or inspect full run history.
- Configure whether users supply own connection or use provided connection as supported—least privilege and licensing.
- Production service ownership avoids flow orphaning when employee leaves; governance and credentials still required.

Child flows

Reusable sub-process invoked by parent. Solution-aware setup and compatible trigger/action required. Use for repeated logic, central error contract, and maintainability. Pass explicit inputs/outputs; avoid circular dependencies.

Classic workflow vs Cloud Flow vs BPF

Feature	Best use	Runtime
Cloud Flow	Cross-service, approvals, async/manual/schedule	Power Automate run
Classic background workflow	Legacy Dataverse async process/waits	System Job
Classic real-time workflow	Legacy synchronous Dataverse operation/cancel	Transaction
BPF	Human stage guidance	Process instance/table
Low-code plug-in	Reusable/transactional server Power Fx	Dataverse event/action

Convert to agent flow — current module

Current Module 39 includes converting a Flow to an agent flow. Treat as modern official preparation content. Verify eligibility, supported actions, agent inputs/outputs, governance, and current Preview/GA status at exam time. It is not a separately named PL-200 objective, so prioritize Cloud Flow core first.

⚠ PL-200 Exam Traps

- “Every Monday” → Scheduled; not Automated.
- “User clicks button” → Instant.
- “When row created” → Automated.
- Share Flow as Run-only when user should invoke but not edit.
- Flow created outside Solution can be added, but solution-aware design/connection references are essential for ALM.

Scenario-based learning

Scenario: Manager manually requests a customer credit refresh from Canvas App and receives status.
Correct: Instant Cloud Flow with Power Apps trigger/input/output; run-only sharing.

Scenario: Every night send summary of overdue cases.
Correct: Scheduled Cloud Flow + List rows filtered at source + email + error handling.

Scenario: New high-value Opportunity requires all three directors' approval.
Correct: Automated Cloud Flow + Everyone must approve + rejection/timeout branches; do not use BPF alone.

Tiny Details You Should Remember

- Default action run-after is success.
- Flow owner/share does not bypass connector/data permissions.
- Run-only ≠ co-owner.
- Instant Power Apps flow needs app/flow permissions and parameter contract.
- Desktop/agent flows are lower explicit priority than Cloud Flow core for current blueprint.

Exam Memory Notes

- Event → Automated.
- Button → Instant.
- Time → Scheduled.
- Invoke only → Run-only.
- Reuse → Child Flow in Solution.
- Human stages → BPF; background action → Cloud Flow.

Module Knowledge Check — Questions

1. Which type runs every weekday at 8 AM?
2. Which type starts from Canvas App button?
3. Which permission lets a user run but not edit Flow?
4. Which approval option requires all approvers?
5. What ALM components replace hard-coded connection and URL?
6. Which feature shows stages to a user?
7. Is Desktop Flow a current named PL-200 skill objective?
8. Why use a Child Flow?

Answers & Explanations

1. **Scheduled Cloud Flow**.

2. **Instant Cloud Flow** using Power Apps trigger.
3. **Run-only** access.
4. **Everyone must approve**.
5. **Connection Reference** and **Environment Variable**.
6. **Business Process Flow**.
7. **No.** Current blueprint names Cloud Flows; know desktop distinction only.
8. To reuse a solution-aware subprocess and avoid duplicated logic.

Primary sources: Cloud Flow overview, Flow types, Sharing permissions, Solution-aware flows.

Module 40 — Challenge project - Build applications and automation solutions

Your current position: Part VIII — Validation Learning Path and Capstone → Module 40 → 8 official Units/Pages.

Module metadata	Current verified value
Exam relevance	Integrated validation capstone — all domains
Official module	Challenge project - Build applications and automation solutions
Catalog last modified	2026-05-26T17:10:00+00:00
Official units	8
Instructor focus	Requirement decomposition and full Dataverse + apps + flow + ALM capstone validation.

Official Unit-by-Unit Journey

Unit 1 — Introduction

Concept / Why: Requirement decomposition and full Dataverse + apps + flow + ALM capstone validation.

Configuration / How: First verify the learning objectives, prerequisites, required Environment or licence, and exercise setup.

Exam relevance: Foundation — identify the product boundary and the main feature choice in the scenario.

Official page subtopics: Example scenario; What are we doing?; What is the main goal?.

Official Microsoft Learn page: Open Unit 1

Unit 2 — Prepare

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Project specification; Microsoft Dataverse requirements; Model-driven application requirements; Canvas application requirements; Power Automate Cloud Flow requirements.

Official Microsoft Learn page: Open Unit 2

Unit 3 — Exercise - Create Microsoft Dataverse tables, columns, and additional assets

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Specification; Player table columns; Other requirements for the Player table; Scouting report table columns; Contact table columns; Table relationships.

Official Microsoft Learn page: Open Unit 3

Unit 4 — Exercise - Compose a model-driven app

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Considerations; Check your work.

Official Microsoft Learn page: Open Unit 4

Unit 5 — Exercise - Build a canvas app

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Considerations; Check your work.

Official Microsoft Learn page: Open Unit 5

Unit 6 — Exercise - Automate with a cloud flow

Concept / Why: This is a hands-on configuration page. Focus on dependencies and the expected result, not only the click sequence.

Configuration / How: Follow the official steps in a Development or Developer Environment, then test ordinary-user security and at least one failure path.

Exam relevance: Practice — an exam question can ask for the configuration outcome or the correct next step.

Official page subtopics: Considerations; Check your work.

Official Microsoft Learn page: Open Unit 6

Unit 7 — Module assessment

Concept / Why: This page is the official self-assessment. Do not memorise an answer; explain the decision rule that makes it correct.

Configuration / How: Attempt it without hands-on notes. For every incorrect option, write down exactly why it does not fit the requirement.

Exam relevance: Review — use the question style for practice and use the previous units as the content source.

Official page subtopics: Check your knowledge.

Official Microsoft Learn page: Open Unit 7

Unit 8 — Summary

Concept / Why: This recap page brings the module learning objectives together.

Configuration / How: Compare it with this guide’s Memory Notes and revisit any Knowledge Check answer you missed.

Exam relevance: Review — last-pass memory consolidation.

Official Microsoft Learn page: Open Unit 8

Instructor Deep Dive — Capstone challenge: build, test, explain

Official challenge project combines Dataverse tables/columns/relationships/assets, Model-driven App, Canvas App, and Cloud Flow. Exam value is not memorizing exercise clicks; it is turning requirement sentences into platform decisions.

Requirement decomposition worksheet

Requirement question	Artifact
What business concepts/rows?	Tables + ownership
What values and constraints?	Columns, Choice/Lookup, keys, formula/rollup
How concepts relate?	1:N/N:N/junction + cascade
Who can do what?	BU, roles, teams, sharing, column security
Which internal data-first UX?	Model-driven forms/views/navigation

Requirement question	Artifact
Which task/mobile UX?	Canvas screens/galleries/forms/Power Fx
Which event action?	Cloud Flow trigger/actions/errors
How deploy?	Solution, publisher, environment vars, connection refs

Capstone implementation order

1. Create custom Publisher/Solution.
2. Design tables, ownership, keys, relationships, cascade.
3. Configure columns, choices, forms/views, Security Roles.
4. Build Model-driven App and test roles/forms/process.
5. Build Canvas App; handle navigation, formulas, delegation, errors.
6. Build solution-aware Flow; filter, idempotency, Run After, sharing.
7. Checker, dependency, import test, negative security tests.

Oral-defense test

For every artifact, be able to answer:

- Why this feature, not nearest alternative?
- Which identity executes it?
- Which Security Role/permission is required?
- Which dependency moves in Solution?
- What happens on failure/retry?
- What happens at 100,000 rows?
- Can a user bypass the UI and still be safely blocked?

⚠ PL-200 Exam Trap

Exercise completion does not prove exam readiness. You must explain design choice and failure/security behavior. If you cannot explain why Choice vs Lookup or Access Team vs Owner Team, rebuild the requirement map.

Scenario-based learning

Scenario: Scout submits report in mobile UI; manager manages master players in data-centric app; approval notification is automated.
Correct composition: Canvas App for scout task; Model-driven App for data management; Dataverse shared model/security; Cloud Flow for approval; all in Solution.

Tiny Details You Should Remember

- Start in Solution, not default customization area.
- Test maker and ordinary roles.

- Import into clean test environment catches hidden dependencies.
- Delegation, trigger loops, and row security are capstone failure hotspots.

Exam Memory Notes

- Requirement → data → security → UX → automation → ALM.
- One platform can combine app types.
- Negative test every security boundary.
- Explain why alternatives fail.

Module Knowledge Check — Questions

1. What should be designed before app screens?
2. Where should production components be created for ALM?
3. Which test catches missing dependencies?
4. Which app fits a focused mobile task?
5. Which app fits relational back-office management?
6. Which issue can make Canvas results incomplete without obvious error?
7. Which issue can make a Dataverse-triggered Flow run forever?
8. What is the final exam-reasoning question for every choice?

Answers & Explanations

1. **Data model, ownership, relationships, security, and requirements.**
2. In a custom **Solution** with correct publisher.
3. Import into a **clean Test environment** plus dependency/checker review.
4. **Canvas App**.
5. **Model-driven App**.
6. **Nondelegable query** over a large source.
7. Updating the same row without trigger guard.
8. “Why is this the supported, least-complex, secure choice—and why are distractors worse?”

Primary source: Official challenge project.

PL-200 Final Revision

Most Important Topics

1. Solve Dataverse security by checking both the action privilege and access to the specific row.
2. Table ownership, Teams, Business Units, Sharing, Column/Hierarchy Security.
3. Relationship direction, Lookup location, cascade, N:N vs junction.
4. Formula/Calculated/Rollup/Business Rule/Flow/plug-in feature selection.
5. Model-driven form/view/navigation/process selection.
6. Canvas Power Fx, Forms vs Patch, state scope, delegation, data security.
7. Power Pages Web Role + Page/Table Permission + Contact/Account scope.
8. Cloud Flow triggers, filters, controls, errors, sharing, solution awareness.
9. Managed/Unmanaged, layers, dependencies, Environment Variables, Connection References.
10. Email/SharePoint/Word template interoperability and separate permission layers.

Security Cheat Sheet

Record access algorithm

1. Environment/license/enabled/security-group gate.
2. App shared/role access.
3. Table action privilege exists?
4. Privilege depth reaches row through owner/BU/team/share/hierarchy?
5. Column Security allows field?
6. Relationship operation has Append + Append To?

Signal	Correct concept
All allowed users see same reference rows	Organization-owned table + Organization privilege
Each rep owns rows	User/Team-owned + User depth
Group collectively owns rows	Owner Team or Entra Group Team
Changing collaborators, selected row, no ownership	Access Team
One-off row access	Sharing
All rows in a BU	BU-depth Security Role

Signal	Correct concept
Manager/subordinate extension	Hierarchy Security + base Read
Hide sensitive column in all channels	Column Security + profile

Security truth: Grants are cumulative. Dataverse has no simple “deny role” to subtract broad permission. UI hiding is not security.

Dataverse Cheat Sheet

Requirement	Feature
External data, no copy	Virtual Table, after limitation review
Human-readable generated ID	Autonumber
Integration natural key/upsert	Alternate Key
Fixed option list	Choice
Option has attributes/relationships	Lookup to table
Related child aggregate	Rollup
Same-row Power Fx result	Formula Column
Duplicate warning/find	Duplicate Detection
Uniqueness enforcement	Alternate Key
Change history	Auditing
Scheduled transform/load	Dataflow + Power Query
Saved deployable query	System View

Model-driven App Cheat Sheet

Need	Component
Full edit/read	Main Form
Fast minimal create	Quick Create Form
Related read-only fields	Quick View Form
Related rows	Subgrid
Activities/notes history	Timeline
Human stages	BPF
App-level custom Canvas experience	Custom Page

Need	Component
Form-context custom Canvas	Embedded Canvas App
Button action/visibility	Modern Command + Power Fx
Operational visual	Chart/Dashboard
Cross-source analytics	Power BI

Canvas App Cheat Sheet

- Many rows → Gallery; one record → Form.
- Standard create/edit → SubmitForm; complex partial/custom update → Patch.
- Filter/Search return tables; LookUp first record.
- Set global; UpdateContext screen; ClearCollect local table.
- Save success actions in Form.OnSuccess; failures in OnFailure/IfError.
- Delegation warning = incomplete/wrong result risk at scale.
- Share app + data source/roles/connections separately.

Power Automate Cheat Sheet

Signal	Feature
Event	Automated Flow
Button/user call	Instant Flow
Time/recurrence	Scheduled Flow
Two branches	Condition
Many cases	Switch
Array	Apply to each
Failure path	Configure Run After + Scopes
Reuse	Child Flow in Solution
Dataverse key create/update	Upsert
Avoid excess runs	Trigger Filter rows/conditions
Deploy connection	Connection Reference

Solution / ALM Cheat Sheet

- Build in custom Unmanaged Solution in Dev.
- Export/import Managed artifact to Test/Prod.

- Default Solution cannot export; deleting unmanaged container leaves customizations.
- Active unmanaged layer overrides managed layers.
- Dependency must be added; “missing dependency” is not solved by turning Flow off.
- Environment Variable = config value; Current > Default.
- Connection Reference = deployable pointer to target Connection.
- Patch incremental and cannot delete; Upgrade handles removal path.
- Run App Checker/Solution Checker, then still import/runtime/security test.

Important Comparisons

A vs B	Deciding difference
Canvas vs Model-driven	UX-first/free layout vs data-first/Dataverse metadata
Business Rule vs Power Fx	Model-driven/table declarative field rule vs app/formula behavior
Business Rule vs Cloud Flow	Immediate field logic vs async/cross-service automation
BPF vs Cloud Flow	Guided human stages vs automated actions
Formula vs Rollup	Same-row/fetch formula vs related aggregate job
Calculated vs Formula	Classic condition/action calculation vs modern Power Fx formula
Choice vs Lookup	Option value vs related record
1:N vs N:N	Child Lookup vs association; junction if relationship attributes
Owner Team vs Access Team	Owns+roles vs record-specific access/no ownership/no roles
Owner Team vs Entra Group Team	Manual membership vs Entra-managed membership; both can own/roles
User-owned vs Organization-owned	Owner/depth/share vs org/none uniform access
Security Role vs Sharing	Reusable broad rule vs selected row grant
Business Unit vs Team	Security hierarchy boundary vs user group/owner/access principal
Managed vs Unmanaged	Deployable controlled layer vs editable source
Environment Variable vs Connection Reference	Value vs authenticated connection pointer

A vs B	Deciding difference
Main vs Quick Create vs Quick View	Full/role-based vs minimal create vs related read-only
Personal vs System View	User-owned vs deployable shared metadata
Command Bar vs BPF	User action vs stage guidance
Instant plug-in vs Automated plug-in	Called with parameters/results vs Dataverse event/transaction

Important Terminology

Term	One-line meaning
Principal	User or Team that can receive security access
Privilege	Action permission on a table
Access depth	Which owned rows privilege reaches
Publisher prefix	Unique schema prefix for Solution components
Active layer	Current unmanaged customization above managed layers
Connection	Credentials/authenticated connector session
Connector	API definition with triggers/actions
Connection Reference	Solution-aware pointer to Connection
Environment Variable	Deployable configuration definition/value
Delegation	Sending query processing to data source
Trigger	Event/manual/time starting a Flow
Dynamic Content	Previous action output token
Web Role	Power Pages authorization group for Contacts
Table Permission	Power Pages table operations + row scope

Important Verified Limitations

- Canvas nondelegation local processing: default 500, configurable up to 2,000; larger query may be wrong.
- BPF: current docs say 10 active per table, 30 stages per process, up to 5 tables.
- Rollup: default 200 per environment / 50 per table; online calculation up to 50,000 related rows; hierarchy depth 10; no N:N or rollup-on-rollup.

- Quick Create: no subgrid/Quick View/web resource/iframe/notes/Bing Maps; one form for users; no role assignment.
- Quick View: read-only, dependent on populated Lookup, no role assignment/scripts.
- Virtual Tables: many Dataverse capabilities unsupported, including several audit/search/calculation/process/offline features—always check current list.
- Custom table ownership cannot be changed after creation.
- Alternate Key cannot use secured columns; NULL uniqueness and special-character API caveats exist.
- Instant low-code plug-in current module: inputs cannot be optional; display name immutable after save; default scope Global.

Current, modern, legacy, and deprecated-awareness map

Category	Treatment for current PL-200
Core current	Dataverse, Model-driven, Canvas, Power Pages, Cloud Flows, Solutions/ALM
Current and explicitly named	Low-code plug-ins, Power Fx commanding, custom pages, classic workflows, BPF
Current prep, not separately named objective	Prompt columns, Power Apps Cards, Copilot-generated artifacts, Power Pages AI agent, agent-flow conversion
Legacy but still explicit exam objective	Classic Dataverse background/real-time workflows
Awareness only, not named current objective	Desktop Flows, dedicated AI Builder/Copilot Studio curriculum

When old course material conflicts with current docs, use current terminology and current supported designer. Example: Entity → Table, Field → Column, Option Set → Choice; legacy UI terms can still appear in older component names/questions.

Common Exam Traps

1. App/Form/View visibility is not data security.
2. Environment Maker is not Dataverse row access.
3. Write privilege without row reach cannot edit.
4. Broader role cannot be cancelled by narrower role.
5. Access Team cannot own or take a Security Role.
6. Lookup on N side; 1:N and N:1 are viewpoints.
7. N:N with attributes needs junction table.
8. Rollup is not real-time and not N:N.

9. Form Role only selects Main Form; it does not authorize data.
10. Power Pages View filter/hidden navigation is not authorization.
11. Canvas app sharing does not share data.
12. Nondelegation is correctness, not only performance.
13. Condition after trigger wastes run compared with trigger filtering.
14. Flow retry can duplicate side effects.
15. Environment Variable and Connection Reference solve different ALM problems.
16. Active unmanaged production layer can hide managed update.

Last-Day Revision Checklist

- ☐ Official retirement/date/schedule still valid; exam booked in correct timezone.
- ☐ Four domains and weights memorized.
- ☐ I have practised solving scenarios with the security algorithm.
- ☐ I can explain Append compared with Append To.
- ☐ I can compare Team types and record ownership.
- ☐ I can choose between Choice and Lookup, relationship and cascade behaviours, and Formula and Rollup columns.
- ☐ I understand form types, View types, and their security limitations.
- ☐ I understand Canvas formulas, state, `Patch`, `SubmitForm`, and delegation.
- ☐ I understand the Power Pages authentication and authorisation chain.
- ☐ I can reason about Cloud Flow types, controls, errors, and Dataverse triggers.
- ☐ I can compare a Business Process Flow, classic workflow, Cloud Flow, and plug-in.
- ☐ I can explain Managed and Unmanaged Solutions, layers, Environment Variables, and Connection References.
- ☐ I have completed the Microsoft exam sandbox and the official Practice Assessment.
- ☐ I revised core objectives without over-prioritising preview or modern-awareness units.

Official Skills Measured Coverage Matrix

Official domain	Objective group	Exact current skill	Covered?	Document section
Configure Microsoft Dataverse	Manage the data model	Create or modify standard, activity, or virtual tables	**Yes**	Modules 2–3
Configure Microsoft Dataverse	Manage the data model	Create new tables or modify existing tables	**Yes**	Modules 2–3
Configure Microsoft Dataverse	Manage the data model	Determine which type of relationship to implement, including one-to-many and many-to-many	**Yes**	Module 5
Configure Microsoft Dataverse	Manage the data model	Configure table relationships including behaviors and cascading rules	**Yes**	Module 5
Configure Microsoft Dataverse	Manage the data model	Create new or modify existing columns, including rollup and formula	**Yes**	Modules 3, 6
Configure Microsoft Dataverse	Manage the data model	Configure table properties	**Yes**	Module 2
Configure Microsoft Dataverse	Manage Dataverse	Configure Dataverse search, and manage the search index	**Yes**	Module 9
Configure Microsoft Dataverse	Manage Dataverse	Manage auditing	**Yes**	Module 9
Configure Microsoft Dataverse	Manage Dataverse	Describe options for importing and exporting data	**Yes**	Module 7
Configure Microsoft Dataverse	Manage Dataverse	Configure duplicate detection settings	**Yes**	Module 9
Configure Microsoft Dataverse	Manage Dataverse	Configure bulk deletion	**Yes**	Module 9

Official domain	Objective group	Exact current skill	Covered?	Document section
Configure Microsoft Dataverse	Configure security settings	Manage business units	**Yes**	Module 8
Configure Microsoft Dataverse	Configure security settings	Create and manage security roles	**Yes**	Module 8
Configure Microsoft Dataverse	Configure security settings	Create and manage users and teams	**Yes**	Module 8
Configure Microsoft Dataverse	Configure security settings	Create and manage column security	**Yes**	Module 8
Configure Microsoft Dataverse	Configure security settings	Configure hierarchy security	**Yes**	Module 8
Configure Microsoft Dataverse	Configure security settings	Configure Microsoft Entra ID group teams	**Yes**	Module 8
Create apps by using Microsoft Power Apps	Create model-driven apps	Create and configure multiple form types	**Yes**	Modules 11–12
Create apps by using Microsoft Power Apps	Create model-driven apps	Use controls in the form designer	**Yes**	Modules 12–13
Create apps by using Microsoft Power Apps	Create model-driven apps	Create and configure views	**Yes**	Modules 7, 11–12
Create apps by using Microsoft Power Apps	Create model-driven apps	Configure custom pages	**Yes**	Module 11
Create apps by using Microsoft Power Apps	Create model-driven apps	Configure modern commanding by using Power Fx	**Yes**	Module 14
Create apps by using Microsoft Power Apps	Create model-driven apps	Embed a canvas app in a model-driven app form	**Yes**	Module 13
Create apps by using Microsoft Power Apps	Create model-driven apps	Add Microsoft Power BI dashboards and reports in a model-driven app	**Yes**	Modules 12, 35

Official domain	Objective group	Exact current skill	Covered?	Document section
Create apps by using Microsoft Power Apps	Describe canvas apps	Describe canvas app structure	**Yes**	Modules 15–16
Create apps by using Microsoft Power Apps	Describe canvas apps	Describe form navigation, formulas, variables and collections, and error handling	**Yes**	Modules 16–18
Create apps by using Microsoft Power Apps	Describe canvas apps	Describe how Microsoft Power Automate flows are called from a canvas app	**Yes**	Module 39
Create apps by using Microsoft Power Apps	Build Microsoft Power Pages	Configure pages, forms, lists, and navigation	**Yes**	Modules 21–24
Create apps by using Microsoft Power Apps	Build Microsoft Power Pages	Describe advanced Power Pages features, including document management, search, multi-step forms, and Power BI	**Yes**	Modules 24–25
Create apps by using Microsoft Power Apps	Build Microsoft Power Pages	Configure website security including web roles and page access	**Yes**	Modules 26–28
Create apps by using Microsoft Power Apps	Build Microsoft Power Pages	Describe use cases for templates	**Yes**	Modules 21, 23
Create apps by using Microsoft Power Apps	Build Microsoft Power Pages	Describe authentication options	**Yes**	Module 28
Create and manage logic and process automation	Create and manage cloud flows	Describe types of flows, use cases, and flow components, including when to use a classic workflow	**Yes**	Modules 32, 39
Create and manage logic and process automation	Create and manage cloud flows	Describe components of a connector	**Yes**	Modules 19, 39

Official domain	Objective group	Exact current skill	Covered?	Document section
Create and manage logic and process automation	Create and manage cloud flows	Implement logic controls including branches, loops, conditions, error handling, and variables	**Yes**	Modules 29, 31, 39
Create and manage logic and process automation	Create and manage cloud flows	Implement dynamic content and expressions	**Yes**	Module 31
Create and manage logic and process automation	Create and manage cloud flows	Work with the Dataverse connector	**Yes**	Module 30
Create and manage logic and process automation	Create and manage cloud flows	Manage and troubleshoot flows	**Yes**	Modules 29, 39
Create and manage logic and process automation	Create and manage business process flows	Configure a business process flow	**Yes**	Module 13; Blueprint Bridge after Module 34
Create and manage logic and process automation	Create and manage business process flows	Add stages, workflows, and flow steps to a business process flow	**Yes**	Module 13; Blueprint Bridge after Module 34
Create and manage logic and process automation	Create and manage business process flows	Manage the business process flow table for a business process flow	**Yes**	Module 13; Blueprint Bridge after Module 34
Create and manage logic and process automation	Create and manage classic Dataverse workflows	Configure a workflow	**Yes**	Module 32 Objective Bridge
Create and manage logic and process automation	Create and manage classic Dataverse workflows	Troubleshoot workflows	**Yes**	Module 32 Objective Bridge
Create and manage logic and process automation	Create and manage classic Dataverse workflows	Manage workflow logs	**Yes**	Module 32 Objective Bridge
Create and manage logic and process automation	Implement low-code logic	Configure low-code plug-ins	**Yes**	Modules 33–34

Official domain	Objective group	Exact current skill	Covered?	Document section
Create and manage logic and process automation	Implement low-code logic	Create and manage Power Fx functions and formulas	**Yes**	Modules 14, 18, 33–34
Create and manage logic and process automation	Implement low-code logic	Create and configure business rules	**Yes**	Module 32
Manage environments	Participate in ALM	Describe use cases for app checker and solution checker	**Yes**	Module 36
Manage environments	Participate in ALM	Create and manage Dataverse solutions	**Yes**	Module 36
Manage environments	Participate in ALM	Describe the difference between managed and unmanaged Dataverse solutions	**Yes**	Module 36
Manage environments	Participate in ALM	Manage Microsoft Power Platform environments for development	**Yes**	Modules 1, 36
Manage environments	Participate in ALM	Import and export Dataverse solutions	**Yes**	Module 36
Manage environments	Manage interoperability with other services	Configure email integration	**Yes**	Module 37
Manage environments	Manage interoperability with other services	Configure Microsoft SharePoint integration	**Yes**	Modules 25, 37
Manage environments	Manage interoperability with other services	Describe options for document management	**Yes**	Modules 25, 37
Manage environments	Manage interoperability with other services	Work with Microsoft Word templates	**Yes**	Module 38

Coverage result: 56/56 current objective bullets mapped. No uncovered objective remains.

Microsoft Learn Module and Unit Coverage Audit

This guide's build audit compares the live catalog sequence captured 2026-08-17 with the source manifest. Covered means the Module has an instructor lesson, exam trap, scenario, Tiny Details, Memory Notes, Knowledge Check, and every official Unit/Page has an individual heading, focus note, and canonical Microsoft Learn link.

Order	Official module	Official units	Guide coverage
1	Create and manage environments in Dataverse	9	**Covered** — 9/9 unit headings + instructor lesson
2	Create tables in Dataverse	12	**Covered** — 12/12 unit headings + instructor lesson
3	Create and manage columns within a table in Dataverse	11	**Covered** — 11/11 unit headings + instructor lesson
4	Work with choices in Dataverse	6	**Covered** — 6/6 unit headings + instructor lesson
5	Create a relationship between tables in Dataverse	8	**Covered** — 8/8 unit headings + instructor lesson
6	Create and define calculation or rollup columns in Dataverse	9	**Covered** — 9/9 unit headings + instructor lesson
7	Load/export data and create data views in Dataverse	10	**Covered** — 10/10 unit headings + instructor lesson
8	Get started with security roles in Dataverse	10	**Covered** — 10/10 unit headings + instructor lesson

Order	Official module	Official units	Guide coverage
9	Use administration options for Dataverse	11	**Covered** — 11/11 unit headings + instructor lesson
10	How to build your first model-driven app with Dataverse	6	**Covered** — 6/6 unit headings + instructor lesson
11	Get started with model-driven apps in Power Apps	13	**Covered** — 13/13 unit headings + instructor lesson
12	Configure forms, charts, and dashboards in model-driven apps	9	**Covered** — 9/9 unit headings + instructor lesson
13	Use specialized components in a model-driven form	8	**Covered** — 8/8 unit headings + instructor lesson
14	Customize the command bar	7	**Covered** — 7/7 unit headings + instructor lesson
15	Get started with Power Apps canvas apps	9	**Covered** — 9/9 unit headings + instructor lesson
16	Customize a canvas app in Power Apps	9	**Covered** — 9/9 unit headings + instructor lesson
17	Navigation in a canvas app in Power Apps	6	**Covered** — 6/6 unit headings + instructor lesson
18	Create formulas to change behaviors in a Power Apps canvas app	8	**Covered** — 8/8 unit headings + instructor lesson

Order	Official module	Official units	Guide coverage
19	Connect to other data in a Power Apps canvas app	5	**Covered** — 5/5 unit headings + instructor lesson
20	Introduction to Power Apps cards	7	**Covered** — 7/7 unit headings + instructor lesson
21	Core components of Power Pages	7	**Covered** — 7/7 unit headings + instructor lesson
22	Access Dataverse in Power Pages websites	6	**Covered** — 6/6 unit headings + instructor lesson
23	Explore Power Pages design studio	9	**Covered** — 9/9 unit headings + instructor lesson
24	Integrate Power Pages websites with Dataverse	8	**Covered** — 8/8 unit headings + instructor lesson
25	Integrate Power Pages with web-based technologies	6	**Covered** — 6/6 unit headings + instructor lesson
26	Explore Power Pages design studio data and security features	6	**Covered** — 6/6 unit headings + instructor lesson
27	Set up Power Pages security	6	**Covered** — 6/6 unit headings + instructor lesson
28	Authentication and user management in Power Pages	8	**Covered** — 8/8 unit headings + instructor lesson

Order	Official module	Official units	Guide coverage
29	Best practices for error handling in Power Automate flows	5	**Covered** — 5/5 unit headings + instructor lesson
30	Use Dataverse triggers and actions in Power Automate	7	**Covered** — 7/7 unit headings + instructor lesson
31	Introduction to expressions in Power Automate	8	**Covered** — 8/8 unit headings + instructor lesson
32	Define and create business rules in Dataverse	6	**Covered** — 6/6 unit headings + instructor lesson
33	Set up low-code plug-ins	9	**Covered** — 9/9 unit headings + instructor lesson
34	Use common low-code plug-in scenarios	9	**Covered** — 9/9 unit headings + instructor lesson
35	Get started building with Power BI	6	**Covered** — 6/6 unit headings + instructor lesson
36	Introduction to solutions for Microsoft Power Platform	6	**Covered** — 6/6 unit headings + instructor lesson
37	Use Microsoft 365 services with model-driven apps and Microsoft Dataverse	10	**Covered** — 10/10 unit headings + instructor lesson
38	Use Microsoft Word and Excel templates with Dataverse	7	**Covered** — 7/7 unit headings + instructor lesson

Order	Official module	Official units	Guide coverage
39	Get started with Power Automate	9	**Covered** — 9/9 unit headings + instructor lesson
40	Challenge project - Build applications and automation solutions	8	**Covered** — 8/8 unit headings + instructor lesson

Audit result: 40/40 modules and 319/319 units/pages covered in official order.

Source and verification notes

- Every Module/Unit page is linked at its Unit heading; this is the complete current official page-level source ledger.
- Key technical claims include direct links in each instructor lesson.
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